

Tensor Analysis with Applications in Engineering

ME 5121 @ NCKU ME

Lecture notes

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Chapter 1

Reviews on Euclidean space

Definition 1.0.1. A Euclidean space is defined as an N -dimensional vector space, denoted by E_N , in which

- (a) The inner product operation between two vectors is defined.
- (b) The metric (or norm), that is, the square of the line element of the space, obtained by applying inner product on vectors themselves, is a positive definite quadratic form (i.e. always real and positive, vanishing only when the line elements vanishes).
- (c) It is possible to express the metric of n squares with constant coefficients by performing a suitable coordinate transformation.

Note. If (c) is not permitted, the vector space is called *N-dimensional Riemannian space*.

Chapter 2

Elements of Cartesian Tensor Analysis

2.1 Indicical notation

Consider $\underline{u}$ and $\underline{v}$ which are elements of E_3

$$\underline{u} \cdot \underline{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 = \sum_{i=1}^3 u_i v_i \stackrel{*}{=} u_i v_i$$

*Einstein's summation convention

2.2. RANGE CONVECTION

2.2 Range convection

- All lower cases, Latin indices (both subscripts and superscripts) have the range 1, 2, 3 or $i, j \in 1, 2, 3$
- All lower cases, Greek indices (both subscripts and superscripts) have the range 1, 2 or $\alpha, \beta \in 1, 2$

e.g. 1) v_i – 3 elements

e.g. 2) v_{ij} – 3^2 elements

e.g. 3) v_{ijk} – 3^3 elements

e.g. 4) $\underbrace{v_{ij\dots k}}_N$ – 3^N elements

2.3 Summation convection

If in a monomial the same index appears *twice* then summation over this index is implied unless otherwise stated.

e.g. 1)

$$v_{ii} = v_{11} + v_{22} + v_{33}$$

v_{ii} (no sum) either v_{11} or v_{22} or v_{33}

e.g. 2)

$$u_i v_i = u_1 v_1 + u_2 v_2 + u_3 v_3 = u_k v_k = u_j v_j$$

thus repeated indices are *dummy*.

CHAPTER 2. ELEMENTS OF CARTESIAN TENSOR ANALYSIS

e.g. 3)

$$a_{ij}x_j = a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3$$

indicial equation

$$a_{ij}x_j = b_i \iff \begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

i.e. 3 equations for $i = 1, 2, 3$

$$a_{ij}x_j = c_{kki} = c_{11i} + c_{22i} + c_{33i}$$

e.g. 4) NONSENSE

$$a_i c_{ji} = b_k$$

(free or non-repeated index on either side is not same)

$$a_{ikk}x_k = c_i$$

(can not have more than two repeated indices)

Rules

- (a) The same index cannot appear more than twice *in the same monomial*.

2.4. KRONECKER DELTA

- (b) Free (unrepeated) indices of either side of an indicial equation must agree.
- (c) Both free and repeated indices are *dummy* and may be replaced by other lower case Latin letters subject to restriction (a) and (b).

2.4 Kronecker delta

Set of a real numbers, denoted by δ_{ij} defined as

$$\delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

e.g. 1) $\delta_{ii} = 3$

e.g. 2) δ_{ii} (no sum) = 1

e.g. 3) $\delta_{ij} = \delta_{ji}$

e.g. 4) $\delta_{ij}v_j = v_i, \delta_{ik}v_{kj} = v_{ij}$

e.g. 5) $\delta_{ij}\delta_{ij} = \delta_{ii} = \delta_{jj} = 3$

e.g. 6) $\delta_{ij}v_{ij} = v_{ii} = v_{jj} = v_{kk}$

e.g. 7) $\delta_{ik}\delta_{jr}v_{kr} = v_{ij}$

CHAPTER 2. ELEMENTS OF CARTESIAN TENSOR ANALYSIS

2.5 Alternating symbol

The 27 real numbers ϵ_{ijk} defined as

$$\epsilon_{ijk} = 1$$

if (ijk) are an *even* permutation of the ordered triplet $(1\ 2\ 3)$.

$$\epsilon_{ijk} = -1$$

if (ijk) are an *odd* permutation of the ordered triplet $(1\ 2\ 3)$.

$$\epsilon_{ijk} = 0$$

otherwise, i.e. at least two are the same.

e.g. 1) $\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$

e.g. 2) $\epsilon_{213} = -1$

e.g. 3) $\epsilon_{121} = 0$

e.g. 4) $\epsilon_{333} = 0$

e.g. 5) $\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij} = -\epsilon_{jik} = -\epsilon_{ikj} = -\epsilon_{kji}$

e.g. 6) $\epsilon_{ijk}\epsilon_{ijk} = \underbrace{\epsilon_{1jk}\epsilon_{1jk}}_{\epsilon_{123}\epsilon_{123} + \epsilon_{132}\epsilon_{132}=2} + \epsilon_{2jk}\epsilon_{2jk} + \epsilon_{3jk}\epsilon_{3jk} = 6$

2.6. REPRESENTATION OF 3×3 DETERMINANTS

2.6 Representation of 3×3 determinants

$$\epsilon_{1jk}v_jw_k = \epsilon_{123}v_2w_3 + \epsilon_{132}v_3w_2 = v_2w_3 - v_3w_2 = \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix}$$

and

$$\epsilon_{2jk}v_jw_k = \epsilon_{213}v_1w_3 + \epsilon_{231}v_3w_1 = -v_1w_3 + v_3w_1 = -\begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix}$$

then

$$\epsilon_{ijk}u_iv_jw_k = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \quad (*)$$

Notation.

$$\text{matrix } \mathbf{V}^\dagger = [V_{ij}] = \begin{bmatrix} V_{11} & V_{12} & V_{13} \\ V_{21} & V_{22} & V_{23} \\ V_{31} & V_{32} & V_{33} \end{bmatrix}$$

$$\det[V_{ij}] = \begin{vmatrix} V_{11} & V_{12} & V_{13} \\ V_{21} & V_{22} & V_{23} \\ V_{31} & V_{32} & V_{33} \end{vmatrix} = \det \mathcal{V}$$

[†]In here, we use $\mathcal{V}$ rather than $\mathbf{V}$ for readability, but in writing, we always use $\mathbf{V}$ to represent tensor. Usually (but not always), the uppercase mean the second order tensor, lowercase mean the first order tensor.

CHAPTER 2. ELEMENTS OF CARTESIAN TENSOR ANALYSIS

Claim 2.6.1.

$$\epsilon_{ijk}\epsilon_{pqr} \det \mathcal{V} = \begin{vmatrix} V_{ip} & V_{iq} & V_{ir} \\ V_{jp} & V_{jq} & V_{jr} \\ V_{kp} & V_{kq} & V_{kr} \end{vmatrix} \quad (**)$$

Note the following properties

$$\det \mathcal{V} \underset{\substack{\text{from } (*) \\ \text{replace} \\ u_i \rightarrow V_{1i} \\ v_j \rightarrow V_{2j} \\ w_k \rightarrow V_{3k}}}{=} \epsilon_{ijk} V_{1i} V_{2j} V_{3k} \underset{\text{transpose of } \det \mathcal{V}}{=} \epsilon_{ijk} V_{i1} V_{j2} V_{k3}$$

Also

$$\epsilon_{pqr} \det \mathcal{V} = \epsilon_{ijk} V_{pi} V_{qj} V_{rk} = \epsilon_{ijk} V_{ip} V_{jq} V_{kr}$$

(take care of the permutation of rows or columns of $\det \mathcal{V}$)

$$\epsilon_{pqr}\epsilon_{pqr} \det \mathcal{V} = \epsilon_{ijk}\epsilon_{pqr} V_{ip} V_{jq} V_{kr}$$

$$\boxed{\det \mathcal{V} = \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} V_{ip} V_{jq} V_{kr}}$$

$$\epsilon_{ijk}\epsilon_{pqr} \det \mathbb{1} \stackrel{(**)}{=} \begin{vmatrix} \delta_{ip} & \delta_{iq} & \delta_{ir} \\ \delta_{jp} & \delta_{jq} & \delta_{jr} \\ \delta_{kp} & \delta_{kq} & \delta_{kr} \end{vmatrix}$$

where $\mathbb{1} = \text{identity matrix} = [\delta_{ij}]$

$$\stackrel{i=p}{\Rightarrow} \boxed{\epsilon_{ijk}\epsilon_{iqr} = \delta_{jq} = \delta_{kr} - \delta_{jr}\delta_{kq}} \quad (\text{proof: Homework})$$

2.7. CONNECTION WITH VECTOR ALGEBRA

if $j = q$ further

$$\epsilon_{ijk} = \epsilon_{ijr} = \delta_{jj}\delta_{kr} - \delta_{jr}\delta_{kj} = 3\delta_{kr} - \delta_{kr} = 2\delta_{kr}$$

2.7 Connection with vector algebra

Consider vectors $\underline{u}$, $\underline{v}$, and $\underline{w}$ in E_3

dot product $\underline{v} \cdot \underline{u}$

$$\underline{v} \cdot \underline{u} = |\underline{v}| |\underline{u}| \cos \theta$$

cross product $\underline{v} \wedge \underline{u}$ ($\underline{v} \times \underline{u}$)

$$|\underline{v} \wedge \underline{u}| = |\underline{v}| |\underline{u}| \sin \theta$$

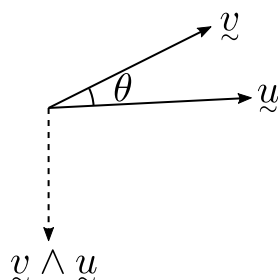


Figure 2.1: Direction of cross product

or in determinant form

$$\underline{v} \wedge \underline{u} = \begin{vmatrix} \underline{e}_1 & \underline{e}_2 & \underline{e}_3 \\ v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \end{vmatrix} = \underline{e}_1 \begin{vmatrix} v_2 & v_3 \\ u_2 & u_3 \end{vmatrix} - \underline{e}_2 \begin{vmatrix} v_1 & v_3 \\ u_1 & u_3 \end{vmatrix} + \underline{e}_3 \begin{vmatrix} v_1 & v_2 \\ u_1 & u_2 \end{vmatrix} \quad (1)$$

CHAPTER 2. ELEMENTS OF CARTESIAN TENSOR ANALYSIS

Let $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\}$ be a rectangular Cartesian coordinate frame with origin O and with unit vector $\underline{e}_i$.

$\mathcal{F}$: the class of all frames

$\mathcal{F}_+$: the class of all such frames that are also right-handed

Properties of $\mathcal{F}$ and $\mathcal{F}_+$

if $X \in \mathcal{F} \Rightarrow \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$

if $X \in \mathcal{F}_+ \Rightarrow \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$ and $(\underline{e}_1 \wedge \underline{e}_2) \cdot \underline{e}_3 = +1$

Let $X \in \mathcal{F}_+$, then the three real numbers defined as

$$u_i^X = \underline{u} \cdot \underline{e}_i^X$$

scalar components of vector $\underline{u}$ on X

Note. No need to put X to the superscript of $\underline{u}$.

$$\underline{u} = u_i \underline{e}_i = u_1 \underline{e}_1 + u_2 \underline{e}_2 + u_3 \underline{e}_3$$

$$\begin{aligned} \underline{u} \cdot \underline{v} &= (u_i \underline{e}_i) \cdot (v_k \underline{e}_k) = u_i v_k \underline{e}_i \cdot \underline{e}_k = u_i v_k \delta_{ik} \\ &= u_i v_i = u_k v_k \end{aligned}$$

$$\therefore \boxed{\underline{u} \cdot \underline{v} = u_i v_i}$$

$$\begin{aligned} \underline{v} \wedge \underline{u} &\stackrel{(1)}{=} \epsilon_{ijk} \underline{e}_i v_j u_k \\ &\stackrel{(*)}{=} (\epsilon_{ijk} v_j u_k) \underline{e}_i = (\underline{v} \wedge \underline{u})_i \underline{e}_i \end{aligned} \tag{2}$$

2.7. CONNECTION WITH VECTOR ALGEBRA

$$\therefore \boxed{(\underline{v} \wedge \underline{u})_i = \epsilon_{ijk} v_j u_k} \quad (3)$$

Note. The expression of equations must be consistent, either in vector form (as (2)) or in indicial form (as (3)).

$$\begin{aligned} (\underline{u} \wedge \underline{v}) \cdot \underline{w} &= (\epsilon_{ijk} u_j v_k \underline{e}_i) \cdot (w_l \underline{e}_l) = \epsilon_{ijk} u_j v_k w_l \underbrace{(\underline{e}_i \cdot \underline{e}_l)}_{\delta_{il}} \\ &= \epsilon_{ijk} w_i u_j v_k \end{aligned}$$

If we start with indicial form

$$(\underline{u} \wedge \underline{v}) \cdot \underline{w} = (\underline{u} \wedge \underline{v})_k w_k = \epsilon_{kij} u_i v_j w_k = \epsilon_{ijk} u_i v_j w_k$$

Example. $\underline{a}$ and $\underline{b}$ are any vectors in E_3 , show that $(\underline{a} \wedge \underline{b}) \cdot (\underline{a} \wedge \underline{b}) + (\underline{a} \cdot \underline{b})^2 = |\underline{a}|^2 |\underline{b}|^2$.

Solution.

$$\begin{aligned} &(\underline{a} \wedge \underline{b}) \cdot (\underline{a} \wedge \underline{b}) + (\underline{a} \cdot \underline{b})^2 \\ &= \epsilon_{ijk} a_j b_k \epsilon_{ilm} a_l b_m + (a_j b_j) (a_k b_k) \\ &= \epsilon_{ijk} \epsilon_{ilm} a_j b_k a_l b_m + a_j b_j a_k b_k \\ &= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) a_j b_k a_l b_m + a_j b_j a_k b_k \\ &= a_j b_k a_j b_k - \underline{a_j b_k a_k b_j} + \underline{a_j b_j a_k b_k} \\ &= |\underline{a}|^2 + |\underline{b}|^2 \end{aligned}$$

Example. Show that

CHAPTER 2. ELEMENTS OF CARTESIAN TENSOR ANALYSIS

$$(\underline{a} \wedge \underline{b}) \cdot (\underline{c} \wedge \underline{d}) = \begin{vmatrix} \underline{a} \cdot \underline{c} & \underline{b} \cdot \underline{c} \\ \underline{a} \cdot \underline{d} & \underline{b} \cdot \underline{d} \end{vmatrix}$$

Solution.

$$\begin{aligned} (\underline{a} \wedge \underline{b}) \cdot (\underline{c} \wedge \underline{d}) &= \epsilon_{ijk} a_j b_k \epsilon_{ilm} c_l d_m \\ &= \epsilon_{ijk} \epsilon_{ilm} a_j b_k c_l d_m \\ &= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) a_j b_k c_l d_m \\ &= a_j c_j b_k d_k - a_j d_j b_k c_k \\ &= (\underline{a} \cdot \underline{c}) (\underline{b} \cdot \underline{d}) - (\underline{a} \cdot \underline{d}) (\underline{b} \cdot \underline{c}) \\ &= \begin{vmatrix} \underline{a} \cdot \underline{c} & \underline{b} \cdot \underline{c} \\ \underline{a} \cdot \underline{d} & \underline{b} \cdot \underline{d} \end{vmatrix} \end{aligned}$$

2.7. CONNECTION WITH VECTOR ALGEBRA

Chapter 3

Introduction to Cartesian tensor

3.1 Transformation of coordinate frames (Definition of Cartesian tensor)

Consider two rectangular Cartesian frames

$$X = \{0; e_1, e_2, e_3\}, \quad X' = \{0; e'_1, e'_2, e'_3\}$$

3.1. TRANSFORMATION OF COORDINATE FRAMES (DEFINITION OF CARTESIAN TENSOR)

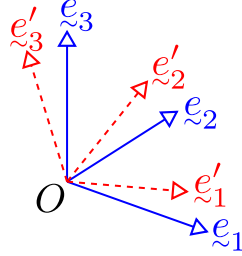


Figure 3.1: Scheme of two rectangular Cartesian frames

Define

$$A_{ij} = \underline{e}'_i \cdot \underline{e}_j = \cos(\underline{e}'_i, \underline{e}_j) = (\underline{e}'_i)^X_j = (\underline{e}_j)^{X'}_i$$

$$\underline{A} = [A_{ij}] = \begin{bmatrix} \underline{e}'_1, \underline{e}_1 & \underline{e}'_1, \underline{e}_2 & \underline{e}'_1, \underline{e}_3 \\ \underline{e}'_2, \underline{e}_1 & \underline{e}'_2, \underline{e}_2 & \underline{e}'_2, \underline{e}_3 \\ \underline{e}'_3, \underline{e}_1 & \underline{e}'_3, \underline{e}_2 & \underline{e}'_3, \underline{e}_3 \end{bmatrix}$$

$$A_{ij} \neq A_{ji}$$

Observed properties

$$\underline{e}'_p \cdot \underline{e}'_q = \delta_{pq}$$

then

$$A_{pi} A_{qi} = (\underline{e}'_p)^X_i (\underline{e}'_q)^X_i = \underline{e}'_p \cdot \underline{e}'_q$$

$$\Rightarrow \boxed{A_{pi} A_{qi} = \delta_{pq}}$$

Similarly

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

$$A_{pi}A_{pj} = (\mathfrak{e}_i)_p^{X'} (\mathfrak{e}_j)_p^{X'} = \mathfrak{e}_i \cdot \mathfrak{e}_j$$

$$\Rightarrow \boxed{A_{pi}A_{pj} = \delta_{ij}}$$

in matrix form

$$\mathfrak{A}\mathfrak{A}^T = \mathfrak{A}^T\mathfrak{A} = \mathfrak{1} \Rightarrow \mathfrak{A} = \mathfrak{A}^{-1}$$

Also

$$\begin{aligned} \det \mathfrak{A} &= \epsilon_{ijk} A_{1i} A_{2j} A_{3k} = \epsilon_{ijk} (\mathfrak{e}'_1)_i^X (\mathfrak{e}'_2)_j^X (\mathfrak{e}'_3)_k^X \\ &= (\mathfrak{e}'_1 \wedge \mathfrak{e}'_2) \cdot \mathfrak{e}'_3 = \begin{cases} +1 & \text{right-handed frame} \\ -1 & \text{left-handed frame} \end{cases} \end{aligned}$$

$$\therefore \det \mathfrak{A} = \pm 1$$

$\mathfrak{A}$ is an *orthogonal* (orthonormal) matrix.

If $X \in \mathcal{F}_+$, then $\det \mathfrak{A} = +1$, $\mathfrak{A}$ is *proper* matrix.

Notation.

The change in coordinate frame $X \rightarrow X'$ which is governed by $\mathfrak{A}$ will be represented as

$$\mathfrak{A}: X \rightarrow X'$$

Consider a vector $\mathfrak{v}$

$$\mathfrak{v} = v_p^X \mathfrak{e}_p = v_p^{X'} \mathfrak{e}'_p$$

3.1. TRANSFORMATION OF COORDINATE FRAMES (DEFINITION OF CARTESIAN TENSOR)

where

$v_p^X \cdots$ scalar components of $\underline{v}$ with respect to frame X

$v_p^{X'} \cdots$ scalar components of $\underline{v}$ with respect to frame X'

$$\underline{v} \cdot \underline{e}'_i = v_i^{X'} = \left(v_p^X \underline{e}_p \right) \cdot \underline{e}'_i = v_p^X A_{ip}$$

$$\Rightarrow \boxed{v_i^{X'} = A_{ip} v_p^X}$$

Similarly

$$\boxed{v_i^X = A_{pi} v_p^{X'}}$$

which are *transformation laws*.

or in matrix form $\underline{v} = \begin{bmatrix} v_1^X \\ v_2^X \\ v_3^X \end{bmatrix}$

$\left. \begin{array}{l} \underline{v}' = \underline{A} \underline{v} \\ \underline{v} = \underline{A}^T \underline{v}' \end{array} \right\}$ gives confusion, the transformation law only change the *components* of a vector not the vector itself.

Definition 3.1.1. Definition of vector

A vector is a function defined on all X such that it satisfies the above transformation laws.

Definition 3.1.2. Definition of a Cartesian tensor

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

Let $N \geq 0$ be an integer. A Cartesian tensor of order N is a function on $\mathcal{F}$ (all X) that assigns to each $X \in \mathcal{F}$, an ordered set of 3^N real numbers. We will denote

$$\underbrace{V_{ij \dots k}^X}_N \cdots \text{components of } \mathcal{V} \text{ in } X$$

such that for every $\mathcal{A}: X \rightarrow X'$ (for every orthogonal transformation which transforms X to X')

$$\underbrace{V_{ij \dots k}^X}_N = A_{ip} A_{jq} \cdots A_{kr} \underbrace{V_{pq \dots r}^X}_N \quad (*)$$

Further, if $\mathcal{U}$ and $\mathcal{V}$ are both tensors of order N and c is a real number then define

- (i) $\mathcal{U} = \mathcal{V}$ if $U_{ij \dots k}^X = V_{ij \dots k}^X \quad \forall X \in \mathcal{F}$
- (ii) $\mathcal{W} = \mathcal{U} + \mathcal{V}$ if $W_{ij \dots k}^X = U_{ij \dots k}^X + V_{ij \dots k}^X \quad \forall X \in \mathcal{F}$
- (iii) $\mathcal{W} = c\mathcal{U}$ if $W_{ij \dots k}^X = cU_{ij \dots k}^X \quad \forall X \in \mathcal{F}$

Note.

- (1) $V_{ij \dots k}$ is not a tensor. They are the components of a tensor in a given frame X .
- (2) $\mathcal{W}$ defined in (ii) and (iii) is itself a tensor of order N .
- (3) If $N = 0$, $\mathcal{V}$ is called a scalar and you write V in place of $\mathcal{V}$.

3.1. TRANSFORMATION OF COORDINATE FRAMES (DEFINITION OF CARTESIAN TENSOR)

(4) If $N = 1$, $\underline{V}$ is a vector.

(5) Equalities needs to be tested for a single frame only.

Definition 3.1.3.

A tensor $\underline{V}$ of order $N \geq 2$ is symmetric/skew symmetric with respect to a pairs of indices of its components if the components of $\underline{V}$ remain unchanged/change sign under the interchange of these two indices.

Definition 3.1.4.

A tensor is fully symmetric if its is symmetric with respect to any pairs of indices (so as skew symmetric).

Property

If the component of a tensor are symmetric in one frame then they are symmetric in any other frame

Proof.

Let

$$V_{\dots i \dots j \dots}^X = V_{\dots j \dots i \dots}^X \quad (\star)$$

$$\begin{aligned} V_{\dots p \dots q \dots}^{X'} &= \dots A_{pi} \dots A_{qj} \dots V_{\dots i \dots j \dots}^X \\ &\stackrel{(\star)}{=} \dots A_{qj} \dots A_{pi} \dots V_{\dots j \dots i \dots}^X \\ &\stackrel{(*)}{=} V_{\dots q \dots p \dots}^{X'} \end{aligned}$$

□

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

Definition 3.1.5.

A tensor $\mathcal{V}$ of order N is *isotropic* if its components $V_{ij\dots k}$ are independent of choice of frame.

Note.

- (1) A scalar is an isotropic tensor of order $N = 0$.
- (2) The only isotropic vector is the null vector $\mathbf{0}$.

Claim 3.1.1.

- (a) Let $V_{ij}^X = \delta_{ij} \quad \forall X \in \mathcal{F}$. V_{ij}^X are the components in frame X of an isotropic second order ($N = 2$) tensor. We write $\mathcal{V} = \mathbf{1}$ (identity tensor).
- (b) $V_{ijk}^X = \epsilon_{ijk} \quad \forall X \in \mathcal{F}_+$. Then V_{ijk}^X are components of an isotropic tensor $\mathcal{V}$ ($N = 3$).

Proof.

Sufficient to show that V_{ij} in (a) and V_{ij} in (b) are tensor component, i.e. satisfy (*).

(a)

$$A_{ip}A_{jq}V_{pq}^X = A_{ip}A_{jq}\delta_{ij} = A_{ip}A_{jq} = \delta_{ij} \stackrel{\text{by def.}}{=} V_{ij}^{X'}$$

* $\forall$, the symbol gives the isotropic property by definition

3.1. TRANSFORMATION OF COORDINATE FRAMES (DEFINITION OF CARTESIAN TENSOR)

(b)

$$\begin{aligned} A_{ip}A_{jq}A_{kr}V_{pqr}^X &= A_{ip}A_{jq}A_{kr}\epsilon_{pqr} = \epsilon_{ijk} \det A \\ &\stackrel{\dagger}{=} \epsilon_{ijk} = V_{ijk}^{X'} \end{aligned}$$

□

From this point on , change notation, denote $\mathcal{F}_+$ by $\mathcal{F}$, i.e. handle only *proper* orthogonal transformation only.

Claim 3.1.2.

- (1) *Every isotropic tensor of 2^{nd} order is a scalar multiple of $\underline{1}$.*
- (2) *Every isotropic tensor of 3^{rd} order is a scalar multiple of $\underline{\epsilon}$.*
- (3) *Every isotropic high order tensor can be constructed from $\underline{1}$ and $\underline{\epsilon}$.
e.g. The most general 4^{th} order isotropic tensor has the form of components*

$$C_{ijkl} = \alpha\delta_{ij}\delta_{kl} + \beta\delta_{ik}\delta_{jl} + \gamma\delta_{il}\delta_{jk}$$

where α , β , and γ are scalars.

[†]If $X \in \mathcal{F}_+$, $\det A = +1$

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

3.2 Outer multiplication of tensors

Let $\mathcal{U}$ and $\mathcal{V}$ be tensor of order M and N respectively.

Define

$$\underbrace{W_{ij \dots k}^X}_M \underbrace{mn \dots r}_N = \underbrace{U_{ij \dots k}^X}_M \underbrace{V_{mn \dots r}^X}_N \quad \forall X \in \mathcal{F}$$

then $W_{ij \dots k mn \dots r}$ are the components of tensor $\mathcal{W}$ of order $M + N$ called the outer product of $\mathcal{U}$ and $\mathcal{V}$ (in that order)

Proof.

$$\begin{aligned} W_{ij \dots k mn \dots r} &\stackrel{\text{by def.}}{=} U_{ij \dots k}^{X'} V_{mn \dots r}^{X'} \\ &\stackrel{(*)}{=} A_{iI} A_{jJ} \dots A_{kK} A_{mM} A_{nN} \dots A_{rR} \underbrace{U_{IJ \dots K}^X V_{MN \dots R}^X}_{\stackrel{\text{by def.}}{=} W_{IJ \dots K MN \dots R}^X} \end{aligned}$$

□

3.3 Contraction of tensors

Let $\mathcal{U}$ be a tensor of order $N \geq 2$. Let $U_{ij \dots k}^X$ be its components with $X \in \mathcal{F}$.

Define the 3^{N-2} real numbers as

$$\underbrace{V_{ij \dots k}^X}_{N-2} = \underbrace{U_{ij \dots r \dots r \dots k}^X}_N \quad X \in \mathcal{F}$$

3.4. OPERATION BETWEEN TENSORS

Then $V_{ij\dots k}^X$ are components of a new tensor $\mathcal{V}$ formed by contraction of $\mathcal{U}$.

Proof.

Similar to the outer product.

□

Let $(N = 2)$ U_{ij}^X are the components of a 2nd tensor $\mathcal{U}$, then

$$\mathcal{V} = U_{rr}^X = U_{11}^X + U_{22}^X + U_{33}^X = \text{tr } \mathcal{U}$$

e.g. If $\mathcal{U} = \mathcal{\sigma}$, the stress tensor, then

$$\text{tr } \mathcal{\sigma} = \sigma_{11}^X + \sigma_{22}^X + \sigma_{33}^X = I_1$$

the 1st invariant of stress tensor.

3.4 Operation between tensors

Let c be a scalar and $\mathcal{U}, \mathcal{V}$ be tensors.

Define

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

Definition $\mathcal{W}$	order $\mathcal{U}$	order $\mathcal{V}$	order $\mathcal{W}$	Absolute (coordinate-free) notation	Name
$W = U_i V_i$	1	1	0	$\mathcal{W} = \mathcal{U} \cdot \mathcal{V}$	dot product between two vectors
$W_i = \epsilon_{ijk} U_j V_k$	1	1	1	$\mathcal{W} = \mathcal{U} \wedge \mathcal{V}$	cross product
$W_i = U_{ij} V_j$	2	1	1	$\mathcal{W} = \mathcal{U} \mathcal{V}$	product
$W_{ijk} = U_{ij} V_k$	2	1	3	(no notation)	outer product
$W_{ij} = U_{ik} V_{kj}$	2	2	2	$\mathcal{W} = \mathcal{U} \mathcal{V}$	2 nd order tensor product
$W_{ij} = U_i V_j$	1	1	2	$\mathcal{W} = \mathcal{U} \otimes \mathcal{V}$	<i>Dyadic</i> or outer product of two vectors

3.5 Inner product or fully-contracted outer product

If $\mathcal{U}$ and $\mathcal{V}$ are both of order $N > 0$, define

$$\mathcal{U} \cdot \mathcal{V} = U_{ij \dots k} V_{ij \dots k}$$

(no symbol of reference frame needed)

Define

3.5. INNER PRODUCT OR FULLY-CONTRACTED OUTER PRODUCT

$$|\mathcal{U}| = \sqrt{\mathcal{U} \cdot \mathcal{U}}$$

the *norm* of a tensor $\mathcal{U}$.

Claim 3.5.1. Quotient Rule

Let 9 real numbers $W_{ij\dots k}^X$ be defined for all $X \in \mathcal{F}$ (these numbers are not known aprior to be tensor components) and suppose that for any vector $\mathcal{v} \neq \mathcal{0}$, the 3 real numbers v_i^X defined by

$$v_i^X = W_{ij}^X u_j^X \quad \forall X \in \mathcal{F}$$

are the components in X of a vector $\mathcal{v}$. Then W_{ij}^X are components in X of a tensor $\mathcal{W}$.

Proof.

By hypothesis, if $\mathcal{A}: X \rightarrow X'$ since both $\mathcal{U}$ and $\mathcal{V}$ are vectors

$$v_i^X \stackrel{(*)}{=} A_{qi} v_q^{X'} \stackrel[\text{def.}]{=} W_{ij}^X u_j^X \stackrel{(*)}{=} W_{ij}^X (A_{qj} u_q^{X'})$$

multiply by A_{pi}

$$\begin{aligned} A_{pi} A_{qi} v_q^{X'} &= A_{pi} W_{ij}^X A_{qj} u_q^{X'} \Rightarrow \delta_{pq} v_q^{X'} = W_{ij}^X A_{pi} A_{qj} u_q^{X'} \\ &\Rightarrow v_p^{X'} = W_{ij}^X A_{pi} A_{qj} u_q^{X'} \end{aligned} \tag{1}$$

Also by definition

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$$v_p^{X'} = W_{pq}^{X'} u_q^{X'} \quad (2)$$

(1) and (2)

$$\left(W_{pq}^{X'} - A_{pi} A_{qj} W_{ij}^X \right) u_q^{X'} = 0$$

Choose $\underline{u}$ such that $u_q^{X'} = \delta_{qs}$ (s -fixed)[‡]

$$\Rightarrow W_{p1}^{X'} - A_{pi} A_{1j} W_{ij}^X = 0$$

$\begin{matrix} 2 \\ 3 \end{matrix}$

$$\Rightarrow W_{ps}^{X'} - A_{pi} A_{sj} W_{ij}^X = 0$$

i.e. W_{ij}^X obeys transformation law.

□

3.6 Product of a second order tensor and a vector

Let $\underline{u}$ and $\underline{v}$ be vectors, $\underline{W}$ be a 2nd order tensor.
Then

$$\underline{W} (\underline{u} + \underline{v}) = \underline{W} \underline{u} + \underline{W} \underline{v}$$

If $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$, then

$$\text{‡choose } \begin{cases} s = 1 \rightarrow q = 1 \\ s = 2 \rightarrow q = 2 \\ s = 3 \rightarrow q = 3 \end{cases}$$

3.7. CONNECTION BETWEEN SECOND ORDER TENSORS AND LINEAR VECTOR FUNCTION (L.V.F.)

$$W_{ij}^X = \underline{e}_i \cdot \left(\widetilde{W} \underline{e}_j \right)$$

the way to find the components of $\widetilde{W}$.

Proof.

$$\begin{aligned} \underline{e}_i \cdot \left(\widetilde{W} \underline{e}_j \right) &= (\underline{e}_i)_k \left(\widetilde{W} \underline{e}_j \right)_k = \delta_{ik} \left[W_{kl} (\underline{e}_j)_l \right] \\ &= \delta_{ik} W_{kl} \delta_{jl} = W_{ij} \end{aligned}$$

□

3.7 Connection between second order tensors and linear vector function (l.v.f.)

A linear vector function $\mathcal{L}(\bullet)$ is a mapping that assigns to each vector belongs to E_3 and other vector $\underline{v} = \mathcal{L}(\underline{u}) \in E_3$ subjected to the following requirements.

- (i) $\mathcal{L}(\underline{u}^{(1)} + \underline{u}^{(2)}) = \mathcal{L}(\underline{u}^{(1)}) + \mathcal{L}(\underline{u}^{(2)})$
- (ii) $\mathcal{L}(c\underline{u}) = c\mathcal{L}(\underline{u}) \quad \forall c \in \mathbb{R}$

Claim 3.7.1. *Connection of second order tensor to l.v.f.*

- (a) *Let $\widetilde{W}$ be a 2^{nd} order tensor and define a function $\mathcal{L}(\bullet)$ as*

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

$$\underline{v} = \mathcal{L}(\underline{u}) = \mathcal{W}\underline{u} \quad \forall \underline{u} \in E_3 \quad (\star)$$

Then $\mathcal{L}$ is a l.v.f.

(b) Conversely, given a linear vector function $\mathcal{L}(\underline{u})$, there exists a unique 2^{nd} order tensor $\mathcal{W}$ such that

$$\mathcal{L}(\underline{u}) = \mathcal{W}\underline{u} \quad \forall \underline{u} \in E_3$$

Proof.

(a) Recall that $\mathcal{W}(\underline{u} + \underline{v}) = \mathcal{W}\underline{u} + \mathcal{W}\underline{v}$

So

$$\mathcal{W}(\underline{u}^{(1)} + \underline{u}^{(2)}) = \mathcal{W}\underline{u}^{(1)} + \mathcal{W}\underline{u}^{(2)} \quad (\text{i})$$

$$\mathcal{W}(c\underline{u}) = c\mathcal{W}\underline{u} \quad (\text{ii})$$

(b) Let $\mathcal{L}(\bullet)$ be given. Choose a frame $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$. Let W_{ji}^X be the j^{th} $\mathcal{L}(\underline{e}_i)$ in frame X .[§]

$$\mathcal{L}(\underline{e}_i) = W_{ji}^X \underline{e}_j \quad (\star\star)$$

[§]Method to find the component of $\mathcal{W}$.

3.7. CONNECTION BETWEEN SECOND ORDER TENSORS AND LINEAR VECTOR FUNCTION (L.V.F.)

Choose an arbitrary vector $\underline{u} \in E_3$. Then

$$\underline{v} = \underline{\mathcal{L}}(\underline{u}) = \underline{\mathcal{L}}(u_i^X \underline{e}_i) = u_i^X \underline{\mathcal{L}}(\underline{e}_i) \stackrel{(\star\star)}{=} u_i^X W_{ji}^X \underline{e}_j$$

But $\underline{v} = v_j^X \underline{e}_j$

$$\Rightarrow v_j^X = u_i^X W_{ji}^X = W_{ji}^X u_i^X \stackrel{i \leftrightarrow j}{=} W_{ij}^X u_j^X$$

By quotient rule, W_{ij}^X are the components of a 2nd order tensor $\underline{W}$

$$\underline{v} = \underline{W}\underline{u} = \underline{\mathcal{L}}(\underline{u})$$

□

Uniqueness

Let's assume that exists $\underline{W}^{(1)}$ and $\underline{W}^{(2)}$ such that

$$\left. \begin{array}{l} \underline{v} = \underline{\mathcal{L}}(\underline{u}) = \underline{W}^{(1)}\underline{u} \\ \underline{v} = \underline{\mathcal{L}}(\underline{u}) = \underline{W}^{(2)}\underline{u} \end{array} \right\} \Rightarrow \left(\underline{W}^{(1)} - \underline{W}^{(2)} \right) \underline{u} = \underline{0} \quad \forall \underline{u} \in E_3$$

or

$$\left(W_{ij}^{(1)} - W_{ij}^{(2)} \right) u_j = 0 \Rightarrow \underline{W}^{(1)} = \underline{W}^{(2)}$$

Note. Sometimes people define second order tensor by l.v.f.

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3.8 Transpose of a second order tensor

Define

$$\underline{V} = \underline{U}^T \quad \text{if} \quad V_{ij} = U_{ji}$$

Symmetric 2nd order tensor as one

$$\underline{U} = \underline{U}^T \quad \text{or} \quad U_{ij} = U_{ji}$$

Skew-symmetric 2nd order tensor as one

$$\underline{U} = -\underline{U}^T \quad \text{or} \quad U_{ij} = -U_{ji}$$

If $\underline{W}$ is a 2nd order tensor and $\underline{u}, \underline{v}$ are vectors, then

$$\boxed{\underline{u} \cdot (\underline{W} \underline{v}) = (\underline{W}^T \underline{u}) \cdot \underline{v}}$$

Proof.

$$\begin{aligned} \underline{u} \cdot (\underline{W} \underline{v}) &= u_i (\underline{W} \underline{v})_i = u_i (W_{ij} v_j) \stackrel{i \leftrightarrow j}{=} (W_{ji} u_j) v_i \\ &= (\underline{W}^T \underline{u})_i v_i = (\underline{W}^T \underline{u}) \cdot \underline{v} \end{aligned}$$

□

If $\underline{y} = \underline{V} \underline{x}$, $\underline{z} = \underline{U} \underline{y}$ ($\underline{x}, \underline{y}, \underline{z} \in E_3$, $\underline{U}, \underline{V}$ are 2nd order tensors)

$$\underline{z} = \underline{U} \underline{V} \underline{x} \quad \text{or} \quad z_i = U_{ij} V_{jk} x_k$$

3.9. PRODUCT BETWEEN TWO SECOND ORDER TENSORS

3.9 Product between two second order tensors

If $\underline{U}$ and $\underline{V}$ are 2nd order tensors, then define

$$\underline{W} = \underline{U}\underline{V} \quad \text{if} \quad W_{ij} = U_{ik}V_{kj}$$

as tensorial product of $\underline{U}$ and $\underline{V}$.

(i) distribute

$$\underline{U} \left(\underline{V}^{(1)} + \underline{V}^{(2)} \right) = \underline{U}\underline{V}^{(1)} + \underline{U}\underline{V}^{(2)}$$

(ii) associate

$$(\underline{U}\underline{V})\underline{W} = \underline{U}(\underline{V}\underline{W})$$

(iii) non-commutative

$$\underline{U}\underline{V} \neq \underline{V}\underline{U}$$

If $\underline{W} = \underline{U}\underline{V}$, then

$$\underline{W}^T = \underline{V}^T \underline{U}^T$$

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Proof.

$$\begin{aligned}
 W_{ij} = U_{ik}V_{kj} &\stackrel{i \leftrightarrow j}{\Rightarrow} W_{ji} = U_{jk}V_{ki} \\
 &= V_{ki}U_{jk} \\
 &= V_{ik}^T U_{kj}^T \\
 &= \left(\underset{\sim}{V}^T \underset{\sim}{U}^T \right)_{ij}
 \end{aligned}$$

but $W_{ji} = W_{ij}^T$

$$\therefore \underset{\sim}{W}^T = \underset{\sim}{V}^T \underset{\sim}{U}^T$$

□

3.10 Tensor (or dyadic) product between two vectors

If $\underset{\sim}{u}$ and $\underset{\sim}{v}$ belong to E_3 , define

$$\underset{\sim}{W} = \underset{\sim}{u} \otimes \underset{\sim}{v} \quad \text{if} \quad W_{ij} = u_i v_j$$

where $\underset{\sim}{W}$ is a 2nd order tensor.

Note.

(1)

$$\underset{\sim}{u} \otimes \underset{\sim}{v} \neq \underset{\sim}{v} \otimes \underset{\sim}{u}$$

(2)

$$\underset{\sim}{u} \otimes \left(\underset{\sim}{v}^{(1)} + \underset{\sim}{v}^{(2)} \right) = \underset{\sim}{u} \otimes \underset{\sim}{v}^{(1)} + \underset{\sim}{u} \otimes \underset{\sim}{v}^{(2)}$$

3.10. TENSOR (OR DYADIC) PRODUCT BETWEEN TWO VECTORS

(3)

$$\underline{W} = \underline{u} \otimes \underline{v} \Leftrightarrow \underline{W}\underline{a} = (\underline{v} \cdot \underline{a}) \underline{u} \quad \forall \underline{a} \in E_3$$

Proof.

$\Rightarrow$

$$\underline{W}\underline{a} = (W_{ik}a_k) \underline{e}_i = (u_i v_k a_k) \underline{e}_i = (v_k a_k) u_i \underline{e}_i = (\underline{v} \cdot \underline{a}) \underline{u}$$

$\Leftarrow$

$$\underline{W}\underline{a} = (\underline{v} \cdot \underline{a}) \underline{u} \quad \forall \underline{a} \in E_3$$

or

$$W_{ij}a_j = v_j a_j u_i$$

$$\therefore (W_{ij} - u_i v_j) a_j = 0$$

Since $\underline{a}$ is arbitrary

$$W_{ij} = u_i v_j \quad \text{or} \quad \underline{W} = \underline{u} \otimes \underline{v}$$

□

If $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$, then

$$\underline{W} = W_{ij}^X \underline{e}_i \otimes \underline{e}_j$$

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Proof.

$$\mathcal{V} = W_{ij}^X \mathcal{e}_i \otimes \mathcal{e}_j = W_{11} \mathcal{e}_1 \otimes \mathcal{e}_1 + W_{12} \mathcal{e}_1 \otimes \mathcal{e}_2 + \cdots$$

$$\begin{aligned} V_{pq} &= \left(W_{ij}^X \mathcal{e}_i \otimes \mathcal{e}_j \right)_{pq} = W_{ij}^X \left(\mathcal{e}_i \otimes \mathcal{e}_j \right)_{pq} \\ &= W_{ij}^X (\mathcal{e}_i)_p (\mathcal{e}_j)_q \\ &= W_{ij}^X \delta_{ip} \delta_{jq} \\ &= W_{pq}^X \end{aligned}$$

$$\therefore \left(W_{ij}^X \mathcal{e}_i \otimes \mathcal{e}_j \right)_{pq} = W_{pq}^X$$

or

$$W_{ij}^X \mathcal{e}_i \otimes \mathcal{e}_j = \mathcal{W}$$

representation theorem of $\mathcal{W}$ in terms of a sum of unit vectors of a frame admits components in the same frame.

Trace of a 2nd order tensor $\mathcal{U}$ as

$$\text{tr } \mathcal{U} = U_{ii} = U_{11} + U_{22} + U_{33}$$

Let $\mathcal{U}$ and $\mathcal{V}$ be 2nd order tensors, then

(1)

$$\mathcal{U} \cdot \mathcal{V} = U_{ij} V_{ij} = \text{tr}(\mathcal{U} \mathcal{V}^T)$$

(2)

$$|\mathcal{U}| = \sqrt{\mathcal{U} \cdot \mathcal{U}} = \sqrt{\text{tr}(\mathcal{U} \mathcal{U}^T)}$$

3.11. ORTHOGONAL TENSORS

(3)

$$\det(\underset{\sim}{U}\underset{\sim}{V}) = \det \underset{\sim}{U} \det \underset{\sim}{V}$$

Proof. exercise

□

□

3.11 Orthogonal tensors

A 2nd order tensor $\underset{\sim}{Q}$ is orthogonal if $\underset{\sim}{Q}\underset{\sim}{Q}^T = \underset{\sim}{1}$, the identity tensor.

$$\underbrace{\underset{\sim}{Q}^{-1}\underset{\sim}{Q}}_{\underset{\sim}{1}}\underset{\sim}{Q}^T = \underset{\sim}{Q}^{-1} \Rightarrow \underset{\sim}{Q}^T = \underset{\sim}{Q}^{-1}$$

or

$$\det(\underset{\sim}{Q}\underset{\sim}{Q}^T) = \det \underset{\sim}{1} = 1 \stackrel{(3)}{=} \det \underset{\sim}{Q} \det \underset{\sim}{Q}^T = \left(\det \underset{\sim}{Q}\right)^2$$

$$\Rightarrow \det \underset{\sim}{Q} = \pm 1$$

If $\underset{\sim}{Q}\underset{\sim}{Q}^T = \underset{\sim}{1}$ and $\det \underset{\sim}{Q} = +1$, then $\underset{\sim}{Q}$ is called *proper orthogonal*.

3.12 Fundamental scalar invariant of a second order tensor

Let $\underset{\sim}{V}$ be a 2nd order tensor. Then define $I_i(\underset{\sim}{V})$ as

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$$I_1(\mathcal{V}) = \text{tr } \mathcal{V} = V_{ii} = V_{11} + V_{22} + V_{33}$$

$$I_2(\mathcal{V}) = \frac{1}{2} (V_{ii}V_{jj} - V_{ij}V_{ij}) = \frac{1}{2} \left[(\text{tr } \mathcal{V})^2 - \text{tr } \mathcal{V}^2 \right]$$

$$I_3(\mathcal{V}) = \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} V_{ip} V_{jq} V_{kr} = \det \mathcal{V}$$

$I_i(\mathcal{V})$ are scalars produced by outer multiplication and contraction and as such as scalars independent of frame.

Definition 3.12.1.

A 2nd order tensor $\mathcal{V}$ is non-singular if $\det \mathcal{V} \neq 0$; otherwise, a 2nd order tensor $\mathcal{V}$ is singular if there exists a vector $\underline{u} \neq \underline{0}$ such that $\mathcal{V}\underline{u} = \underline{0}$ or $V_{ij}u_j = 0$, then by Cramer's rule $\det[V_{ij}] = 0$.

Claim 3.12.1.

- (a) Let $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$. Also let $\mathcal{Q}$ be a proper orthogonal tensor and define vectors $\underline{e}'_i = \mathcal{Q}\underline{e}_i$. Then vector trait $\underline{e}'_i$ are base vectors for an other proper orthogonal frame $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} \in \mathcal{F}$ and the matrix $\underline{A}$ of direction cosines that transforms X to X' ($\underline{A}: X \rightarrow X'$) is given by $A_{ij} = Q_{ji}^X$.
- (b) Let $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ and $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} \in \mathcal{F}$ and also let $\underline{A}: X \rightarrow X'$ then there exists an unique (proper orthogonal) such that $\underline{e}'_i = \mathcal{Q}\underline{e}_i$ and $\mathcal{Q}$ is determined by $Q_{ij}^X = A_{ji}$.

Proof.

3.13. PRINCIPAL VALUE AND PRINCIPAL DIRECTION OF SECOND ORDER TENSORS DEFINITION

$$\underline{e}'_i \cdot \underline{e}'_j = \left(\underline{Q} \underline{e}_i \right) \cdot \left(\underline{Q} \underline{e}_j \right) \stackrel{\P}{=} \underbrace{\underline{Q}^T \underline{Q}}_{\underline{1}} \underbrace{\underline{e}_i \cdot \underline{e}_j}_{\delta_{ij}} = \delta_{ij}$$

By definition of $\underline{A}$

$$\underline{e}'_i \cdot \underline{e}_j = A_{ij}$$

$$\therefore A_{ij} = \underline{e}'_i \cdot \underline{e}_j = \left(\underline{Q} \underline{e}_i \right) \cdot \underline{e}_j \stackrel{\P}{=} Q_{ji}^X$$

$$\det \underline{A} = \det \underline{Q}^T = \det \underline{Q} = +1$$

□

3.13 Principal value and principal direction of second order tensors definition

Let $\underline{V}$ be a second order tensor, a real number $\lambda = \lambda(\underline{V})$ is a principal value (or an eigenvalue) of $\underline{V}$ if there exists a vector $\underline{n}$ such that

$$\underline{V} \underline{n} = \lambda \underline{n} \qquad \text{and} \qquad \underline{n} \cdot \underline{n} = 1$$

$$\begin{aligned} \P \underline{u} \cdot (\underline{W} \underline{v}) &= (\underline{W}^T \underline{u}) \cdot \underline{v} \\ \P W_{ij}^X &= \underline{e}_i \cdot (\underline{W} \underline{e}_j) \end{aligned}$$

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or

$$V_{ij}n_j = \lambda n_i \quad \text{and} \quad n_i n_i = 1$$

If so, $\underline{n}$ is called a principal direction (or eigenvector) of $\underline{V}$ corresponding to eigenvalue λ

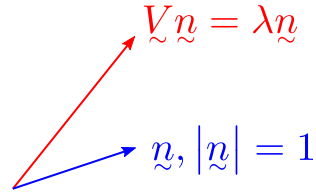


Figure 3.2: Scheme of non-principal direction vector

1. if $\underline{n}$ is *not* a principal direction vector.

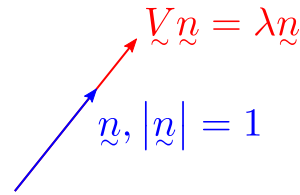


Figure 3.3: Scheme of principal direction vector

2. if $\underline{n}$ is a principal direction vector.

Also $\underline{n}$ and $-\underline{n}$ define a principal direction or principal axis belonging to eigenvalue λ .

Claim 3.13.1.

3.13. PRINCIPAL VALUE AND PRINCIPAL DIRECTION OF SECOND ORDER TENSORS DEFINITION

Let $\underline{V}$ be a 2^{nd} order tensor. Then λ is a principal value of $\underline{V}$ if and only if it is a real root of the cubic equation

$$F(\lambda) = \det(\underline{V} - \lambda \underline{1}) = -\lambda^3 + I_1(\underline{V})\lambda^2 - I_2(\underline{V})\lambda + I_3(\underline{V}) = 0$$

which is characteristic equation of $\underline{V}$.

If λ is a principal value of $\underline{V}$, then the principal direction vector belonging to λ are the solutions for $\underline{n}$ of the equations

$$\begin{cases} (V_{ij} - \lambda \delta_{ij}) n_j = 0 \\ n_i n_i = 1 \end{cases}$$

Note. Every 2^{nd} order tensor has at least one real eigenvalue (since a cubic of real coefficient has at least one real root). If the other roots are complex, they are complex conjugate.

Proof.

Let λ be eigenvalue, then there exists 3 real numbers n_i , not all zeros such that

$$V_{ij}n_j - \lambda n_i = 0$$

or

$$(V_{ij} - \lambda \delta_{ij}) n_j = 0$$

From Cramer's Rule

$$\det(V_{ij} - \lambda \delta_{ij}) = 0$$

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(for non-trivial solution)

$$\det(V_{ij} - \lambda \delta_{ij}) = \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} (V_{ip} - \lambda \delta_{ip}) (V_{jq} - \lambda \delta_{jq}) (V_{kr} - \lambda \delta_{kr}) = 0$$

(by definition of invariants)

$$\Rightarrow -\lambda^3 + I_1(\mathcal{V})\lambda^2 - I_2(\mathcal{V})\lambda + I_3(\mathcal{V}) = 0$$

□

3.14 Symmetric tensors for second order tensors

Theorem 3.14.1.

Let $\mathcal{V} = \mathcal{V}^T$, then all three roots $(\lambda_1, \lambda_2, \lambda_2)$ of the characteristic equation are real.

$$F(\lambda) = \det(\mathcal{V} - \lambda \mathbf{1}) = -\lambda^3 + I_1\lambda^2 - I_2\lambda + I_3 = 0 \quad (\text{i})$$

Further, there exists at least one set of three mutually perpendicular principal axes of $\mathcal{V}$ and if

$X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$ is a principal frame for $\mathcal{V}$ then one has

$$[V_{ij}^{X'}] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \text{or} \quad V_{ij}^{X'} = \lambda_i \delta_{ij} \quad (\text{no sum})$$

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provided that

$$\underline{V}\underline{e}'_i = \lambda_i \underline{e}'_i \quad (\text{no sum})$$

i.e. $\underline{e}'_i$ are principal direction vectors.

Case I $(\lambda_1, \lambda_2, \lambda_3)$ are distinct roots of (i) (simple roots) then there exists a unique principal axis belonging to each principal value λ_i and the three principal axes are mutually perpendicular.

Case II Say $\lambda_1 \neq \lambda_2 = \lambda_3 = \lambda_*$, i.e. (i) has one simple root and a double root. Then there exists a unique principal axis belonging to λ_1 and every axis which is perpendicular to the former is a principal axis belonging to λ_* .

Case III $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_*$, i.e. (i) has a triple root. Then every axis is a principal axis belonging to λ_* and

$$\underline{V} = \lambda_* \underline{1} \quad \text{or} \quad V_{ij} = \lambda_* \delta_{ij} \quad \forall X \in \mathcal{F}$$

so $\underline{V}$ is an isotropic tensor.

Lemma 3.14.1.

If $\underline{n}^{(1)}$ and $\underline{n}^{(2)}$ are principal direction vectors belonging to two distinct principal values λ_1 and λ_2 , then $\underline{n}^{(1)} \perp \underline{n}^{(2)}$.

Proof.

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$$\begin{cases} \underline{V}\underline{n}^{(1)} = \lambda_1 \underline{n}^{(1)} \\ \underline{V}\underline{n}^{(2)} = \lambda_2 \underline{n}^{(2)} \end{cases} \text{ where } \underline{V} = \underline{V}^T \quad (\textcircled{\text{S}})$$

$$\underline{n}^{(2)} \cdot \lambda_1 \underline{n}^{(1)} \stackrel{(\textcircled{\text{S}})}{=} \underline{n}^{(2)} \cdot \underline{V}\underline{n}^{(1)} \stackrel{**}{=} \left(\underline{V}^T \underline{n}^{(2)} \right) \cdot \underline{n}^{(1)}$$

$$\stackrel{\underline{V}^T = \underline{V}}{=} \underline{V}\underline{n}^{(2)} \cdot \underline{n}^{(1)} \stackrel{(\textcircled{\text{S}})}{=} \lambda_2 \underline{n}^{(2)} \cdot \underline{n}^{(1)}$$

$$\Rightarrow (\lambda_1 - \lambda_2) \underline{n}^{(2)} \cdot \underline{n}^{(1)} = 0$$

$$\stackrel{\lambda_1 \neq \lambda_2}{\Rightarrow} \underline{n}^{(2)} \cdot \underline{n}^{(1)} = 0 \text{ or } \underline{n}^{(2)} \perp \underline{n}^{(1)}$$

□

Lemma 3.14.2.

Let λ be a principal value of a symmetric, 2^{nd} order tensor $\underline{V}$ and $\underline{n}$ be a principal vector belonging to λ . For fixed k ($k = 1, 2, 3$), choose $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ in such a way that $\underline{e}_k = \underline{n}$, i.e. $\underline{V}\underline{n} = \lambda \underline{n}$, $\underline{V}\underline{e}_k = \lambda \underline{e}_k$. Then

$$V_{ki}^X = V_{ik}^X = \lambda \delta_{ik}$$

e.g. $k = 1$

$$V_{ki}^X = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & V_{22}^X & V_{23}^X \\ 0 & V_{32}^X & V_{33}^X \end{bmatrix}$$

$$** \underline{u} \cdot (\underline{W}\underline{v}) = (\underline{W}^T \underline{u}) \cdot \underline{v}$$

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Proof.

Choose and fix k . By hypothesis

$$\begin{aligned} V_{ij} \mathfrak{e}_k = \lambda \mathfrak{e}_k &\Rightarrow V_{ij} (\mathfrak{e}_k)_j = \lambda (\mathfrak{e}_k)_i \Rightarrow V_{ij} \delta_{kj} = \lambda \delta_{ki} \\ &\Rightarrow V_{ki}^X = \lambda \delta_{ki} \end{aligned}$$

□

Proof. Proof of theorem 3.14.1

Let $\mathfrak{n}^{(1)}$ be a principal direction vector belonging to λ_1 . Choose frame $X' = \{0; \mathfrak{e}'_1, \mathfrak{e}'_2, \mathfrak{e}'_3\} \in \mathcal{F}$ that $\mathfrak{e}'_1 = \mathfrak{n}^{(1)}$. Then by lemma 3.14.2

$$V_{1i}^{X'} = \lambda_1 \delta_{i1} \quad \text{or} \quad V_{11}^{X'} = \lambda_1, V_{12}^{X'} = V_{21}^{X'} = V_{13}^{X'} = V_{31}^{X'} = 0 \quad (\text{a})$$

Thus

$$\begin{aligned} \det(V_{ij}^{X'} - \lambda \delta_{ij}) &= \det \begin{bmatrix} \lambda_1 - \lambda & 0 & 0 \\ 0 & V_{22}^{X'} - \lambda & V_{23}^{X'} - \lambda \\ 0 & V_{32}^{X'} - \lambda & V_{33}^{X'} - \lambda \end{bmatrix} = 0 \\ &\Rightarrow (\lambda_1 - \lambda) \left\{ \lambda^2 - (V_{22}^{X'} + V_{33}^{X'}) \lambda + V_{22}^{X'} V_{33}^{X'} - (V_{23}^{X'})^2 \right\} = 0 \quad (\text{b}) \end{aligned}$$

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$$\begin{aligned}
 \therefore \lambda &= \lambda_1 \\
 \lambda &= \begin{pmatrix} \lambda_2 \\ \lambda_3 \end{pmatrix} \\
 &= \frac{V_{22}^{X'} + V_{33}^{X'}}{2} \\
 &\quad \pm \underbrace{\sqrt{\frac{\left(V_{22}^{X'} + V_{33}^{X'}\right) - 4 \left[V_{22}^{X'} V_{33}^{X'} - \left(V_{23}^{X'}\right)^2\right]}{4}}}_{\sqrt{\frac{\left(V_{22}^{X'} - V_{33}^{X'}\right)^2}{4} + \left(V_{23}^{X'}\right)^2}}
 \end{aligned} \tag{c}$$

the roots are real roots.

Case I $\lambda_1, \lambda_2, \lambda_3$ are distinct

Let $\underline{n}^{(i)}$ be the three principal axes belonging to λ_i by lemma 3.14.1, $\underline{n}^{(i)}$ are mutually perpendicular. Let $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} \in \mathcal{F}$ such that $\underline{n}^{(1)} = \underline{e}'_1$, $\underline{n}^{(2)} = \underline{e}'_2$, $\underline{n}^{(3)} = \underline{e}'_3$. By lemma 3.14.2 for $k = 1, 2, 3$

$$\left. \begin{aligned} V_{i1}^{X'} &= \lambda_1 \delta_{i1} \\ V_{i2}^{X'} &= \lambda_2 \delta_{i2} \\ V_{i3}^{X'} &= \lambda_3 \delta_{i3} \end{aligned} \right\} V_{ij}^{X'} = \lambda_j \delta_{ij} \quad (\text{no sum})$$

i.e.

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$$[V_{ij}^{X'}] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Case II $\lambda_1 \neq \lambda_2 = \lambda_3 = \lambda_*$

Choose $X' = \{0; \mathfrak{e}'_1, \mathfrak{e}'_2, \mathfrak{e}'_3\} \in \mathcal{F}$ such that $\mathfrak{e}'_1 = \mathfrak{n}^{(1)}$ then

(a) holds, i.e. $V_{11}^{X'} = \lambda_1, V_{12}^{X'} = V_{21}^{X'} = V_{13}^{X'} = V_{31}^{X'} = 0$. In addition, $\lambda_2 = \lambda_3 = \lambda_*$

$$\stackrel{(c)}{\Rightarrow} V_{22}^{X'} = V_{33}^{X'} = \lambda_*, V_{23}^{X'} = V_{32}^{X'} = 0 \quad (d)$$

i.e.

$$[V_{ij}^{X'}] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_* & 0 \\ 0 & 0 & \lambda_* \end{bmatrix}$$

To show that any choice of $\mathfrak{e}'_2, \mathfrak{e}'_3$ ($\perp \mathfrak{n}^{(1)}$) is principal axis

$$(\mathcal{V}\mathfrak{e}'_2)_i = V_{ij}^{X'} (\mathfrak{e}'_2)_j = V_{ij}^{X'} \delta_{2j} = V_{i2}^{X'} = \lambda_* \delta_{i2} = \lambda_* (\mathfrak{e}'_2)_i$$

$$\therefore \mathcal{V}\mathfrak{e}'_2 = \lambda_* \mathfrak{e}'_2$$

Similarly

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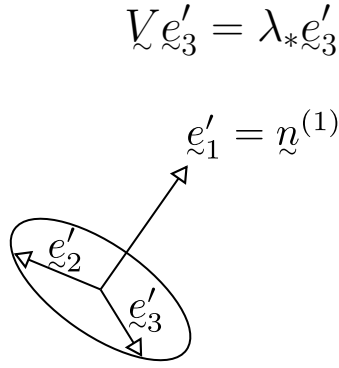


Figure 3.4: Scheme of eigenvectors of $\lambda_1 \neq \lambda_2 = \lambda_3 = \lambda_*$

infinite choice of $\underline{e}'_2$ and $\underline{e}'_3$.

Case III

$\lambda_1 = \lambda_2 = \lambda_3 = \lambda_*$

Choose $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} \in \mathcal{F}$ such that $\underline{e}'_1 = \underline{n}^{(1)}$ (one eigenvector belonging to $\lambda_1 = \lambda_*$). Then (a) holds and (d) holds, i.e.

$$[V_{ij}^{X'}] = \begin{bmatrix} \lambda_* & 0 & 0 \\ 0 & \lambda_* & 0 \\ 0 & 0 & \lambda_* \end{bmatrix} \quad \text{or} \quad V_{ij}^{X'} = \lambda_* \delta_{ij} \quad \text{or} \quad \underline{V} = \lambda_* \underline{1}$$

$$\Rightarrow \underline{V}\underline{n} = \lambda_* \underline{1}\underline{n} = \lambda_* \underline{n}$$

where $\underline{n}$ is any unit vector vector.

Every unit vector $\underline{n}$ is a principal direction vector belonging to λ_* and every frame $X \in \mathcal{F}$ is a principal frame for $\underline{V}$.

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□

Theorem 3.14.2. *Spectral Representation theorem*

Let $X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$ is a principal frame for a symmetric 2^{nd} tensor $\mathcal{V}$ and let λ_i be the eigenvalues. Then

$$\mathcal{V} = \sum_{i=1}^3 \lambda_i \mathbf{e}'_i \otimes \mathbf{e}'_i = \lambda_1 \mathbf{e}'_1 \otimes \mathbf{e}'_1 + \lambda_2 \mathbf{e}'_2 \otimes \mathbf{e}'_2 + \lambda_3 \mathbf{e}'_3 \otimes \mathbf{e}'_3$$

Proof.

Recall that for all $X \in \mathcal{F}$, not necessary principal

$$\mathcal{V} = V_{ij}^X \mathbf{e}_i \otimes \mathbf{e}_j$$

if we choose a principal frame $X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$, then

$$V_{ij}^{X'} = \lambda_i \delta_{ij} \quad (\text{no sum})$$

So

$$\begin{aligned} \mathcal{V} &= V_{11}^{X'} \mathbf{e}'_1 \otimes \mathbf{e}'_1 + V_{12}^{X'} \mathbf{e}'_1 \otimes \mathbf{e}'_2 + V_{13}^{X'} \mathbf{e}'_1 \otimes \mathbf{e}'_3 + \cdots \\ &= \lambda_1 \mathbf{e}'_1 \otimes \mathbf{e}'_1 + 0 + 0 + \cdots \\ &= \lambda_1 \mathbf{e}'_1 \otimes \mathbf{e}'_1 + \lambda_2 \mathbf{e}'_2 \otimes \mathbf{e}'_2 + \lambda_3 \mathbf{e}'_3 \otimes \mathbf{e}'_3 \end{aligned}$$

□

If $\lambda_1 \neq \lambda_2 = \lambda_3 = \lambda_*$, then

$$\mathcal{V} = (\lambda_1 - \lambda_*) \mathbf{e}'_1 \otimes \mathbf{e}'_1 + \lambda_* \mathbf{1}^{\dagger\dagger}$$

$$\mathbf{1}^{\dagger\dagger} \cdot \mathbf{1} = \sum_{j=1}^3 \mathbf{e}'_j \otimes \mathbf{e}'_j \quad (\text{Homework})$$

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If $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_*$, then

$$\mathbb{V}_{\lambda_*} \mathbb{1}$$

3.15 Skew-symmetric tensors for second order tensors

$$\mathbb{V} = -\mathbb{V}^T \quad \text{or} \quad V_{ij} = -V_{ji}$$

$$[V_{ij}] = \begin{bmatrix} 0 & V_{12} & V_{13} \\ -V_{12} & 0 & V_{23} \\ -V_{13} & -V_{23} & 0 \end{bmatrix} \Rightarrow \det \mathbb{V} = 0$$

every skew-symmetric tensor is *singular*.

Three number V_{12} , V_{13} , V_{23} are needed to define a skew symmetric tensor.

Define a vector w as follows

$$w = \text{dual } \mathbb{V} \quad \text{if} \quad w_i = \frac{1}{2} \epsilon_{ijk} V_{kj} \quad (\star)$$

The component V_{ij} of $\mathbb{V}$ are related to w by

$$V_{ij} = \epsilon_{ikj} w_k \quad \text{or} \quad [V_{ij}] = \begin{bmatrix} 0 & -\omega_3 & -\omega_2 \\ \omega_3 & 0 & -\omega_1 \\ \omega_2 & \omega_1 & 0 \end{bmatrix}$$

Proof.

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From (★)

$$w_i = \frac{1}{2} \epsilon_{ijk} V_{kj}$$

multiply both side by $-\epsilon_{ipq}$

$$\begin{aligned} -\epsilon_{ipq} w_i &= -\frac{1}{2} \epsilon_{ipq} \epsilon_{ijk} V_{kj} \\ &= -\frac{1}{2} (\delta_{pj} \delta_{qk} - \delta_{pk} \delta_{qj}) V_{kj} \\ &= -\frac{1}{2} (V_{qp} - V_{pq}) = V_{pq} \end{aligned}$$

$$\therefore V_{pq} = \epsilon_{piq} w_i \xrightarrow[p \rightarrow i, q \rightarrow j]{i \rightarrow k} V_{ij} = \epsilon_{ikj} w_k$$

□

If $\underline{V}$ is skew-symmetric and $\underline{w} = \text{dual } \underline{V}$, then

$$\boxed{\underline{V} \underline{u} = \underline{w} \wedge \underline{u} \quad \forall \underline{u} \in E_3}$$

Proof.

$$(\underline{w} \wedge \underline{u})_i = \underbrace{\epsilon_{ikj} w_k}_{V_{ij}} u_j = V_{ij} u_j = (\underline{V} \underline{u})_i$$

□

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3.16 Decomposition Theorem for second order tensors

Claim 3.16.1.

Every 2^{nd} order tensor $\mathcal{W}$ admits a unique decomposition such that

$$\mathcal{W} = \mathcal{U} + \mathcal{V} \quad (*)$$

where

$$\begin{aligned} \mathcal{U} & : \text{symmetric} & (\mathcal{U} = \mathcal{U}^T) \\ \mathcal{V} & : \text{skew-symmetric} & (\mathcal{V} = -\mathcal{V}^T) \end{aligned}$$

Further this resolution is

$$\begin{cases} \mathcal{U} = \frac{1}{2} (\mathcal{W} + \mathcal{W}^T) \\ \mathcal{V} = \frac{1}{2} (\mathcal{W} - \mathcal{W}^T) \end{cases} \quad (**)$$

or

$$\begin{cases} U_{ij} = \frac{1}{2} (W_{ij} + W_{ji}) \\ V_{ij} = \frac{1}{2} (W_{ij} - W_{ji}) \end{cases}$$

$$\mathcal{U} = \text{sym } \mathcal{W} \qquad \mathcal{V} = \text{skew } \mathcal{W}$$

Claim 3.16.2. *Isotropic tensors of 1^{st} and 2^{nd} order tensor*

3.16. DECOMPOSITION THEOREM FOR SECOND ORDER TENSORS

- (a) If v is an isotropic vector, then $v = 0$.
- (b) If V is an isotropic skew-symmetric 2^{nd} order tensor, then $V = 0$.
- (c) If V is an isotropic 2^{nd} order tensor, then $V = \alpha 1$, $\alpha \in \mathbb{R}$.

Proof.

- (a) v is an isotropic vector, choose $X = \{0; e_1, e_2, e_3\} \in \mathcal{F}$,
 $v = v_i^X e_i$. Construct
 $X' = \{0; e'_1, e'_2, e'_3\} = \{0; e_1, e_3, -e_2\}$.

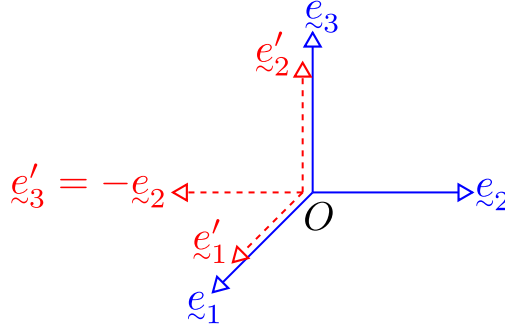


Figure 3.5: Scheme of frame X'

X' obtained from X by rotation about e_1 axis (angle: $\pi/2$)
 Define $X'' = \{0; e''_1, e''_2, e''_3\} \in \mathcal{F}$ by rotation of X through $\pi/2$ about e_1 axis, i.e.

$$X'' = \{0; e''_1, e''_2, e''_3\} = \{0; -e_3, e_2, e_1\}.$$

By the definition of isotropic tensor

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$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_X = \begin{bmatrix} \underline{v} \cdot \underline{e}'_1 = \underline{v} \cdot \underline{e}_1 = v_1 \\ \underline{v} \cdot \underline{e}'_2 = \underline{v} \cdot \underline{e}_3 = v_3 \\ \underline{v} \cdot \underline{e}'_3 = \underline{v} \cdot (-\underline{e}_2) = -v_2 \end{bmatrix}_{X'} = \begin{bmatrix} -v_3 \\ v_2 \\ v_1 \end{bmatrix}_{X''}$$

$$\Rightarrow \begin{cases} v_1 = -v_3 \\ v_3 = v_2 \\ -v_2 = v_1 \end{cases}$$

$$\Rightarrow v_1 = v_2 = v_3 = 0$$

(b) Let $\underline{V} = -\underline{V}^T$ and be isotropic

$$\underline{w} = \text{dual } \underline{V}$$

or

$$w_i = \frac{1}{2} \underbrace{\epsilon_{ijk}}_{\text{isotropic}} \underbrace{V_{kj}}_{\substack{\text{isotropic} \\ \text{by hypothesis}}} \Rightarrow \underline{w} \text{ is isotropic} \stackrel{(a)}{=} \underline{w} = 0$$

$$\Rightarrow \underline{V} = \underline{0}$$

(c) Let $\underline{V} \neq 0$ be an isotropic tensor, then both $\text{sym } \underline{V}$ and $\text{skew } \underline{V}$ are isotropic. By (b) $\text{skew } \underline{V} = \underline{0}$, thus

3.16. DECOMPOSITION THEOREM FOR SECOND ORDER TENSORS

$$\underset{\sim}{V} = \text{sym } \underset{\sim}{V} \quad \text{or} \quad \underset{\sim}{V} = \underset{\sim}{V}^T$$

i.e. any non-zero isotropic tensor is symmetric.

Thus there exists a frame $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ such that

$$[V_{ij}^X] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \text{and} \quad \underset{\sim}{V} \underline{e}_i = \lambda_i \underline{e}_i \quad (\text{no sum})$$

Introduce $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} = \{0; \underline{e}_2, \underline{e}_3, \underline{e}_1\} \in \mathcal{F}$

$$\begin{aligned} V_{ij}^{X'} &= \underline{e}'_i \cdot \left(\underset{\sim}{V} \underline{e}'_j \right) \\ &\stackrel[\text{representation thm.}]{\text{spectral}} = \underline{e}'_i \cdot \left(\lambda_1 \underline{e}_1 \otimes \underline{e}_1 + \lambda_2 \underline{e}_2 \otimes \underline{e}_2 + \lambda_3 \underline{e}_3 \otimes \underline{e}_3 \right) \underline{e}'_j \end{aligned}$$

$$\because \underline{e}'_1 = \underline{e}_2, \underline{e}'_2 = \underline{e}_3, \underline{e}'_3 = \underline{e}_1$$

$$\begin{aligned} V_{11}^{X'} &= \underline{e}'_1 \cdot \left(\lambda_1 \underline{e}_1 \otimes \underline{e}_1 + \lambda_2 \underline{e}_2 \otimes \underline{e}_2 + \lambda_3 \underline{e}_3 \otimes \underline{e}_3 \right) \underline{e}'_1 \\ &= \underline{e}_2 \cdot \left(\lambda_1 \underline{e}_1 \otimes \underline{e}_1 + \lambda_2 \underline{e}_2 \otimes \underline{e}_2 + \lambda_3 \underline{e}_3 \otimes \underline{e}_3 \right) \underline{e}_2 \\ &\stackrel{\dagger\dagger}{=} \underline{e}_2 \cdot \left[\lambda_1 \underline{e}_1 (\underline{e}_1 \cdot \underline{e}_2) + \lambda_2 \underline{e}_2 (\underline{e}_2 \cdot \underline{e}_2) + \lambda_3 \underline{e}_3 (\underline{e}_3 \cdot \underline{e}_2) \right] \\ &= \lambda_2 \end{aligned}$$

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Similarly

$$V_{22}^{x'} = \lambda_3, \quad V_{33}^{X'} = \lambda_1 \quad \text{and} \quad V_{ij}^{X'} = 0, i \neq j$$

$$\therefore [V_{ij}^{X'}] = \begin{bmatrix} \lambda_2 & 0 & 0 \\ 0 & \lambda_3 & 0 \\ 0 & 0 & \lambda_1 \end{bmatrix} \xrightarrow[\substack{\text{isotropic} \\ [V_{ij}^X]=[V_{ij}^{X'}]}}{\lambda_1 = \lambda_2, \lambda_2 = \lambda_3, \lambda_3 = \lambda_1}$$

i.e. $\lambda_1 = \lambda_2 = \lambda_3 = \alpha$ or $\mathbb{V} = \alpha \mathbb{1}$.

□

Claim 3.16.3.

Every $\mathcal{Q}^{nd}$ order tensor $\mathbb{W}$ admits a unique additive decomposition such that

$$\mathbb{W} = \mathbb{U} + \mathbb{V} \tag{*}$$

where

$$\text{tr } \mathbb{U} = 0, \quad \mathbb{V} \text{ is isotropic}$$

The resolution is given by

$$\mathbb{V} = \frac{1}{3} (\text{tr } \mathbb{W}) \mathbb{1}, \quad \mathbb{U} = \mathbb{W} - \mathbb{V} = \mathbb{W} - \frac{1}{3} (\text{tr } \mathbb{W}) \mathbb{1}$$

$^{\ddagger\ddagger}(\underline{u} \otimes \underline{v}) \underline{a} = (\underline{v} \cdot \underline{a}) \underline{u}$

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or in component form

$$V_{ij} = \frac{1}{3}W_{kk}\delta_{ij}, \quad U_{ij} = W_{ij} - \frac{1}{3}W_{kk}\delta_{ij} \quad (**)$$

Proof.

Assume the existence of the decomposition (*), then

$$\underline{U} = \underline{W} - \underline{V} \quad \text{and} \quad \underline{V} = \alpha \underline{1} \quad (\because \underline{V} \text{ is isotropic})$$

$$\Rightarrow \text{tr } \underline{W} = \text{tr } \underline{U} + \text{tr } \underline{V} = 0 + \alpha \text{tr } \underline{1} = 3\alpha$$

$$\Rightarrow \alpha = \frac{1}{3} \text{tr } \underline{W}$$

or

$$\underline{V} = \frac{1}{3} (\text{tr } \underline{W}) \underline{1}$$

$$\underline{U} = \underline{W} - \frac{1}{3} (\text{tr } \underline{W}) \underline{1}$$

$\underline{U}$ is called deviatoric (or traceless) part of $\underline{W}$.

$\underline{V}$ is called the hydrostatic (or spherical) or isotropic part of $\underline{W}$. □

Lemma 3.16.1. *Square root of positive definite symmetric 2^{nd} order tensor*

Definition 3.16.1.

$\underline{S}$ is *positive definite* if $\underline{v} \cdot (\underline{S}\underline{v}) > 0 \quad \forall \underline{v} \in E_3, \underline{v} \neq 0$.

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Let $\mathcal{S}$ be a symmetric, positive definite 2^{nd} order tensor. Then there exists a unique, symmetric, positive definite 2^{nd} order tensor $\mathcal{U}$ such that

$$\mathcal{U}\mathcal{U} = \mathcal{U}^2 = \mathcal{S} \quad \text{or} \quad U_{ik}U_{kj} = S_{ij}$$

One calls $\mathcal{U}$ the square root of $\mathcal{S}$ and denote $\mathcal{U} = \sqrt{\mathcal{S}}$. One can show that if λ_i are the eigenvalues of $\mathcal{S}$, then $\lambda_i > 0$, and $\alpha_i = \sqrt{\lambda_i}$ are the eigenvalue of $\mathcal{U} = \sqrt{\mathcal{S}}$, i.e. if $X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$ is the principal frame for $\mathcal{S}$, the X' is also principal frame for $\mathcal{U} = \sqrt{\mathcal{S}}$ and

$$[S_{ij}^{X'}] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}, \quad [U_{ij}^{X'}] = \begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \sqrt{\lambda_2} & 0 \\ 0 & 0 & \sqrt{\lambda_3} \end{bmatrix}$$

Proof.

$\mathcal{S}$ is symmetric so there exists $X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$ such that

$$[S_{ij}^{X'}] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Let $\mathbf{v} = \mathbf{e}'_1$, since $\mathcal{S}$ is positive definite, i.e. $\mathbf{e}'_1 \cdot (\mathcal{S}\mathbf{e}'_1) > 0$

$$\Rightarrow S_{11}^{X'} > 0 \Rightarrow \lambda_1 > 0$$

Similarly, let $\mathbf{v} = \mathbf{e}'_2$

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$$\Rightarrow \lambda_2 > 0$$

and

$$\mathfrak{v} = \mathfrak{e}'_3 \Rightarrow \lambda_3 > 0$$

Now construct a symmetric 2nd order tensor $\mathcal{U}$ such that $\mathcal{U}$ has the same principal frame as $\mathcal{S}$, i.e. X' , and in that form

$$[U_{ij}^{X'}] = \begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \sqrt{\lambda_2} & 0 \\ 0 & 0 & \sqrt{\lambda_3} \end{bmatrix} = \begin{bmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ 0 & 0 & \alpha_3 \end{bmatrix}; \alpha_i = \sqrt{\lambda_i} > 0$$

(S)

The $\mathcal{U}$ constructed this way satisfies

$$\mathcal{U}\mathcal{U} = \mathcal{S}$$

and $\mathcal{U}$ is positive definite and symmetric $\Rightarrow$ identified *one* square root of $\mathcal{S}$.

One has to show now that $\mathcal{U}$ given by (S) is *unique*.

If $\mathcal{S}\mathfrak{e} = \mathcal{U}^2\mathfrak{e} = \mu\mathfrak{e}, |\mathfrak{e}| = 1, \mu > 0$, then we can prove $\mathcal{U}\mathfrak{e} = \sqrt{\mu}\mathfrak{e}$ where $\mathfrak{e}$ is a principal direction vector of $\mathcal{S}$.

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$$\Rightarrow \left(\underline{U}^2 - \mu \underline{1} \right) \underline{e} = \underline{0}$$

$$\Rightarrow \left(\underline{U} + \sqrt{\mu} \underline{1} \right) \underbrace{\left(\underline{U} - \sqrt{\mu} \underline{1} \right) \underline{e}}_{\equiv \underline{v}} = \underline{0}$$

Assume $\underline{v} \neq \underline{0}$, then

$$\begin{aligned} \left(\underline{U} + \sqrt{\mu} \underline{1} \right) \underline{v} = \underline{0} &\Rightarrow \underline{U} \underline{v} = -\sqrt{\mu} \underline{v} \\ &\Rightarrow \underline{v} \cdot (\underline{U} \underline{v}) = -\sqrt{\mu} (\underline{v} \cdot \underline{v}) < 0 \end{aligned}$$

which contradicts the assumption that $\underline{U}$ is positive definite. So

$$\underline{v} = \underline{0} \Rightarrow \underline{U} \underline{e} = \sqrt{\mu} \underline{e}$$

□

Theorem 3.16.1. *Polar decomposition*

Let $\underline{W}$ be a non-singular 2^{nd} order tensor ($\det \underline{W} \neq 0$). Then there exists unique 2^{nd} order tensor $\underline{U}$, $\underline{V}$, $\underline{Q}$, and $\underline{Q}'$ such that

$$\begin{cases} \underline{W} = \underline{Q} \underline{U} = \underline{V} \underline{Q}'^{\S\S} \\ \underline{Q}, \underline{Q}': \text{orthogonal}, \underline{U}, \underline{V}: \text{symmetric definite} \end{cases} \quad (*)$$

and

$$\underline{U} = \sqrt{\underline{W}^T \underline{W}}, \quad \underline{V} = \sqrt{\underline{W} \underline{W}^T}$$

$\S\S \underline{Q} \underline{U}$: right decomposition, $\underline{V} \underline{Q}'$: left decomposition.

3.16. DECOMPOSITION THEOREM FOR SECOND ORDER TENSORS

$$\underset{\sim}{Q} = \underset{\sim}{W}\underset{\sim}{U}^{-1}, \quad \underset{\sim}{Q}' = \underset{\sim}{V}^{-1}\underset{\sim}{W}$$

Further

$$\underset{\sim}{Q} = \underset{\sim}{Q}', \quad \underset{\sim}{V} = \underset{\sim}{Q}\underset{\sim}{U}\underset{\sim}{Q}^T, \quad \underset{\sim}{U} = \underset{\sim}{Q}^T\underset{\sim}{V}\underset{\sim}{Q}$$

If in addition $\det \underset{\sim}{W} > 0$, then $\underset{\sim}{Q}$ is proper orthogonal, i.e. $\det \underset{\sim}{Q} = +1$.

Proof.

1. If $\underset{\sim}{Q} = \underset{\sim}{Q}'$, then $(*) \Rightarrow \underset{\sim}{V} = \underset{\sim}{Q}\underset{\sim}{U}\underset{\sim}{Q}^T, \underset{\sim}{U} = \underset{\sim}{Q}^T\underset{\sim}{V}\underset{\sim}{Q}$.
2. Concentrate existence and uniqueness of right decomposition, i.e.

$$\underset{\sim}{W} = \underset{\sim}{Q}\underset{\sim}{U}$$

Assume that there exists $\underset{\sim}{Q}$ and $\underset{\sim}{U}$ such that

$$\underset{\sim}{W} = \underset{\sim}{Q}\underset{\sim}{U}, \underset{\sim}{Q}\underset{\sim}{Q}^T = \underset{\sim}{1}, \underset{\sim}{U}^T = \underset{\sim}{U}, \underset{\sim}{U}: \text{positive definite} \quad (1)$$

Let

$$\underset{\sim}{S} = \underset{\sim}{W}^T\underset{\sim}{W}, \quad S_{ij} = W_{ki}W_{kj} \quad (2)$$

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

$$\mathcal{S} = \mathcal{U}^T \underbrace{\mathcal{Q}^T \mathcal{Q}}_{\mathcal{I}} \mathcal{U} = \mathcal{U}^T \mathcal{U} = \mathcal{U} \mathcal{U} = \mathcal{U}^2$$

So

$$\mathcal{S} = \mathcal{S}^T \quad (3)$$

$$(2) \& (3) \Rightarrow \mathcal{U} = \sqrt{\mathcal{W}^T \mathcal{W}} \quad (4)$$

$$\text{From (1), } \mathcal{Q} = \mathcal{W} \mathcal{U}^{-1} \quad (5)$$

Now need to prove that $\mathcal{U} = \sqrt{\mathcal{W}^T \mathcal{W}}$ is meaningful, i.e. $\mathcal{S} = \mathcal{W}^T \mathcal{W}$ is symmetric and positive definite and $\mathcal{Q}$ from (5) is orthogonal.

$$\because \mathcal{S} = \mathcal{W}^T \mathcal{W}$$

$$\Rightarrow \mathcal{S}^T = \left(\mathcal{W}^T \mathcal{W} \right)^T = \mathcal{W}^T \left(\mathcal{W}^T \right)^T = \mathcal{W}^T \mathcal{W} = \mathcal{S}$$

Also, $\det \mathcal{S} = \det \mathcal{W}^T \det \mathcal{W} = (\det \mathcal{W})^2 > 0$
 $(\because \det \mathcal{W} \neq 0)$

3.16. DECOMPOSITION THEOREM FOR SECOND ORDER TENSORS

$$\begin{aligned} \underline{x} \cdot \underline{S} \underline{x} &= \underline{x} \cdot \left(\underline{W}^T \underline{W} \right) \underline{x} = \underline{x} \cdot \underline{W}^T \left(\underline{W} \underline{x} \right) \\ &\stackrel{\P}{=} \left(\underline{W} \underline{x} \right) \cdot \left(\underline{W} \underline{x} \right) > 0 \end{aligned}$$

Then $\underline{S}$ is symmetric and positive definite.

From (5), $\underline{Q} = \underline{W} \underline{U}^{-1}$

$$\begin{aligned} \underline{Q} \underline{Q}^T &= \underline{W} \underline{U}^{-1} \underbrace{\left(\underline{U}^{-1} \right)^T}_{\left(\underline{U}^T \right)^{-1}} \underline{W}^T = \underline{W} \left(\underline{U}^T \underline{U} \right)^{-1} \underline{W}^T \\ &= \underline{W} \underbrace{\left(\underline{U}^2 \right)^{-1}}_{\underline{S}^{-1}} \underline{W}^T \\ &= \underline{W} \left(\underline{W}^T \underline{W} \right)^{-1} \underline{W}^T \\ &= \underline{W} \underline{W}^{-1} \left(\underline{W}^T \right)^{-1} \underline{W}^T \\ &= \underline{1} \end{aligned}$$

3. The left decomposition proof is left as exercise.

4. Let $\det \underline{W} > 0 \Rightarrow \det \underline{W} \stackrel{(*)}{=} \det \underline{Q} \det \underline{U} > 0$. From the fact $\underline{U} = \sqrt{\underline{S}}$, $\underline{S}$ is symmetric positive definite

$$\P \underline{u} \cdot (\underline{F} \underline{v}) = \left(\underline{F}^T \underline{u} \right) \cdot \underline{v}$$

CHAPTER 3. INTRODUCTION TO CARTESIAN TENSOR

$$\therefore \det \underline{U} > 0 \Rightarrow \det \underline{Q} > 0$$

$$\Rightarrow \det \underline{Q} = +1, \text{ i.e. } \underline{Q} \text{ is proper orthogonal}$$

□

3.16. DECOMPOSITION THEOREM FOR SECOND ORDER TENSORS

Chapter 4

Tensor Calculus

Let $S \in E_3$ be a point set. A tensor field of order N on S is an N -th order tensor valued function of position defined on S . Before the next section, we need to review some concepts in vector calculus in appendix A.

4.1 Smoothness of tensor fields

Let R be a region in E_3 . Let $\underline{V}$ be a tensor field of order N on R . $\underline{V}$ is continuous on R and we will write $\underline{V} \in C(R)$ if the functions $\underbrace{V_{ij \dots k}^X}_N$ are continuous scalar functions on R

$\forall X \in \mathcal{F}$. We say that $\underline{V}$ is M times continuously differentiable on R (M is positive integer) and write $\underline{V} \in C^M(R)$ if $\underline{V} \in C(R)$

and the partial derivatives $\underbrace{\frac{\partial^M V_{ij \dots k}^X}{\partial x_p \partial x_q \dots \partial x_r}}_M = \underbrace{V_{ij \dots k}^X}_N \underbrace{pq \dots r}_M$

4.1. SMOOTHNESS OF TENSOR FIELDS

exist on $\mathring{R} \forall X \in \mathcal{F}$ and they coincide with continuous function.

Theorem 4.1.1. *Let R be a domain (open region) and $\mathcal{V} \in C^M(R)$ ($M \geq 1$) be a tensor field of order N . Define*

$$W_{ij\dots ks}^X(\mathfrak{x}) = \frac{\partial}{\partial x_s} \left\{ V_{ij\dots k}^X(\mathfrak{x}) \right\}, \forall \mathfrak{x} \in R, \forall X \in \mathcal{F}$$

where x_s are the coordinates in X of points in R .

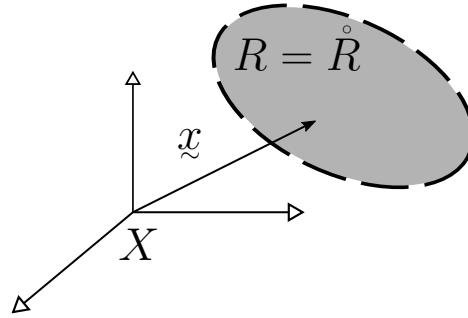


Figure 4.1: Scheme for R

Then $W_{ij\dots ks}^X$ are the 3^{N+1} components in X of a tensor field $\mathcal{W}$ of order $N + 1$ and $\mathcal{W} \in C^{M-1}(R)$. One calls $\mathcal{W}$ the gradient of $\mathcal{V}$ and writes

$$\mathcal{W} = \text{grad } \mathcal{V} = \nabla \mathcal{V}$$

CHAPTER 4. TENSOR CALCULUS

Also

$$W_{ij\dots ks}^X = V_{ij\dots k,s}^{X*}$$

Proof.

Consider $X = \{0; \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \in \mathcal{F}$, $X' = \{0; \mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\} \in \mathcal{F}$ and the matrix A of direction cosine that transforms X to X' ($A: X \rightarrow X'$).

$$\begin{cases} \mathfrak{x} = x_i \mathbf{e}_i = x'_m \mathbf{e}'_m \\ x'_m = A_{mi} x_i, x_i = A_{mi} x'_m \end{cases}$$

$$\begin{aligned} W_{pq\dots rs}^{X'}(\mathfrak{x}) &= \frac{\partial}{\partial x'_s} \left\{ V_{pq\dots r}^{X'}(\mathfrak{x}) \right\} = \frac{\partial}{\partial x'_s} \left\{ A_{pi} A_{qj} \cdots A_{rk} V_{ij\dots k}^X(\mathfrak{x}) \right\} \\ &= \frac{\partial}{\partial x_l} \left\{ A_{pi} A_{qj} \cdots A_{rk} V_{ij\dots k}^X(\mathfrak{x}) \right\} \frac{\partial x_l}{\partial x'_s} \end{aligned}$$

where

$$\frac{\partial x_l}{\partial x'_s} = \frac{\partial (A_{ml} x'_m)}{\partial x'_s} = A_{ml} \frac{\partial x'_m}{\partial x'_s} = A_{ml} \delta_{ms} = A_{sl}$$

$$\begin{aligned} \therefore W_{pq\dots rs}^{X'}(\mathfrak{x}) &= A_{pi} A_{qj} \cdots A_{rk} A_{sl} \frac{\partial}{\partial x_l} \left\{ V_{pq\dots r}^X(\mathfrak{x}) \right\} \\ &= A_{pi} A_{qj} \cdots A_{rk} A_{sl} W_{ij\dots kl}^X(\mathfrak{x}) \end{aligned}$$

satisfies the transformation law □

*partial derivative with respect to x_s

4.2. RELATION TO VECTOR CALCULUS

4.2 Relation to vector calculus

Let R be a domain. Let ϕ and $\underline{v}$ be suitably smooth scalar and vector field respectively.

Define

$$\begin{aligned}\underline{w} = \text{grad } \phi &\equiv \nabla \phi && \text{if } w_i = \phi_{,i} = \frac{\partial \phi}{\partial x_i} \\ \psi = \text{div } \underline{v} &\equiv \nabla \cdot \underline{v} && \text{if } \psi = v_{i,i} = \frac{\partial v_i}{\partial x_i} \\ \underline{w} = \text{curl } \underline{v} &\equiv \nabla \wedge \underline{v} && \text{if } w_i = \epsilon_{ijk} \frac{\partial}{\partial x_j} v_k = \epsilon_{ijk} v_{k,j} \\ \psi = \text{Lap } \phi &\equiv \nabla^2 \phi && \text{if } w_i = \phi_{,ii} = \frac{\partial^2 \phi}{\partial x_1^2} + \frac{\partial^2 \phi}{\partial x_2^2} + \frac{\partial^2 \phi}{\partial x_3^2} \\ \underline{w} = \text{Lap } \underline{v} &\equiv \nabla^2 \underline{v} && \text{if } w_i = v_{i,kk}\end{aligned}$$

Notation.

- $\nabla \phi \neq \phi_{,i}$ but $(\nabla \phi)_i = \phi_{,i}$ or $\nabla \phi = \phi_{,i} \underline{e}_i$
- $\nabla^2 \underline{v} = v_{i,kk} \underline{e}_i$

Example. $\underline{u} \in C^1(R)$ is a vector field. Show that

$$\underline{u} \wedge (\nabla \wedge \underline{u}) = \frac{1}{2} \nabla (\underline{u} \cdot \underline{u}) - (\underline{u} \cdot \nabla) \underline{u}$$

which is important relation for continuum mechanics.

CHAPTER 4. TENSOR CALCULUS

Proof.

$$\begin{aligned}
 \left[\underline{u} \wedge (\nabla \wedge \underline{u}) \right]_i &= \epsilon_{ijk} u_j (\nabla \wedge \underline{u})_k = \epsilon_{ijk} u_j \epsilon_{klm} u_{m,l} \\
 &= \epsilon_{ijk} \epsilon_{klm} u_j u_{m,l} = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) u_j u_{m,l} \\
 &= u_j u_{j,i} - u_j u_{i,j} = u_j \frac{\partial u_j}{\partial x_i} - u_j \frac{\partial u_i}{\partial x_j} \\
 &= \frac{1}{2} \frac{\partial (u_j u_j)}{\partial x_i} - u_j \frac{\partial u_i}{\partial x_j} \\
 &= \frac{1}{2} \left[\nabla (\underline{u} \cdot \underline{u}) \right]_i - \left[(\underline{u} \cdot \nabla) \underline{u} \right]_i \\
 \therefore \underline{u} \wedge (\nabla \wedge \underline{u}) &= \frac{1}{2} \nabla (\underline{u} \cdot \underline{u}) - (\underline{u} \cdot \nabla) \underline{u}
 \end{aligned}$$

□

Example. Show that

$$\nabla \wedge (\nabla \wedge \underline{u}) = \nabla (\nabla \cdot \underline{u}) - \nabla^2 \underline{u}$$

Proof.

$$\begin{aligned}
 \left[\nabla \wedge (\nabla \wedge \underline{u}) \right]_i &= \epsilon_{ijk} (\nabla \wedge \underline{u})_{k,j} = \epsilon_{ijk} (\epsilon_{klm} u_{m,l})_{,j} \\
 &= \epsilon_{kij} \epsilon_{klm} u_{m,lj} = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) u_{m,lj} \\
 &= u_{j,ij} - u_{i,jj} = \underbrace{\frac{\partial}{\partial x_i} (\nabla \cdot \underline{u})}_{\left[\nabla (\nabla \cdot \underline{u}) \right]_i} - (\nabla^2 \underline{u})_i
 \end{aligned}$$

$$\therefore \nabla \wedge (\nabla \wedge \underline{u}) = \nabla (\nabla \cdot \underline{u}) - \nabla^2 \underline{u}$$

4.3. DIVERGENCE THEOREM

□

Let R be a domain and $\mathcal{W}$ be a suitably smooth 2^{nd} order tensor. Define

$$\begin{aligned}
 \mathcal{T} = \text{grad } \mathcal{W} &\equiv \nabla \mathcal{W} && \text{if } T_{ijk} = W_{ij,k} \quad (\mathcal{T}: 3^{rd} \text{ order}) \\
 \mathcal{v} = \text{div } \mathcal{W} &\equiv \nabla \cdot \mathcal{W} && \text{if } v_i = W_{ij,j} \quad (\mathcal{v}: \text{vector}) \\
 \mathcal{U} = \text{curl } \mathcal{W} &\equiv \nabla \wedge \mathcal{W} && \text{if } U_{ij} = \epsilon_{ipq} \frac{\partial}{\partial x_p} W_{jq} \quad (\mathcal{U}: 2^{nd} \text{ order}) \\
 \mathcal{U} = \text{Lap } \mathcal{W} &\equiv \nabla^2 \mathcal{W} && \text{if } U_{ij} = W_{ij,kk} \quad (\mathcal{U}: 2^{nd} \text{ order})
 \end{aligned}$$

4.3 Divergence theorem

Let R be a bounded closed region. Let $\mathcal{v} \in C(R) \cap C^1(\mathring{R})$ be a vector field and suppose $\nabla \cdot \mathcal{v} \in C(R)$.

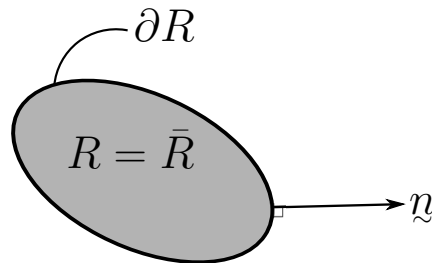


Figure 4.2: Scheme for R

Then

CHAPTER 4. TENSOR CALCULUS

$$\int_{\partial R} \underline{v} \cdot \underline{n} dA = \int_R \nabla \cdot \underline{v} dV$$

or

$$\int_{\partial R} v_i n_i dA = \int_R v_{i,i} dV \quad (*)$$

where $\underline{n}$ is the outer unit normal vector on ∂R .

Further, the indicial version $(*)$ holds if v_i are merely a triplet of real valued functions, rather than the components of a vector field.

$$\Rightarrow \int_{\partial R} (\bullet) n_i dA = \int_R (\bullet)_{,i} dV$$

Example.

- $\int_{\partial R} \phi n_i dA = \int_R \phi_{,i} dV$
- $\int_{\partial R} W_{ij} n_j dA = \int_R W_{ij,j} dV$

Theorem 4.3.1. *Let R be a domain ($R = \mathring{R}$), $\underline{v}$ be a vector field on R and $\underline{\hat{x}}$ be the position vector of a point in R . (R can be simply/multiply connected, bounded or unbounded)*

(a) *If $\phi \in C^{N+1}(R)$ ($N \geq 1$) is a scalar field such that*

$$\underline{v} = \nabla \phi \text{ on } R \quad (i)$$

then

4.3. DIVERGENCE THEOREM

$$\mathfrak{v} \in C^N(R), \nabla \wedge \mathfrak{v} = \mathfrak{Q} \text{ or } \epsilon_{ijk} v_{k,j} = 0 \text{ or } v_{i,j} = v_{j,i} \quad (\text{ii})$$

and the line integral

$$\begin{aligned} \int_{\mathring{x}}^{\mathfrak{x}} \mathfrak{v}(\mathfrak{z}) \cdot d\mathfrak{z} &= \int_{\mathring{x}}^{\mathfrak{x}} v_i(\mathfrak{z}) d\xi_i \\ &= \int_{\mathring{x}}^{\mathfrak{x}} \left[v_1(\mathfrak{z}) d\xi_1 + v_2(\mathfrak{z}) d\xi_2 + v_3(\mathfrak{z}) d\xi_3 \right] \end{aligned}$$

(iii)

is path independent $\forall x \in R$.

(b) Conversely, if R is a simply connected domain and $\mathfrak{v}$ is a vector field satisfying (ii), then the line integral is path independent, i.e. (iii) holds; and further there exists a scalar field $\phi \in C^{N+1}(R)$ unique up to a constant such that (i) holds and

$$\phi(\mathfrak{x}) - \phi(\mathring{x}) = \int_{\mathring{x}}^{\mathfrak{x}} \mathfrak{v}(\mathfrak{z}) \cdot d\mathfrak{z}$$

Theorem 4.3.2. Let R be a domain and $\mathbb{T}$ be a 2^{nd} order tensor field on R .

CHAPTER 4. TENSOR CALCULUS

(a) If $\mathfrak{v} \in C^{N+1}(R)$ ($N \geq 1$) is a vector field such that

$$\mathfrak{T} = \nabla \mathfrak{v} \text{ or } T_{ij} = v_{i,j} \text{ on } R \quad (\text{i})$$

then

$$\mathfrak{T} \in C^N(R), \nabla \wedge \mathfrak{T} = 0^\dagger \text{ on } R \quad (\text{ii})$$

(b) Conversely, if R is simply connected and if $\mathfrak{T}$ satisfies (ii), then exists a vector field $\mathfrak{v} \in C^{N+1}(R)$ such that (i) holds.

Further, if in addition to (ii)

$$\text{tr } \mathfrak{T} = 0 \text{ on } R \quad (\text{iii})$$

Then exists a 2^{nd} order tensor field $\mathfrak{W} \in C^{N+1}(R)$ such that

$$\mathfrak{W}^T = -\mathfrak{W} \text{ and } \mathfrak{T} = \nabla \wedge \mathfrak{W} \text{ on } R \quad (\text{iv})$$

or

$$T_{ij} = \epsilon_{ipq} W_{jq,p}$$

$^\dagger 2^{nd}$ order tensor

4.3. DIVERGENCE THEOREM

Proof.

(a) If $\underline{T} = \nabla \underline{v}$, then

$$\begin{aligned} (\nabla \wedge \underline{T})_{ij} &= (\nabla \wedge \nabla \underline{v})_{ij} = \epsilon_{ipq} (\bullet)_{jq,p} \\ &= \epsilon_{ipq} (v_{j,q})_{,p} = \epsilon_{ipq} v_{j,qp} \\ &= -\epsilon_{iqp} v_{j,qp} \stackrel{p \leftrightarrow q}{=} -\epsilon_{ipq} v_{j,pq} = 0 \end{aligned}$$

(b) First prove that

$$\nabla \wedge (\underline{T}^T \underline{a}) = (\nabla \wedge \underline{T}) \underline{a} \text{ if } \underline{a} = \text{constant vector} \quad (1)$$

$$\begin{aligned} \left[\nabla \wedge (\underline{T}^T \underline{a}) \right]_i &= \epsilon_{ijk} (\underline{T}^T \underline{a})_{k,j} = \epsilon_{ijk} (T_{km}^T a_m)_{,j} \\ &= \epsilon_{ijk} (T_{mk} a_m)_{,j} = \epsilon_{ijk} T_{mk,j} a_m \\ &= (\nabla \wedge \underline{T})_{im} a_m = \left[(\nabla \wedge \underline{T}) \underline{a} \right]_i \end{aligned}$$

Choose a coordinate frame $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ and let $\underline{t}_k$ (denote 3 vectors) defined as $\in \mathcal{F}$ follows

$$\underline{t}_k = \underline{T}^T \underline{e}_k \quad (k = 1, 2, 3)$$

Recalls that $\underline{T}$ is such that $\underline{T} \in C^N(R)$, $\nabla \wedge \underline{T} = \underline{0}$ on R

(ii)

CHAPTER 4. TENSOR CALCULUS

Consider now

$$\nabla \wedge \underline{t}_k = \nabla \wedge \left(\underline{T}^T \underline{e}_k \right) \stackrel{(1)}{=} (\nabla \wedge \underline{T}) \underline{e}_k \stackrel{(ii)}{=} \underline{0}$$

Thus

$$\nabla \wedge \underline{t}_k = \underline{0} \text{ on } R \text{ for } k = 1, 2, 3$$

So from the Theorem 4.3.1

$$\underline{t}_k = \underline{T}^T \underline{e}_k = \nabla \phi_k \text{ on } R, \phi_k \in C^{N+1}(R)$$

Thus

$$\left(\underline{T}^T \right)_{ij}^X (\underline{e}_k)_j = T_{ji}^X \delta_{kj} = \phi_{k,i} \Rightarrow T_{ki}^X = \phi_{k,i} \text{ on } R$$

Define a new vector field $\underline{v} = \phi_k \underline{e}_k$, then

$$T_{ki}^X = \phi_{k,i} = v_{k,i}$$

$$\Rightarrow \underline{T} = \nabla \underline{v} \text{ on } R \text{ and (i) holds}$$

For the 2^{nd} part, assume that $\underline{v}$ is the vector field established above, i.e. $\nabla \underline{v} = \underline{T}$.

Suppose $\underline{T}$ satisfies (ii) and (iii), i.e.

$$\nabla \wedge \underline{T} = \underline{0} \text{ and } \text{tr } \underline{T} = 0 \tag{2}$$

Then

$$\text{tr } \underline{T} = \text{tr } v_{i,j} = v_{i,i} = 0 \tag{3}$$

4.3. DIVERGENCE THEOREM

Define now a 2^{nd} order tensor field $\tilde{W} \in C^{N+1}(R)$ by

$$W_{jq} = \epsilon_{jqk} v_k \text{ on } R \quad (4)$$

Obviously

$$\tilde{W}^T = -\tilde{W}$$

and

$$\begin{aligned} (\nabla \wedge \tilde{W})_{ij} &= \epsilon_{ipq} W_{jq,p} \stackrel{(4)}{=} \epsilon_{ipq} \epsilon_{jqk} v_{k,p} \\ &= -\epsilon_{qip} \epsilon_{qjk} v_{k,p} = -(\delta_{ij} \delta_{pk} - \delta_{ik} \delta_{pj}) v_{k,p} \\ &= -\delta_{ij} \underbrace{v_{p,p}}_{=0 \text{ (by (iii))}} + v_{i,j} = v_{i,j} \\ &\Rightarrow \nabla \wedge \tilde{W} = \nabla v = \tilde{T} \end{aligned}$$

□

Chapter 5

Kinematics of a Continuum

5.1 Terminologies and definitions

- Body

A body occupies regions in E_3 and is composed of material points(or particles) to each of which the density of mass, momentum, and energy etc. are defined throughout the body in a continuously except, possibly, for a finite number of singular points, lines, surface, or volumes of discontinuously such as holes, cracks, voids, and cavities in the body which require special attention.

- Kinematics

Study of motion of bodies regardless the nature of force that cause the motion.

5.1. TERMINOLOGIES AND DEFINITIONS

- Deformations

Let a body occupies a region $R \in E_3$. Every particle can be located by its position vector $\underline{x}$ from some fixed point O . When the body is deformed, every particle takes up new position vector denoted by $\underline{y}$. (referred to the same origin O)

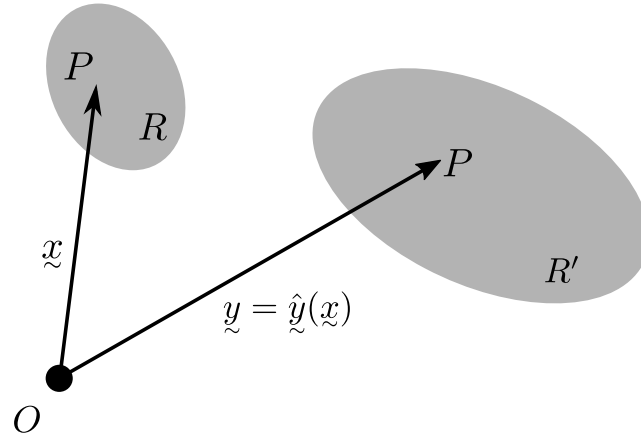


Figure 5.1: Scheme for R

Then the deformation of the body is known if the function

$$\underline{y} = \hat{y}(\underline{x}) \text{ or } y_i = \hat{y}_i(\underline{x}) \quad (1)$$

is known.

Note.

- (1) It is possible that the body can occupy the same in two different states in which particles occupy different positions.

CHAPTER 5. KINEMATICS OF A CONTINUUM

- e.g.a) A body occupies a sphere in E_3 . It is then rotated but occupies the same sphere. The two state should be distinguishable. (an example of *rigid deformation*)
- e.g.b) R is a circular cylinder. Twist a rubble rod occupying R and it is will still do so.

(2) In continuum mechanics, we demand the mapping (1) be

- (a) 1 to 1 (globally), i.e. $\underline{y} = \hat{y}(\underline{x})$, $\underline{x} = \hat{x}(\underline{y})$ and $\hat{y}^{-1}(\underline{x}) = \hat{x}(\underline{y}) = \underline{x} \quad \forall \underline{x} \in R$
- (b) $\hat{y} \in C^1(R)$ and $\hat{y}^{-1} \in C^1(R')$
- (c) $\det \nabla \hat{y} > 0$ on R

Property (a) guarantees that the body not penetrate itself, i.e. distinct particles occupy distinct positions in every state.

Property (b) assumes *no tearing*.

Property (c) represents two things:

- 1 to 1 correspondence of local directions.(curves that intersect each other at nonzero angle are not mapped on to the tangent ones)
- locally the volume of a part of the body(a subregion) per unit original volume is always grater than zero.

5.1. TERMINOLOGIES AND DEFINITIONS

Note that properties (b)+(c) does *not* imply that $\hat{y}$ is globally 1 to 1 on R . They only imply that $\hat{y}$ is locally 1 to 1, i.e. could have $\det \nabla \hat{y} > 0$ with $y = \hat{y}(\underline{x}) = \hat{y}(\underline{x}')$.

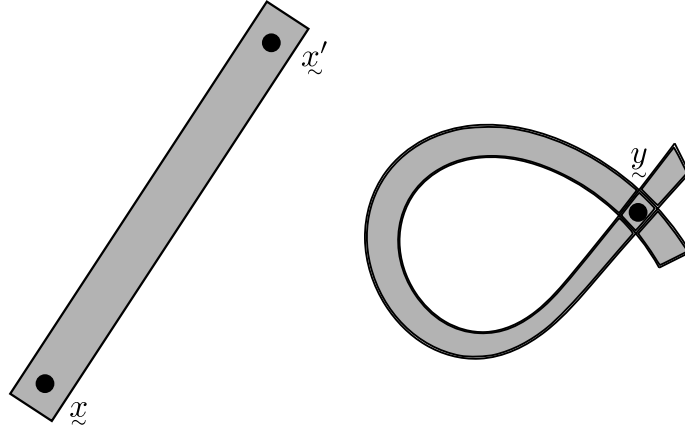


Figure 5.2: Scheme for properties (b)+(c)

- (3) $\underline{y} = \hat{y}(\underline{x})$, $\underline{x} = \hat{x}(\underline{y})$ and $\hat{x} = \hat{y}^{-1}: R' \rightarrow R$
Then

$$\hat{y}(\hat{x}(\underline{y})) = \underline{y} \quad \forall \underline{y} \in R'$$

By chain rule

$$\frac{\partial \hat{y}_i(\underline{x})}{\partial x_k} \frac{\partial \hat{x}_k}{\partial y_j} = \frac{\partial y_i}{\partial y_j} = \delta_{ij}$$

$$\Rightarrow \boxed{\nabla \hat{y}(\underline{x}) \nabla \hat{x}(\underline{y}) = \mathbb{1}} \Rightarrow \det \nabla \hat{y} \det \nabla \hat{x} = 1 \quad (\star)$$

or

$$\det \nabla \hat{x}(\underline{y}) = \frac{1}{\det \nabla \hat{y}(\underline{x})} \quad \forall \underline{x} \in R$$

CHAPTER 5. KINEMATICS OF A CONTINUUM

Definition 5.1.1. We say that $\hat{y}: R \rightarrow R'$ is an *admissible deformation* if

- (a) $\hat{y}$ is 1 to 1 on R
- (b) $\hat{y} \in C^2(R)$
- (c) $\det \nabla \hat{y} > 0$ on R

Note. By the *smooth inverse theorem*^{*}, then $\hat{x} = \hat{y}^{-1}$ on R' is also admissible, i.e. $\hat{x} \in C^2(R')$; while $\det \nabla \hat{x} > 0$ on R' by equation (★).

5.2 Motion of a body

For fixed choice of the region R (called *reference configuration*)

$$\hat{y}(\mathfrak{x}, x): R \rightarrow R_t, t \in [t_0, t_1] \equiv \tau$$

represent the motion of the body.

^{*}referred to advanced calculus

5.2. MOTION OF A BODY

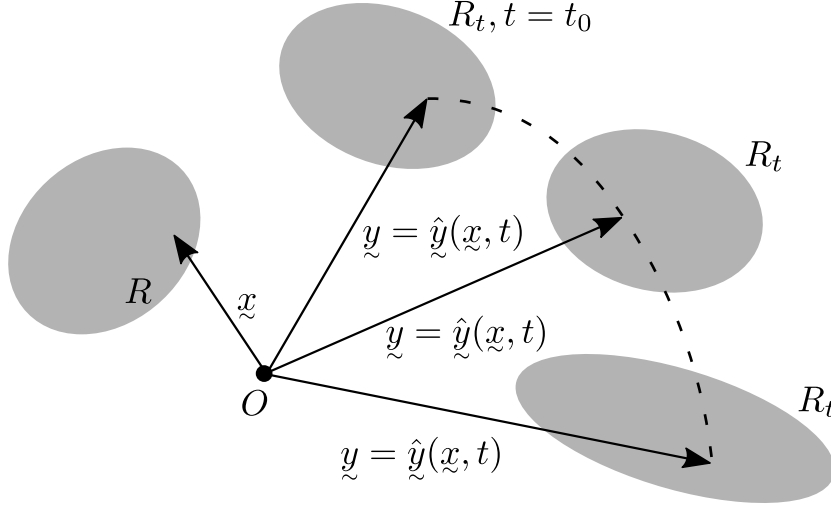


Figure 5.3: the motion of a body

Note. The reference configuration R is arbitrary. It needs not occupied by the body at any time during the motion. You may even choose *fictitious* one if it can help to simplify the formulation of the problem. The technique is often used in plastic mechanics.

A continuous motion is one such that

$$\underline{y} = \hat{y}(\bullet, t) \in C(\tau)$$

Definition 5.2.1. The Jacobian of the motion $\hat{y}(\bullet, \bullet)$ on $R \times \tau$ is

$$J(\underline{x}, t) = \det \nabla \hat{y}(\underline{x}, t) \quad \forall (\underline{x}, t) \in R \times \tau$$

If we choose R_{t_*} as reference configuration where $t_* \in [t_0, t_1]$ then

$$\hat{y}(\underline{x}, t_*) = \underline{x}$$

CHAPTER 5. KINEMATICS OF A CONTINUUM

$$\begin{aligned}\Rightarrow \nabla \hat{y}(\underline{x}, t_*) = \underline{1} &\Rightarrow J(\underline{x}, t_*) = 1 \\ J(\underline{x}, t) &> 0 \quad \forall (\underline{x}, t) \in R \times \tau\end{aligned}$$

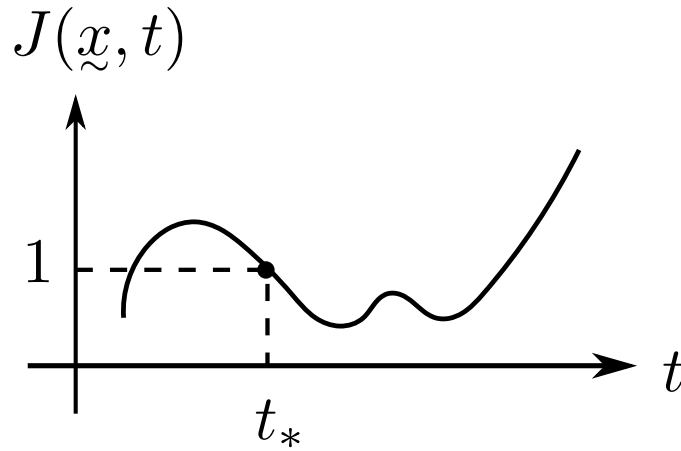


Figure 5.4: Jacobian versus time

Definition 5.2.2. A motion is admissible if $\hat{y} \in C^2(R \times \tau)$

Note.

- (a) Admissible motions are smooth one-parameter families of admissible deformations, $\hat{y}(\underline{x}, \bullet)$.
- (b) $\frac{\partial \hat{y}(\underline{x}, t)}{\partial t}$, $\frac{\partial^2 \hat{y}(\underline{x}, t)}{\partial t^2}$ exist and are continuous $\forall (\underline{x}, t) \in R \times \tau$.
- (c) There exists $y^{-1}(\bullet, t) = \underline{x}(\bullet, t)$ defined on $R_t \quad \forall t \in \tau$.

$$\hat{x}(\bullet, t): R_t \rightarrow R$$

5.2. MOTION OF A BODY

$$\hat{x}(\hat{y}(\underline{x}, t), t) = \underline{x}$$

$$\hat{y}(\hat{x}(\underline{y}, t), t) = \underline{y}$$

$$\hat{x}(\bullet, t) \in C^2(R_t) \quad \forall t \in \tau$$

$$\Rightarrow \frac{\partial \hat{x}}{\partial t}, \frac{\partial^2 \hat{x}}{\partial t^2} \text{ exist and are continuous on } m \triangleq \left\{ (\underline{y}, t), \underline{y} \in R_t, t \in \tau \right\}.$$

- (d) Let P be the particle of the body and $\underline{y} = \hat{y}(\underline{x}, t)$ is the position vector of P at time.

$x_i \quad \dots \quad$ Lagrangian coordinate of P

$y_i \quad \dots \quad$ Eulerian coordinate of P

For fixed $\underline{x}$, $\hat{y}(\underline{x}, t)$ is the path line of P .

- (e) Given a motion let Ψ be a field defined on $R \times \tau$, $\Psi = \Psi(\underline{x}, t), \underline{x} \in R, t \in \tau$, then one may define ψ on m by

$$\psi(\underline{y}, t) = \Psi(\hat{x}(\underline{y}, t), t)$$

where

$$\hat{x}(\bullet, t) = \underline{y}^{-1}(\bullet, t)$$

Equivalently, given $\psi(\underline{y}, t)$ on m one can define $\Psi(\underline{x}, t)$ such that

$$\Psi(\underline{x}, t) = \psi(\hat{y}(\underline{x}, t), t) \quad \forall (\underline{x}, t) \in R \times \tau$$

In this case we refer to $\Psi(\underline{x}, t)$ as a reference (or a material) field and $\psi(\underline{y}, t)$ as spatial field.

CHAPTER 5. KINEMATICS OF A CONTINUUM

Example. Consider the motion of a body

$$y_1 = x_1 + kt x_2$$

$$y_2 = x_2$$

$$y_3 = x_3$$

The Lagrangian coordinate $\mathfrak{x}$ gives the position of a particle at $t = 0$.

(a) Sketch the configuration of a unit cube at time t .

(b) If the temperature field is given by

$$\theta = y_1 + y_2$$

find the material field of temperature.

(c) Find the velocity the rate of change of temperature for particular material particles. Express both in material and spatial description.

Solution.

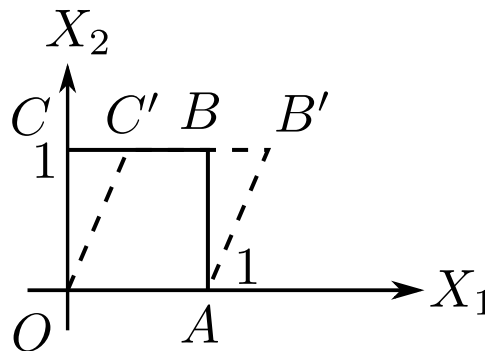


Figure 5.5: Scheme of example

5.2. MOTION OF A BODY

(a) By particle on OA

$$(x_1, x_2, x_3)_{OA} = (x_1, 0, x_3)$$

$$(y_1, y_2, y_3)_{OA} = (x_1, 0, x_3)$$

i.e. OA does not change.

For CB

$$(x_1, x_2, x_3)_{CB} = (x_1, 1, x_3)$$

$$(y_1, y_2, y_3)_{CB} = (x_1 + kt, 1, x_3)$$

i.e. CB move horizontally through a distance kt .

For OC

$$(x_1, x_2, x_3)_{OC} = (0, x_2, x_3)$$

$$(y_1, y_2, y_3)_{OC} = (ktx_2, x_2, x_3)$$

i.e. particle on OC move horizontally through a distance linearly proportional to its height.

Similarly for AB .

(b) $\theta = y_1 + y_2 = x_1 + ktx_2 + x_2 = x_1 + (kt + 1)x_2$

(c) Velocity components of the particle with Lagrangian position vector $\underline{x}$.

$$v_i = \left. \frac{\partial y_i(\underline{x}, t)}{\partial t} \right|_{\underline{x} \text{ fixed}}$$

$$\Rightarrow \begin{cases} v_1 = kx_2, v_2 = 0, v_3 = 0 \cdots \text{material field of velocity} \\ v_1 = ky_2, v_2 = 0, v_3 = 0 \cdots \text{spatial field of velocity} \end{cases}$$

Rate of change of temperature

CHAPTER 5. KINEMATICS OF A CONTINUUM

$$\begin{aligned}\left.\frac{\partial\theta}{\partial t}\right|_{\mathfrak{x} \text{ fixed}} &= kx_2 \cdots \text{material field of temperature} \\ &= ky_2 \cdots \text{spatial field of temperature}\end{aligned}$$

5.2. MOTION OF A BODY

Chapter 6

Study of Deformation

If R is the reference configuration of a body, let $\hat{\underline{y}}(\bullet, t): R \rightarrow R_t$. Choose any $t_d \in \tau$. Let $R_d \equiv R_{t_d} = \hat{\underline{y}}(R, t_d)$, so we can write

$$\hat{\underline{y}}(\bullet, t_d) = \hat{\underline{y}}(\bullet): R \rightarrow R_d$$

where $\hat{\underline{y}}(\bullet)$ is a deformation.

Define a *displacement field* $\underline{u}(\underline{x})$ such that

$$\hat{\underline{y}}(\underline{x}) = \underline{x} + \underline{u}(\underline{x})$$

6.1. DEFORMATION GRADIENT FIELD

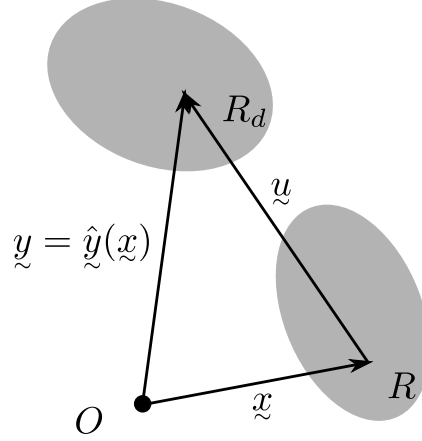


Figure 6.1: Scheme of deformation field

6.1 Deformation gradient field

$$\underline{F}(\underline{x}) = \nabla \hat{y}(\underline{x}) \in \mathcal{L}_+$$

where $\mathcal{L}_+$ is set of all 2^{nd} order tensor with positive determinant
or

$$F_{ij} = \frac{\partial y_i(\underline{x})}{\partial x_j} = y_{i,j} \text{ on } R$$

$$\det \underline{F}(\underline{x}) = \det \nabla \hat{y}(\underline{x}) = J(\underline{x}) > 0$$

$$\nabla \hat{x}(\underline{y}) = \underline{F}^{-1}(\hat{x}(\underline{y})) \quad \forall \underline{y} \in R_d$$

$$\because \hat{y}(\underline{x}) = \underline{x} + \underline{u}(\underline{x})$$

$$\Rightarrow \nabla \hat{y}(\underline{x}) = \underline{1} + \nabla \underline{u}(\underline{x})$$

$$\Rightarrow \underline{F}(\underline{x}) = \underline{1} + \nabla \underline{u}(\underline{x})$$

CHAPTER 6. STUDY OF DEFORMATION

or

$$F_{ij}(\underline{x}) = \delta_{ij} + u_{i,j}(\underline{x})$$

where $\nabla u(\underline{x})$ is called displacement gradient field.

Lemma 6.1.1. *Let $\hat{y}: R \rightarrow R_d$ be a deformation and let $\phi: R_d \rightarrow R$ be continuous. Then for any subregion (compact) $P_d \subset R_d$*

$$\int_{P_d} \phi(\underline{y}) dV_{\underline{y}} = \int_P \phi(\hat{y}(\underline{x})) J(\underline{x}) dV_{\underline{x}}$$

where

$$\begin{aligned} P &= \hat{y}^{-1}(P_d), P_d = \hat{y}(P) \\ J &= \det \nabla \hat{y} \text{ on } R \end{aligned}$$

Definition 6.1.1. $\hat{y}: R \rightarrow R_d$ is *locally volume preserving* (l.v.p.) or *isochoric* if for every compact $P \subset R$

$$\text{volume } P = \text{volume } P_d, P_d = \hat{y}(P)$$

Claim 6.1.1. *A deformation $\hat{y}: R \rightarrow R_d$ is isochoric if and only if $J(\underline{x}) = 1 \quad \forall \underline{x} \in R$*

Proof.

For any $P \subset R$ with $P_* = \hat{y}(P) \subset R_*$

$$\text{volume } (P) = \int_P dV_{\underline{x}}$$

6.1. DEFORMATION GRADIENT FIELD

$$\text{volume } (P_*) = \int_{P_*} dV_{\hat{y}} \stackrel{\substack{\text{Lemma} \\ (6.1.1)}}{=} \int_P J(\mathfrak{x}) dV_{\mathfrak{x}}$$

Hence $\hat{y}$ is l.v.p. if and only if

$$\text{volume } (P) - \text{volume } (P_*) = 0$$

or

$$\begin{aligned} \int_P [J(\mathfrak{x}) - 1] dV_{\mathfrak{x}} &= 0 \quad \forall P \subset R \\ \Rightarrow J(\mathfrak{x}) &= 1 \end{aligned}$$

Define an incompressible body as that possible induced deformation are l.v.p.

□

Definition 6.1.2. Rigid Deformation

An admissible deformation $\hat{y}: R \rightarrow R_*$ is rigid if

$$|\hat{y}(\mathfrak{x}) - \hat{y}(\mathfrak{z})| = |\mathfrak{x} - \mathfrak{z}| \quad \forall \mathfrak{x}, \mathfrak{z} \in R$$

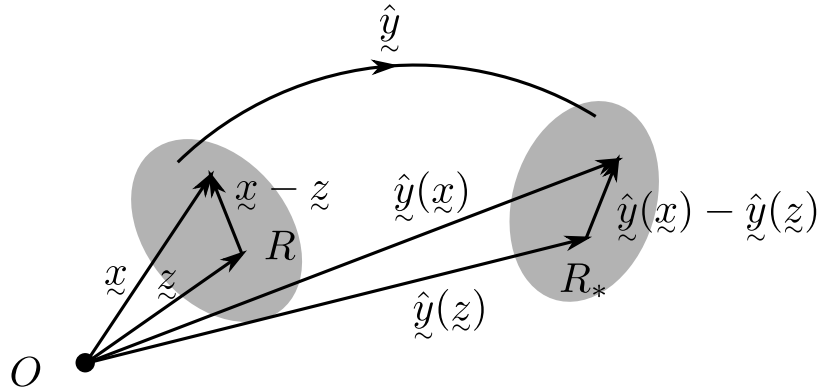


Figure 6.2: Scheme of rigid deformation

CHAPTER 6. STUDY OF DEFORMATION

Claim 6.1.2. $\hat{y}$ is rigid if and only if

$$\underline{y} = \hat{\underline{y}}(\underline{x}) = \underline{Q}\underline{x} + \underline{c} \quad (1)$$

where

$\underline{c}$ is a constant tensor

$\underline{Q}$ is a constant proper orthogonal 2^{nd} order tensor, i.e.

$$\underline{Q}\underline{Q}^T = \underline{1}, \det \underline{Q} = +1$$

Proof.

“ $\Leftarrow$ ” Assume that (1) holds. Then (1) is admissible since

- (1) is 1 to 1
- (1) is $C^\infty(R)$
- From (1)

$$\nabla \hat{\underline{y}} = \underline{F}(\underline{x}) = \underline{Q}$$

or

$$y_{i,j} = \frac{\partial}{\partial x_j} (Q_{ik}x_k + c_i) = Q_{ij}$$

$$\Rightarrow \det \underline{F} = \det \nabla \hat{\underline{y}} = \det \underline{Q} = +1 > 0$$

$$\left| \hat{\underline{y}}(\underline{x}) - \hat{\underline{y}}(\underline{z}) \right| \stackrel{(1)}{=} \left| \underline{Q}(\underline{x} - \underline{z}) \right| \stackrel{\downarrow}{=} |\underline{x} - \underline{z}|$$

$$\text{Recall } \underline{u} \cdot \underline{W}\underline{v} = \left(\underline{W}^T \underline{u} \right) \cdot \underline{v}$$

6.1. DEFORMATION GRADIENT FIELD

$$\begin{aligned}
 \therefore \left| \underset{\sim}{Q}(\underline{x} - \underline{z}) \right| &= \sqrt{\underbrace{\underset{\sim}{Q}(\underline{x} - \underline{z}) \cdot \underset{\sim}{Q}(\underline{x} - \underline{z})}_{\underset{\sim}{u}} \underbrace{\underset{\sim}{Q}(\underline{x} - \underline{z})}_{\underset{\sim}{W}} \underbrace{\underset{\sim}{Q}(\underline{x} - \underline{z})}_{\underset{\sim}{v}}} \\
 &= \sqrt{\underbrace{\underset{\sim}{Q}^T \underset{\sim}{Q}}_{\underset{\sim}{1}}} (\underline{x} - \underline{z}) \cdot (\underline{x} - \underline{z}) \\
 &= |\underline{x} - \underline{z}|
 \end{aligned}$$

“ $\Rightarrow$ ” Assume that $\hat{\underline{y}}$ is rigid then

$$[\hat{\underline{y}}_i(\underline{x}) - \hat{\underline{y}}_i(\underline{z})] [\hat{\underline{y}}_i(\underline{x}) - \hat{\underline{y}}_i(\underline{z})] - (x_i - z_i)(x_i - z_i) = 0$$

Take $\frac{\partial}{\partial x_j}$ of above

$$\begin{aligned}
 2F_{ij}(\underline{x}) [\hat{\underline{y}}_i(\underline{x}) - \hat{\underline{y}}_i(\underline{z})] - 2\delta_{ij}(x_i - z_i) &= 0 \\
 \Rightarrow \underset{\sim}{F}^T(\underline{x}) [\hat{\underline{y}}(\underline{x}) - \hat{\underline{y}}(\underline{z})] &= \underline{x} - \underline{z}
 \end{aligned}$$

or

$$\begin{aligned}
 2F_{ij}(\underline{x}) [\hat{\underline{y}}_i(\underline{x}) - \hat{\underline{y}}_i(\underline{z})] - 2\delta_{ij}(x_i - z_i) &= 0 \\
 \Rightarrow \underset{\sim}{F}^T(\underline{x}) [\hat{\underline{y}}(\underline{z}) - \hat{\underline{y}}(\underline{x})] &= \underline{z} - \underline{x}
 \end{aligned}$$

Take gradient with respect to $\underline{z}$

$$\underset{\sim}{F}^T(\underline{x}) \underset{\sim}{F}(\underline{z}) = \underset{\sim}{1} \quad \forall \underline{x}, \underline{z} \in R \quad (2)$$

Set $\underline{z} = \underline{x}$

CHAPTER 6. STUDY OF DEFORMATION

$$\underline{F}^T(\underline{x})\underline{F}(\underline{x}) = \underline{1} \Rightarrow \underline{F}(\underline{x}) \text{ is orthogonal}$$

Fixed $\underline{x}$ in (2), $\underline{F}(\underline{z})$ is constant tensor.

$\det \underline{F} > 0$ ($\because$ admissible) $\Rightarrow \underline{F}$ is constant and proper orthogonal or $\underline{F} = \underline{Q}$.

□

Note. If $\underline{v} = \underline{x} - \underline{z}$, $\underline{v}_* = \hat{\underline{y}}(\underline{x}) - \hat{\underline{y}}(\underline{z})$, then $\underline{v}_* = \underline{Q}\underline{v}$.

Remark.

- (a) A reflection is distance preserving but not admissible ($\because \det \underline{F} < 0$) and thus not rigid.
- (b) Rigid deformation is l.v.p. since $J = 1$.

$$J = \det \nabla \hat{\underline{y}} = \det \underline{F}(\underline{x}) = \det \underline{Q} = 1$$

Definition 6.1.3. Homogeneous Deformation

A deformation $\hat{\underline{y}}: R \rightarrow R_*$ is homogeneous if $\nabla \hat{\underline{y}} = \text{constant}$ tensor on R , i.e. there exists $\underline{F} = \text{constant} \in \mathcal{L}_+$ (set of all 2^{nd} order tensors with positive determinant, $\det \underline{F} > 0$) and $\underline{c} = \text{constant} \in E_3$ such that

$$\underline{y} = \hat{\underline{y}} = \underline{F}\underline{x} + \underline{c} \quad \forall \underline{x} \in R$$

Any general admissible deformation, $\underline{y} = \hat{\underline{y}}(\underline{x})$, is homogeneous locally.

Let $\underline{y} = \hat{\underline{y}}(\underline{x})$ be a general deformation (admissible)

CHAPTER 6. STUDY OF DEFORMATION

$$\underline{F} = \underline{F}(\underline{\dot{x}}) = \text{constant tensor}, \det \underline{F}(\underline{\dot{x}}) > 0$$

$$\underline{c} \stackrel{\Delta}{=} \hat{y}(\underline{\dot{x}}) - \underline{F}(\underline{\dot{x}})\underline{\dot{x}} = \text{constant}$$

Claim 6.1.3.

(a) *For a homogeneous deformation*

$$\mathcal{H}: y = \hat{y}(x) = \underline{F}x + \underline{c}$$

$\det \underline{F} > 0, x \in R$, the inverse $\hat{y}^{-1}$ is also a homogeneous deformation.

(b) *If further**

$$\mathcal{H}': \hat{y}'(x) = \underline{F}'x + \underline{c}', \det \underline{F}' > 0$$

$$\mathcal{H}'': \hat{y}''(x) = \underline{F}''x + \underline{c}'', \det \underline{F}'' > 0$$

Then $\mathcal{H} = \mathcal{H}'\mathcal{H}''$ defined as

$$\hat{y}(x) = \underline{F}''(\hat{y}'(x)) + \underline{c}''$$

is also homogeneous.

* $\mathcal{H}'$ and $\mathcal{H}''$ are not *prime*, they mean the different deformation but in same motion.

6.1. DEFORMATION GRADIENT FIELD

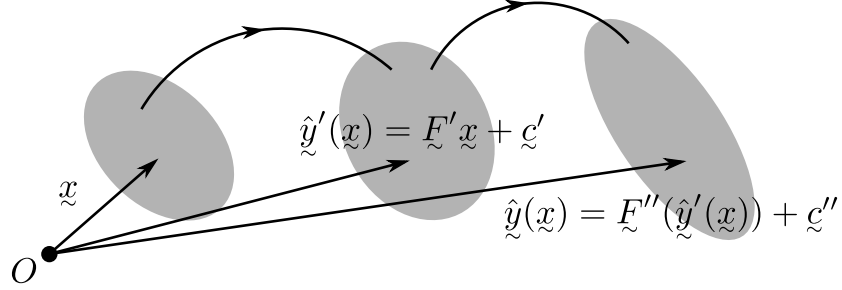


Figure 6.4: Scheme of homogeneous deformation continuously

Proof.

(a)

$$\begin{aligned}\hat{y} &= F \underline{x} + \underline{c} \\ \Rightarrow F^{-1} \hat{y} &= \underline{x} + F^{-1} \underline{c} \\ \therefore \underline{x} = \hat{x}(y) &= F^* y + \underline{c}^*\end{aligned}$$

where

$$\begin{aligned}F^* &= F^{-1} = \text{constant and } \det F^* = \det F^{-1} > 0^\dagger \\ \underline{c}^* &= -F^{-1} \underline{c} = \text{constant}\end{aligned}$$

(b)

$$\hat{y}(\underline{x}) = F'' (F' \underline{x} + \underline{c}') + \underline{c}'' = F \underline{x} + \underline{c}$$

where

[†]: $\det F > 0$

CHAPTER 6. STUDY OF DEFORMATION

$$\begin{aligned} \underline{F} &= \underline{F}'' \underline{F}' = \text{constant}, \det \underline{F} = \det \underline{F}'' \det \underline{F}' > 0 \\ \underline{c} &= \underline{F}'' \underline{c}' + \underline{c}'' = \text{constant} \end{aligned}$$

□

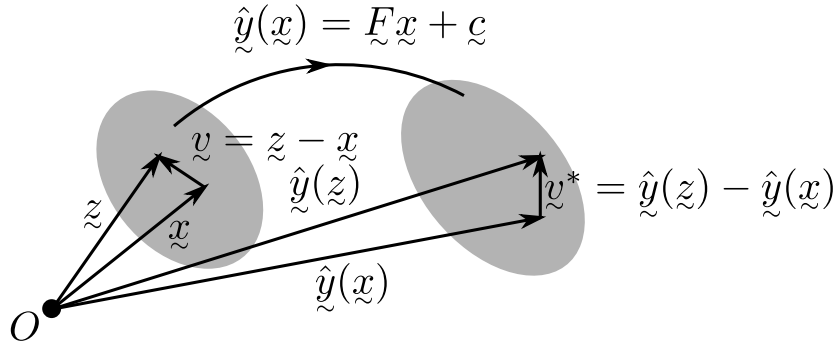


Figure 6.5: Scheme of homogeneous deformation

$$\begin{aligned} v &= z - x \\ v^* &= \hat{y}(z) - \hat{y}(x) \\ &= (\underline{F}z + \underline{c}) - (\underline{F}x + \underline{c}) \\ &= \underline{F} (z - x) = \underline{F}v \end{aligned}$$

$$\boxed{v^* = \underline{F}v \text{ or } v_i^* = F_{ij}v_j} \quad (1)$$

For a rigid deformation, $\underline{F} = \underline{Q}$, $v_i^* = Q_{ij}v_j$, i.e. pure rotation.

Theorem 6.1.1.

Let v, w be vectors in R , v^, w^* be their images under homogeneous deformation*

6.1. DEFORMATION GRADIENT FIELD

$$\mathcal{H}: \hat{y}(\mathfrak{x}) = \mathfrak{F}\mathfrak{x} + \mathfrak{c}$$

Then

$$\mathfrak{v}^* \cdot \mathfrak{w}^* - \mathfrak{v} \cdot \mathfrak{w} = 2\mathfrak{v} \cdot (\mathfrak{E}\mathfrak{w}) = 2v_i E_{ij} w_j = 2E_{ij} v_i w_j \quad (2)$$

where

$$\mathfrak{E} = \frac{1}{2} \left\{ \mathfrak{F}^T \mathfrak{F} - \mathfrak{1} \right\} \cdots \text{Lagrangian (Green) strain tensor} \quad (3)$$

or

$$\begin{aligned} E_{ij} &= \frac{1}{2} \left\{ F_{ik}^T F_{kj} - \delta_{ij} \right\} = \frac{1}{2} \left\{ F_{ki} F_{kj} - \delta_{ij} \right\} \\ &= \frac{1}{2} \left\{ \frac{\partial y_k}{\partial x_i} \frac{\partial y_k}{\partial x_j} - \delta_{ij} \right\} \end{aligned}$$

$\cdots$ Components of Lagrangian (Green) strain tensor

Proof.

$$\begin{aligned} \mathfrak{v}^* \cdot \mathfrak{w}^* - \mathfrak{v} \cdot \mathfrak{w} &\stackrel{(1)}{=} \mathfrak{F}\mathfrak{v} \cdot \mathfrak{F}\mathfrak{w} - \mathfrak{v} \cdot \mathfrak{w} \\ &= {}^\sharp \mathfrak{F}^T \mathfrak{F}\mathfrak{w} \cdot \mathfrak{v} - \mathfrak{v} \cdot \mathfrak{w} \\ &= \mathfrak{v} \cdot \mathfrak{F}^T \mathfrak{F}\mathfrak{w} - \mathfrak{v} \cdot \mathfrak{w} \\ &= \mathfrak{v} \cdot \left(\mathfrak{F}^T \mathfrak{F} - \mathfrak{1} \right) \mathfrak{w} \\ &\stackrel{(2)}{=} 2\mathfrak{v} \cdot \mathfrak{E}\mathfrak{w} \end{aligned}$$

CHAPTER 6. STUDY OF DEFORMATION

□

When $\underline{F} = \underline{Q}$ (rigid deformation) $\Rightarrow \underline{E} = \underline{0}$

Note.

$$\begin{aligned}
 \underline{v}^* \cdot \underline{w}^* - \underline{v} \cdot \underline{w} &\stackrel{(1)}{=} \underline{v}^* \cdot \underline{w}^* - \underline{F}^{-1} \underline{v}^* \cdot \underline{F}^{-1} \underline{w}^* \\
 &= \underline{v}^* \cdot \underline{w}^* - \underline{v}^* \cdot \underline{F}^{-T} \underline{F}^{-1} \underline{w}^* \\
 &= \underline{v}^* \cdot \left(\underline{1} - \underline{F}^{-T} \underline{F}^{-1} \right) \underline{w} \\
 &\triangleq 2 \underline{v}^* \cdot \underline{M} \underline{w}^*
 \end{aligned}$$

where

$$\underline{M} = \frac{1}{2} \left\{ \underline{1} - \underline{F}^{-T} \underline{F}^{-1} \right\} \dots \text{Eulerian (Almansi) strain tensor}$$

or

$$\begin{aligned}
 M_{ij} &= \frac{1}{2} \left\{ \delta_{ij} - F_{ik}^{-T} F_{kj}^{-1} \right\} \stackrel{Hwk}{=} \frac{1}{2} \left\{ \delta_{ij} - F_{ki}^{-1} F_{kj}^{-1} \right\} \\
 &= \frac{1}{2} \left\{ \delta_{ij} - \frac{\partial x_k}{\partial y_i} \frac{\partial x_k}{\partial y_j} \right\}
 \end{aligned}$$

$\dots$ Components of Eulerian (Almansi) strain tensor

- Both $\underline{E}$ and $\underline{M}$ are symmetric.
- For rigid deformation, $\underline{F} = \underline{Q} \Rightarrow \underline{E} = \underline{M} = \underline{0}$

${}^\dagger \underline{F} \underline{w} \cdot \underline{F} \underline{v} = \underline{F}^T \underline{F} \underline{w} \cdot \underline{v}$

6.2. STRAIN TENSOR IN TERMS OF DISPLACEMENT FIELD

6.2 Strain tensor in terms of displacement field

Recall that

$$\begin{aligned}\hat{\underline{y}}(\underline{x}) &= \underline{x} + \underline{u}(\underline{x}) \Rightarrow \underline{F} = \nabla_{\underline{x}} \hat{\underline{y}}(\underline{x}) = \underline{1} + \nabla_{\underline{x}} \underline{u}(\underline{x}) \\ \hat{\underline{x}}(\underline{y}) &= \underline{y} - \underline{u}(\underline{y}) \Rightarrow \underline{F}^{-1} = \nabla_{\underline{y}} \hat{\underline{x}}(\underline{y}) = \underline{1} - \nabla_{\underline{y}} \underline{u}(\underline{y})\end{aligned}$$

$$\begin{aligned}\underline{E} &= \frac{1}{2} \left\{ \underline{F}^T \underline{F} - \underline{1} \right\} \\ &= \frac{1}{2} \left\{ \left[\underline{1} + \nabla_{\underline{x}} \underline{u}(\underline{x}) \right]^T \left[\underline{1} + \nabla_{\underline{x}} \underline{u}(\underline{x}) \right] - \underline{1} \right\} \\ &= \frac{1}{2} \left\{ \nabla_{\underline{x}} \underline{u} + \left(\nabla_{\underline{x}} \underline{u} \right)^T + \left(\nabla_{\underline{x}} \underline{u} \right)^T \left(\nabla_{\underline{x}} \underline{u} \right) \right\}\end{aligned}$$

or

$$E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} + \frac{\partial u_k}{\partial x_i} \frac{\partial u_k}{\partial x_j} \right)$$

$$\begin{aligned}\underline{M} &= \frac{1}{2} \left\{ \underline{1} - \underline{F}^{-T} \underline{F}^{-1} \right\} \\ &= \frac{1}{2} \left\{ \underline{1} - \left[\underline{1} - \nabla_{\underline{y}} \underline{u}(\underline{y}) \right]^T \left[\underline{1} - \nabla_{\underline{y}} \underline{u}(\underline{y}) \right] \right\} \\ &= \frac{1}{2} \left\{ \nabla_{\underline{y}} \underline{u} + \left(\nabla_{\underline{y}} \underline{u} \right)^T - \left(\nabla_{\underline{y}} \underline{u} \right)^T \left(\nabla_{\underline{y}} \underline{u} \right) \right\}\end{aligned}$$

or

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$$M_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial y_j} + \frac{\partial u_j}{\partial y_i} - \frac{\partial u_k}{\partial y_i} \frac{\partial u_k}{\partial y_j} \right)$$

From polar decomposition theorem

$$\begin{aligned} \underline{F} &= \underline{Q} \underline{U} \cdots \text{right decomposition} \\ &= \underline{V} \underline{Q} \cdots \text{left decomposition} \end{aligned}$$

with

$$\underline{U} = \sqrt{\underline{F}^T \underline{F}}, \underline{V} = \sqrt{\underline{F} \underline{F}^T}, \underline{Q} = \underline{F} \underline{U}^{-1} = \underline{V}^{-1} \underline{F}$$

which is proper orthogonal, $\because \det \underline{F} > 0$ for admissible deformation.

Definition 6.2.1.

$\underline{C} \triangleq \underline{F}^T \underline{F}$ called right deformation tensor

$\underline{G} \triangleq \underline{F} \underline{F}^T$ called left deformation tensor

Then

$$\underline{C} = \underline{U}^2, \underline{G} = \underline{V}^2, \underline{E} = \frac{1}{2} \{ \underline{C} - \underline{1} \}$$

According to the polar decomposition theorem, $\underline{C}$ and $\underline{G}$ are both symmetric and positive definite and have the same eigenvalues but different eigenvectors, i.e. different principle frame.

Proof.

Let $\underline{X} = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ be the principal frame for $\underline{G}$ and let $\lambda_i > 0$ be the three eigenvalues

6.3. GEOMETRIC INTERPRETATION OF $\underline{E}$ OR E_{ij}^X

$$\underline{G}\underline{e}_i = \lambda_i \underline{e}_i \text{ (no sum)}$$

Let $X' = \{0; \underline{e}'_1, \underline{e}'_2, \underline{e}'_3\} \in \mathcal{F}$ with $\underline{e}'_i = \underline{Q}^T \underline{e}_i$ such that

$$\underline{C} = \underline{Q}^T \underline{G} \underline{Q}$$

then

$$\underline{C}\underline{e}'_i = \underline{Q}^T \underline{G} \underline{Q} \underline{Q}^T \underline{e}_i = \underline{Q}^T \underline{G} \underline{e}_i = \lambda_i \underline{Q}^T \underline{e}_i = \lambda_i \underline{e}'_i$$

$$\Rightarrow \underline{C}\underline{e}'_i = \lambda_i \underline{e}'_i$$

□

From (2) and (3)

$$\begin{aligned} \underline{v}^* \cdot \underline{w}^* &= \underline{v} \cdot \underline{w} + 2\underline{v} \cdot \underline{E}\underline{w} \\ &= \underline{v} \cdot \underline{w} + \underline{v} \cdot \underline{F}^T \underline{F}\underline{w} - \underline{v} \cdot \underline{w} \end{aligned}$$

$$\Rightarrow \boxed{\underline{v}^* \cdot \underline{w}^* = \underline{v} \cdot \underline{C}\underline{w}} \quad (4)$$

For rigid deformation, $\underline{C} = \underline{G} = \underline{1}$

6.3 Geometric interpretation of $\underline{E}$ or E_{ij}^X

Define

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$$\lambda(\underline{v}) \triangleq \frac{|\underline{v}^*|}{|\underline{v}|} \dots \text{stretch ratio of } \underline{v}$$

$$\epsilon(\underline{v}) \triangleq \frac{|\underline{v}^*| - |\underline{v}|}{|\underline{v}|} \dots \text{extension ratio of } \underline{v}$$

$$\Rightarrow \epsilon(\underline{v}) = \lambda(\underline{v}) - 1$$

If the deformation is homogeneous, then

$$\lambda(\alpha \underline{v}) = \lambda(\underline{v}), \epsilon(\alpha \underline{v}) = \epsilon(\underline{v}), \alpha \in \mathbb{R}, \alpha \neq 0$$

Set $\underline{w} = \underline{v}$ in (2)

$$\begin{aligned} \underline{v}^* \cdot \underline{v}^* - \underline{v} \cdot \underline{v} &= 2\underline{v} \cdot \underline{E}\underline{v} \\ \Rightarrow |\underline{v}^*|^2 - |\underline{v}|^2 &= 2\underline{v} \cdot \underline{E}\underline{v} \\ \Rightarrow \frac{|\underline{v}^*|^2 - |\underline{v}|^2}{|\underline{v}|^2} &= \frac{2\underline{v} \cdot \underline{E}\underline{v}}{|\underline{v}|^2} \\ \Rightarrow \lambda^2(\underline{v}) - 1 &= \frac{2\underline{v} \cdot \underline{E}\underline{v}}{|\underline{v}|^2} \end{aligned}$$

$$\Rightarrow \lambda(\underline{v}) = \sqrt{1 + \frac{2\underline{v} \cdot \underline{E}\underline{v}}{|\underline{v}|^2}} \quad \forall \underline{v} \neq 0$$

Let $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$

6.3. GEOMETRIC INTERPRETATION OF $\underline{E}$ OR E_{ij}^X

$$\begin{aligned}
 \epsilon(\underline{e}_i) &= \lambda(\underline{e}_i) - 1 \\
 &= \sqrt{1 + \frac{2\underline{e}_i \cdot \underline{E}\underline{e}_i}{1}} - 1 \quad (\text{no sum}) \\
 &= \sqrt{1 + 2E_{ii}} - 1 \quad (\text{no sum}) \\
 E_{ii}^X &= \frac{\lambda^2(\underline{e}_i) - 1}{2} \quad (\text{no sum})
 \end{aligned}$$

So given a frame $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$, the diagonal elements $E_{11}^X, E_{22}^X, E_{33}^X$ of the Lagrangian strain tensor in the frame are associated with the length changes of the unit vector of X .

Interpretation of $E_{ij}^X, i \neq j$

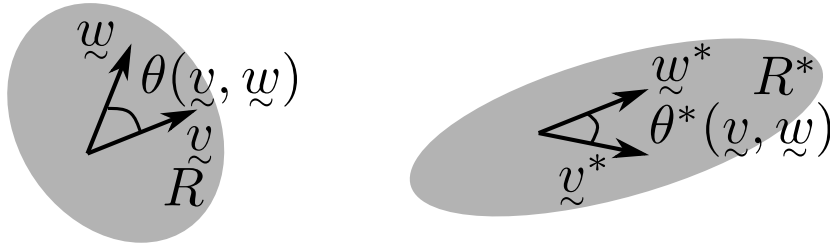


Figure 6.6: Scheme of change of theta

$$\begin{aligned}
 \underline{v}^* \cdot \underline{w}^* &= |\underline{v}^*| |\underline{w}^*| \cos(\theta^*(\underline{v}, \underline{w})) \\
 &\stackrel{(2)}{=} \underline{v} \cdot \underline{w} + 2\underline{v} \cdot \underline{E}\underline{w}
 \end{aligned}$$

Also

$$\begin{aligned}
 |\underline{v}^*| &= |\underline{v}| \lambda(\underline{v}) = |\underline{v}| [1 + \epsilon(\underline{v})] \\
 |\underline{w}^*| &= |\underline{w}| \lambda(\underline{w}) = |\underline{w}| [1 + \epsilon(\underline{w})]
 \end{aligned}$$

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$$\begin{aligned}
 \therefore \cos(\theta^*(\underline{v}, \underline{w})) &= \frac{\underline{v} \cdot \underline{w} + 2\underline{v} \cdot \underline{E}\underline{w}}{|\underline{v}^*| |\underline{w}^*|} \\
 &= \frac{\underline{v} \cdot \underline{w} + 2\underline{v} \cdot \underline{E}\underline{w}}{|\underline{v}| |\underline{w}| [1 + \epsilon(\underline{v})] [1 + \epsilon(\underline{w})]} \quad (*)
 \end{aligned}$$

Take $\underline{v} = \underline{e}_i$, $\underline{w} = \underline{e}_j$, $i \neq j$

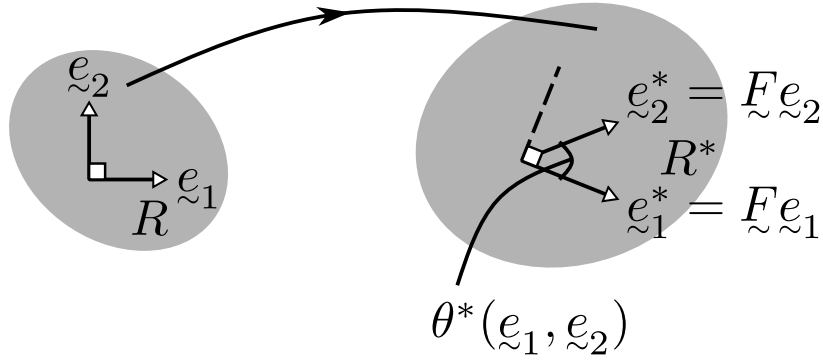


Figure 6.7: Deformation under principal frame

$$\therefore \cos(\theta^*(\underline{e}_i, \underline{e}_j)) = \frac{2\underline{e}_i \cdot \underline{E}\underline{e}_j}{\sqrt{1 + 2E_{ii}} \sqrt{1 + 2E_{jj}}}, i \neq j \text{ (no sum)}$$

Define

$$\mu_{ij} \triangleq \frac{\pi}{2} - \theta^*(\underline{e}_i, \underline{e}_j)$$

Then

$$\sin \mu_{ij} = \frac{2E_{ij}^X}{\sqrt{1 + 2E_{ii}} \sqrt{1 + 2E_{jj}}}, i \neq j \text{ (no sum)}$$

6.3. GEOMETRIC INTERPRETATION OF $\underline{E}$ OR E_{ij}^X

$\therefore$ If $E_{ij}^X = 0, \forall i \neq j$

$$\mu_{ij} = 0 \text{ or } \theta^*(\underline{e}_i, \underline{e}_j) = \frac{\pi}{2}$$

i.e.

$$\underline{e}_i^* \perp \underline{e}_j^*$$

Let $X \in \mathcal{F}$ be principal frame for $\underline{C}$ and therefore principal frame for $\underline{E}$. ($\because \underline{C} = 2\underline{E} + \underline{1}$) Let λ_i^2 be eigenvalue for $\underline{C}$. Then

$$[C_{ij}^X] = \begin{bmatrix} \lambda_1^2 & 0 & 0 \\ 0 & \lambda_2^2 & 0 \\ 0 & 0 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 + 2E_1 & 0 & 0 \\ 0 & 1 + 2E_2 & 0 \\ 0 & 0 & 1 + 2E_3 \end{bmatrix} \quad (*)$$

where E_i are eigenvalue of $\underline{E}$

What's the physical meaning of λ_i ?

From (*)

$$\underbrace{\lambda_i}_{\text{mathematical stuff}} = \sqrt{1 + 2E_i} \stackrel{X: \text{principal frame}}{=} \sqrt{1 + 2E_{ii}^X}$$

$$= \underbrace{\lambda(\underline{e}_i^X)}_{\text{physical meaning}}$$

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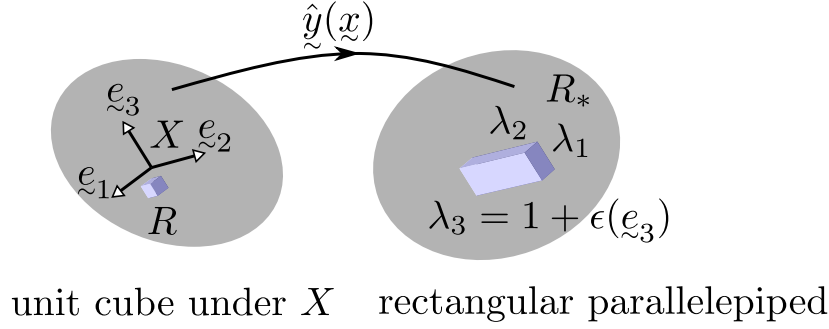


Figure 6.8: Deformation under principal frame

Note. The cube and the parallelepiped may have different orientation.

6.4 Volume change

$$V(R_*) = \int_{R_*} dV_{\tilde{y}} = \int_R J dV_{\tilde{x}} \stackrel[\text{deformation}]{\text{homogeneous}} J \int_R dV_{\tilde{x}} = J dV(R)$$

$$J = \frac{V(R_*)}{V(R)}$$

From previous schematic Figure. 6.8, $J = \lambda_1 \lambda_2 \lambda_3$
 Define *volume dilatation* as

$$\begin{aligned} \Theta &= \frac{V(R_*) - V(R)}{V(R)} \\ &= \lambda_1 \lambda_2 \lambda_3 - 1 = \sqrt{(1 + 2E_1)(1 + 2E_2)(1 + 2E_3)} - 1 \\ &= {}^{\S} \sqrt{1 + 2I_1(\underline{E}) + 4I_2(\underline{E}) + 8I_3(\underline{E})} - 1 \end{aligned}$$

6.4. VOLUME CHANGE

Definition 6.4.1. $\mathcal{H}$ is a *pure* homogeneous deformation if it is of the form

$$\mathcal{H}: y = \hat{y}(\mathfrak{x}) = \tilde{F}\mathfrak{x}, \tilde{F} = \tilde{F}^T \text{ and } \tilde{F} \text{ is positive definite}$$

Properties of the pure homogeneous definition

In general

$$\left. \begin{array}{l} \tilde{E} = \frac{1}{2} \left(\tilde{F}^T \tilde{F} - \mathbb{1} \right) \\ \tilde{C} = \tilde{F}^T \tilde{F} \\ \tilde{G} = \tilde{F} \tilde{F}^T \end{array} \right\} \Rightarrow \left. \begin{array}{l} \tilde{E} = \frac{1}{2} \left(\tilde{F}^2 - \mathbb{1} \right) \\ \tilde{C} = \tilde{F}^2 \\ \tilde{G} = \tilde{F}^2 \end{array} \right\} \Rightarrow \tilde{C} = \tilde{G}$$

i.e.

$\tilde{C}, \tilde{G}, \tilde{E}, \tilde{U}, \tilde{V}$ have same principal axes

$\tilde{C}, \tilde{G} \cdots$ same principal values

$\tilde{U}, \tilde{V} \cdots$ same principal values

Consider $X \in \mathcal{F}$ (principal frame for $\tilde{C}, \tilde{G}, \tilde{E}$)

$$[C_{ij}^X] = [G_{ij}^X] = \begin{bmatrix} \lambda_1^2 & 0 & 0 \\ 0 & \lambda_2^2 & 0 \\ 0 & 0 & \lambda_3^2 \end{bmatrix}, [F_{ij}^X] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$\begin{aligned} \mathcal{H}: y = \tilde{F}\mathfrak{x} &\Rightarrow y_i^X = F_{ij}^X x_j^X \\ &\Rightarrow y_i^X = \lambda_i x_i^X \text{ (no sum)} \end{aligned}$$

^s $I_1(\tilde{E}) = E_1 + E_2 + E_3, I_2(\tilde{E}) = E_1 E_2 + E_2 E_3 + E_1 E_3, I_3(\tilde{E}) = E_1 E_2 E_3$

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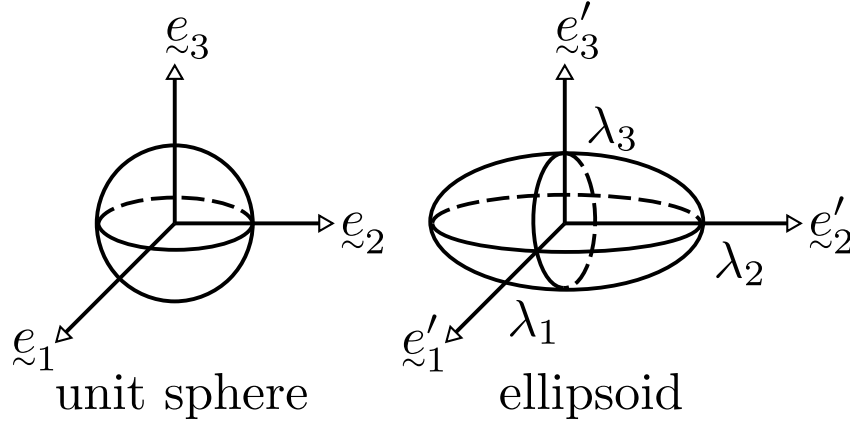


Figure 6.9: Scheme of pure homogeneous deformation

Note. The cube and the rectangular parallelepiped have the *same* orientation.

Theorem 6.4.1. *Resolutions of Arbitrary Homogeneous Deformation*

Let $\underline{y} = \hat{\underline{y}}(\underline{x}) = \underline{F}\underline{x} + \underline{c}$, $\det \underline{F} > 0$

Recall $\underline{F} = \underline{Q}\underline{U} = \underline{V}\underline{Q}$

where $\underline{U}, \underline{V}$ are symmetric positive definite, $\underline{Q}$ is orthogonal.

$\mathcal{H}'_1: \quad \underline{x} \rightarrow \underline{y}': \quad \underline{y}' = \underline{U}\underline{x} \quad (\text{pure homogeneous deformation})$

$\mathcal{H}''_1: \quad \underline{y}' \rightarrow \underline{y}'': \quad \underline{y}'' = \underline{Q}\underline{y}' \quad (\text{rigid deformation about } O)$
(rotation)

$\mathcal{H}'''_1: \quad \underline{y}'' \rightarrow \underline{y}: \quad \underline{y} = \underline{y}'' + \underline{c} \quad (\text{translation})$

Then $\mathcal{H}: \underline{x} \rightarrow \underline{y}: \underline{y} = \underline{F}\underline{x} + \underline{c}$ is given by

6.4. VOLUME CHANGE

$$\mathcal{H} = \mathcal{H}_1''' \mathcal{H}_1'' \mathcal{H}_1' \quad (*)$$

Also suppose

$$\mathcal{H}_2': \quad \underline{x} \rightarrow \underline{y}': \quad \underline{y}' = Q\underline{x} \quad (\text{rigid deformation about } O) \\ (\text{rotation})$$

$$\mathcal{H}_2'': \quad \underline{y}' \rightarrow \underline{y}'': \quad \underline{y}'' = V\underline{y}' \quad (\text{pure homogeneous} \\ \text{deformation})$$

$$\mathcal{H}_2''': \quad \underline{y}'' \rightarrow \underline{y}: \quad \underline{y} = \underline{y}'' + \underline{c} \quad (\text{translation})$$

Then $\mathcal{H}: \underline{x} \rightarrow \underline{y}: \underline{y} = F\underline{x} + \underline{c}$ is given by

$$\mathcal{H} = \mathcal{H}_2''' \mathcal{H}_2'' \mathcal{H}_2' \quad (**)$$

Proof.

$$\begin{aligned} \underline{y}' &= U\underline{x}, \underline{y}'' = Q\underline{y}' \\ \underline{y} &= \underline{y}'' + \underline{c} \\ &= QU\underline{x} + \underline{c} \\ &= F\underline{x} + \underline{c} \Rightarrow (*) \text{ holds} \end{aligned}$$

similar for (**). □

2-D illustration of the two Equivalent Resolution of $\mathcal{H}$

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Let $X \in \mathcal{F}$ be principal frame for $\mathcal{U}, \mathcal{C}, \mathcal{Q}$ and $X' \in \mathcal{F}$ be principal frame for $\mathcal{V}, \mathcal{G}$.

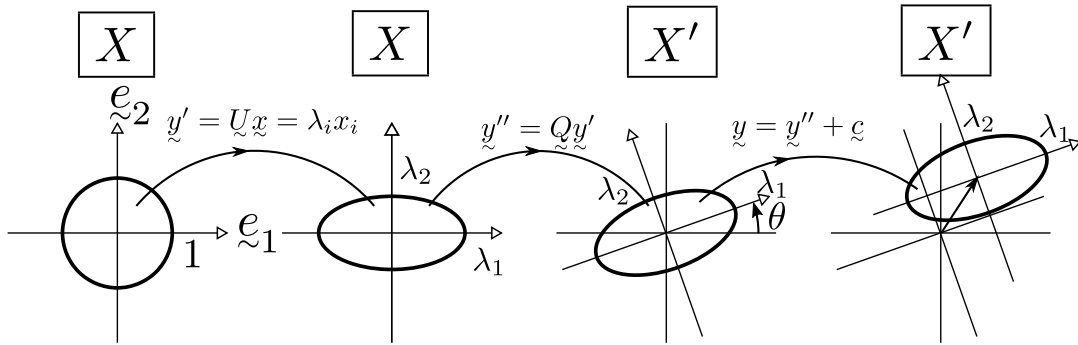


Figure 6.10: Scheme of Resolution of $\mathcal{H}_1$

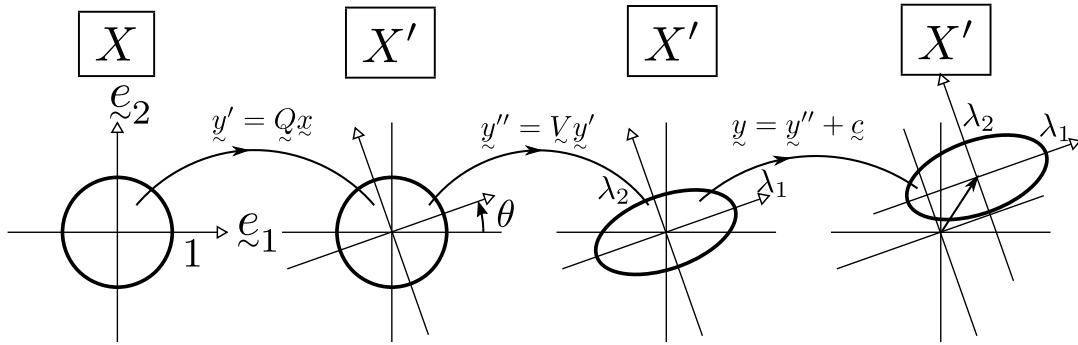


Figure 6.11: Scheme of Resolution of $\mathcal{H}_2$

Recall that $\mathcal{V} = \mathcal{Q}\mathcal{U}\mathcal{Q}^T$, $\mathcal{U} = \mathcal{Q}^T\mathcal{U}\mathcal{Q}$. If $\mathcal{U}$ is symmetric and $\mathcal{Q}$ is proper orthogonal and $X = \{0; e_1, e_2, e_3\} \in \mathcal{F}$ is principal frame of $\mathcal{U}$, $X' = \{0; e'_1, e'_2, e'_3\} \in \mathcal{F}$ where $e'_1 = \mathcal{Q}e_1$ is principal frame for $\mathcal{V} = \mathcal{Q}\mathcal{U}\mathcal{Q}^T$.

6.5. DIRECT GEOMETRICAL INTERPRETATION OF THE LAGRANGIAN STRAIN TENSOR FIELD OF AN ADMISSIBLE DEFORMATION (GENERAL DEFORMATION)

6.5 Direct geometrical interpretation of the Lagrangian Strain tensor field of an admissible deformation (General Deformation)

Let $\hat{y}: R \rightarrow R_d$ be an admissible deformation, $\mathring{x}$ an interior point of R , and $\mathring{\phi} = \phi(\mathring{x})$ the Lagrangian strain tensor of the deformation.

Let further $X = \{0; \mathfrak{e}_1, \mathfrak{e}_2, \mathfrak{e}_3\} \in \mathcal{F}$.

- (a) Suppose Γ_i (i fixed) is a regular arc issuing from $\mathring{x}$, let s be the arc-length along Γ_i measured from $\mathring{x}$ and let $\mathfrak{e}_i$ be tangent to the Γ_i at $\mathring{x}$.

Thus assume:

Parametrise Γ_i

$$\begin{aligned} \Gamma_i: \mathfrak{x} &= \bar{\mathfrak{x}}(s) \quad (0 \leq s \leq l), \bar{\mathfrak{x}} \in \mathfrak{C}^1([0, l]), \\ \bar{\mathfrak{x}}(0) &= \mathring{x}, \bar{\mathfrak{x}}'(0) = \mathfrak{e}_i \end{aligned} \tag{1}$$

Let $\bar{\Gamma}_i^* = \hat{y}(\Gamma_i) \cdots$ deformation image of Γ_i and $\bar{s}^*(s)$ the arc-length along $\bar{\Gamma}_i^*$ measured from $\mathring{y} = \hat{y}(\mathring{x})$. Thus

$$\bar{\Gamma}_i^*: \mathfrak{y} = \hat{y}(\bar{\mathfrak{x}}(s)) = \bar{\mathfrak{y}}(s) \quad (0 \leq s \leq l) \tag{2}$$

$\mathfrak{C}^1$ unique tangent exists

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Then the arc-length along Γ_i^* is

$$\bar{s}(s) = \int_0^s \left\{ \bar{\mathbf{y}}'(\sigma) \cdot \bar{\mathbf{y}}'(\sigma) \right\}^{1/2} d\sigma \quad (3)$$

and

$$\bar{s}'(0) \equiv \left. \frac{d\bar{s}}{ds} \right|_{s=0} = \sqrt{1 + 2\phi_{ii}(\dot{\mathbf{x}})} \quad (\text{no sum})$$

Note also that $\left. \frac{d\bar{s}}{ds} \right|_{s=0} = \lambda(\dot{\mathbf{x}}, \mathbf{e}_i)$, where $\lambda(\dot{\mathbf{x}}, \mathbf{e}_i)$ is the stretch ratio of $\mathbf{e}_i$ in the approximating local homogeneous deformation of $\dot{\mathbf{x}}$.

Proof.

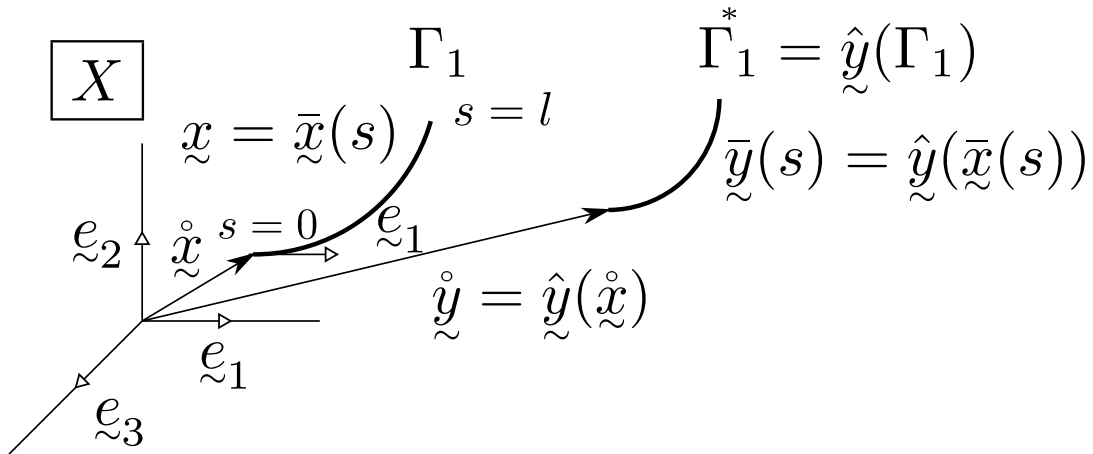


Figure 6.12: Scheme of deformation of curve

From (1), (2) one gets

6.5. DIRECT GEOMETRICAL INTERPRETATION OF THE LAGRANGIAN STRAIN TENSOR FIELD OF AN ADMISSIBLE DEFORMATION (GENERAL DEFORMATION)

$$\begin{aligned}
 ds^*(s) &= \sqrt{d\underline{y} \cdot d\underline{y}} = \sqrt{\underline{y}'(s) \cdot \underline{y}'(s)} ds \\
 \Rightarrow \dot{s}^*(s) &= \sqrt{\underline{y}'(s) \cdot \underline{y}'(s)} \\
 \Rightarrow \dot{s}^*(0) &= \sqrt{\underline{y}'(0) \cdot \underline{y}'(0)}
 \end{aligned}$$

$$\begin{aligned}
 &\stackrel{\text{Chain}}{\underset{\text{Rule}}{=}} \sqrt{\left. \frac{\partial \hat{y}_k}{\partial x_p} \right|_{\underline{x}=\underline{\dot{x}}} \left. \frac{d\bar{x}_p(s)}{ds} \right|_{s=0} \left. \frac{\partial \hat{y}_k}{\partial x_g} \right|_{\underline{x}=\underline{\dot{x}}} \left. \frac{d\bar{x}_g(s)}{ds} \right|_{s=0}} \\
 &= \sqrt{\hat{y}_{k,p}(\underline{\dot{x}}) \bar{x}'_p(0) \hat{y}_{k,g}(\underline{\dot{x}}) \bar{x}'_g(0)} \\
 &= \sqrt{\underline{F}(\underline{\dot{x}}) \bar{\underline{x}}'(0) \cdot \underline{F}(\underline{\dot{x}}) \bar{\underline{x}}'(0)} \\
 &= \sqrt{\left\{ \underline{F}^T(\underline{\dot{x}}) \underline{F}(\underline{\dot{x}}) \underline{e}_i \right\} \cdot \underline{e}_i \text{ (no sum)}}
 \end{aligned}$$

Recall $\bar{\underline{x}}'(0) = \underline{e}_i$ for Γ_i

$$\begin{aligned}
 \dot{s}^*(0) &= \sqrt{\left\{ \left(2\phi(\underline{\dot{x}}) + 1 \right) \underline{e}_i \right\} \cdot \underline{e}_i} = \sqrt{2\phi_{ii}(\underline{\dot{x}}) + 1 \text{ (no sum)}} \\
 &= \|\lambda(\underline{\dot{x}}, \underline{e}_i)
 \end{aligned}$$

□

- (b) Let Γ_i, Γ_j ($i \neq j$), (i, j fixed) be two regular arcs both issuing from $\underline{\dot{x}}$ and having $\underline{e}_i, \underline{e}_j$ as respective tangent vectors at $\underline{\dot{x}}$.

^{||}by result of homogeneous definition

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Let

$$\overset{*}{\Gamma}_i = \hat{y}(\Gamma_i), \overset{*}{\Gamma}_j = \hat{y}(\Gamma_j)$$

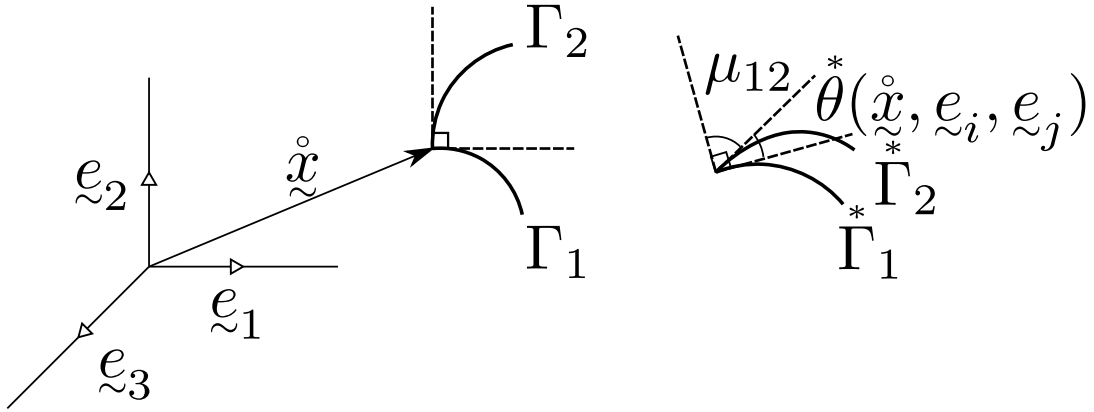


Figure 6.13: Scheme of deformation of curves, $i \neq j$

Call $\overset{*}{\theta}(\overset{\circ}{x}, \mathfrak{e}_i, \mathfrak{e}_j) \in [0, \pi]$ the angle between the tangent vectors of $\overset{*}{\Gamma}_1$ and $\overset{*}{\Gamma}_2$ at $\mathring{y} = \hat{y}(\overset{\circ}{x})$, and set

$$\mathring{\mu}_{ij} \equiv \mu_{ij}(\overset{\circ}{x}) = \frac{\pi}{2} - \overset{*}{\theta}(\overset{\circ}{x}, \mathfrak{e}_i, \mathfrak{e}_j) \quad (i \neq j)$$

Then

$$\sin \mathring{\mu}_{ij} = \frac{2\mathring{\phi}_{ij}}{\sqrt{1 + 2\mathring{\phi}_{ii}} \sqrt{1 + 2\mathring{\phi}_{jj}}} \quad (i \neq j) \text{ (no sum)} \quad (\S\S)$$

Proof equivalent to part(a) [take as exercise]

6.6. INFINITESIMAL DEFORMATION

Note the (§§) coincides with the formula for $\sin \mu_{ij}$ appropriate to the approximately local homogeneous deformation.

Note. We also had proved that any deformation is locally homogeneous. Any deformation in general is *not* in the form of

$$\underline{y} = \hat{\underline{y}}(\underline{x}) = \underline{F}\underline{x} + \underline{c}$$

or

$$\text{grad } \hat{\underline{y}}(\underline{x}) = \nabla \hat{\underline{y}} = \underline{F}^*(\underline{x})$$

i.e. $\nabla \hat{\underline{y}}$ is function of $\underline{x}$

6.6 Infinitesimal deformation

Consider

$$\hat{\underline{y}}(\underline{x}) = \underline{F}\underline{x} + \underline{c} = \underline{x} + (\underline{F} - \underline{1})\underline{x} + \underline{c} = \underline{u}(\underline{x}) + \underline{x}$$

where

$$\underline{u}(\underline{x}) = (\underline{F} - \underline{1})\underline{x} + \underline{c}$$

$$\nabla \underline{u} = \underline{F} - \underline{1} = ** \text{ constant}$$

**by homogeneous definition

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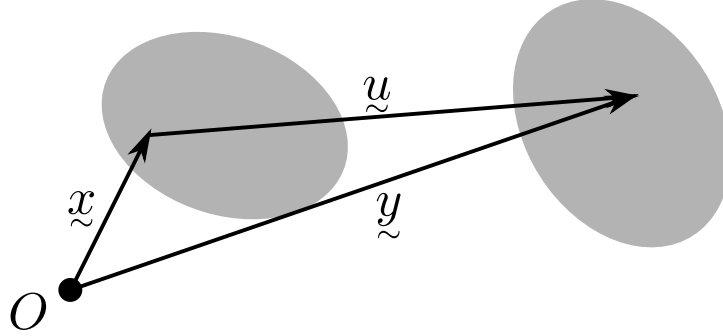


Figure 6.14: Scheme of infinitesimal deformation

define

$$\varepsilon = \frac{1}{2} \left(\nabla \underline{u} + (\nabla \underline{u})^T \right) = \text{sym } \nabla \underline{u}$$

or

$$\varepsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$$

$$\omega = \frac{1}{2} \left(\nabla \underline{u} - (\nabla \underline{u})^T \right) = \text{skew } \nabla \underline{u}$$

or

$$\omega_{ij} = \frac{1}{2} (u_{i,j} - u_{j,i})$$

$$\theta = \text{tr } \varepsilon = \varepsilon_{ii} = u_{i,i} \cdots \text{infinitesimal volume dilatation}$$

Assume we have deformation such that

$$|\nabla \underline{u}| \ll 1 \text{ or } \sqrt{u_{i,j} u_{i,j}} \ll 1$$

i.e., infinitesimal deformation

6.6. INFINITESIMAL DEFORMATION

$$\begin{aligned}
 \underline{E} &= \frac{1}{2} \left(\underline{F}^T \underline{F} - \underline{1} \right) = \frac{1}{2} \left[\left(1 + \nabla \underline{u} \right) \left(1 + (\nabla \underline{u})^T \right) - \underline{1} \right] \\
 &= \frac{1}{2} \left[\nabla \underline{u} + (\nabla \underline{u})^T + \cancel{(\nabla \underline{u})^T (\nabla \underline{u})} \xrightarrow{\text{neglected}} \right] \\
 &\approx \frac{1}{2} \left[\nabla \underline{u} + (\nabla \underline{u})^T \right] = \underline{\varepsilon}
 \end{aligned}$$

Theorem 6.6.1.

$$\epsilon(\underline{e}_i) = [1 + 2E_{ii}]^{1/2} - 1 \quad (\text{no sum})$$

If $|\nabla \underline{u}| \ll 1$, then

$$\epsilon(\underline{e}_i) \sim E_{ii} \sim \varepsilon_{ii} \quad (\text{no sum})$$

Also

$$\sin \mu_{ij} = \frac{2E_{ij}}{\sqrt{1 + 2E_{ii}} \sqrt{1 + 2E_{jj}}}, i \neq j \quad (\text{no sum})$$

then

$$\mu_{ij} \sim 2E_{ij} \sim 2\varepsilon_{ij}, i \neq j$$

and the volume dilatation

CHAPTER 6. STUDY OF DEFORMATION

$$\begin{aligned}
 \Theta &= \frac{V(R_*) - V(R)}{V(R)} = J^{\dagger\dagger} - 1 = \det(\mathbb{1} + \nabla \underline{u}) - 1 \\
 &= \begin{vmatrix} 1 + u_{1,1} & u_{1,2} & u_{1,3} \\ u_{2,1} & 1 + u_{2,2} & u_{2,3} \\ u_{3,1} & u_{3,2} & 1 + u_{3,3} \end{vmatrix} - 1 \\
 &\approx u_{1,1} + u_{2,2} + u_{3,3} = u_{i,i} = \text{tr } \nabla \underline{u} = \text{tr } \underline{\varepsilon}
 \end{aligned}$$

Note. For rigid deformation, $\underline{y} = F\underline{x} = Q\underline{x}$

$$\underline{E} = \frac{1}{2} \left(F^T F - \mathbb{1} \right) = \frac{1}{2} \left(Q^T Q - \mathbb{1} \right) = \underline{0}$$

but

$$\begin{aligned}
 \underline{\varepsilon} &= \frac{1}{2} \left(\nabla \underline{u} + (\nabla \underline{u})^T \right) \\
 &= \frac{1}{2} \left(Q - \mathbb{1} + Q^T - \mathbb{1} \right) \\
 &= \frac{1}{2} \left(Q + Q^T - 2\mathbb{1} \right) \neq \underline{0}
 \end{aligned}$$

in general (except $Q \sim \mathbb{1}$ or $\underline{x} \sim \underline{y}$)

Therefore, $\underline{\varepsilon}$ is not good for rigid deformation. ^{‡‡}

^{‡‡} $J = \det \nabla \hat{y} = \det F$

^{‡‡}In FEM, we always fixed the body because the basic formulas, i.e. shape functions N , are based on *infinitesimal deformation*.

6.6. INFINITESIMAL DEFORMATION

Chapter 7

Kinetics of Continuous Media

Recall the definition of admissible motion

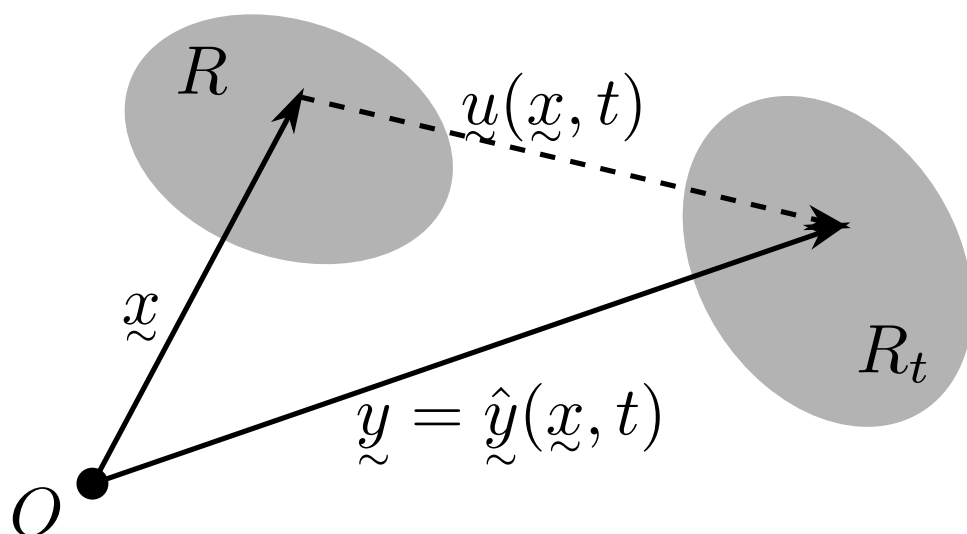


Figure 7.1: Scheme of deformation

7.1. MASS DENSITY, CONSERVATION OF MASS (EQUATION OF CONTINUITY)

$$\hat{\underline{y}}(\bullet, t): R \rightarrow R_t$$

$$t \in \tau = [t_0, t_1]$$

$$\underline{y} = \hat{\underline{y}}(\underline{x}, t) = \underline{x} + \underline{u}(\underline{x}, t)$$

$$\hat{\underline{y}} \in C^2(R \times \tau)$$

$$J = \det \nabla \hat{\underline{y}} = \det (\underline{1} + \nabla \underline{u}) > 0$$

$$\hat{\underline{y}}^{-1} \equiv \hat{\underline{x}}(\bullet, t) \text{ on } R_t, \hat{\underline{x}}(\bullet, t) \in C^2(R_t)$$

7.1 Mass density, conservation of mass (equation of continuity)

Assume that exists a scalar valued function $\rho(\bullet, t)$ on R_t , $\forall t \in [t_0, t_1]$ called the mass density of the body at time t , with the following properties.

$$(i) \quad \rho(\bullet, t) \in C(R_t), \rho(\underline{y}, t) > 0 \quad \forall \underline{y} \in R_t \quad \forall t \in [t_0, t_1]$$

$$(ii) \quad \text{If } P \subset R \text{ is a compact region and if } P_t = \hat{\underline{y}}(P, t), \text{ then}$$

$$m(P_t) = \int_{P_t} \rho(\underline{y}, t) dV_{\underline{y}} \quad \forall t \in \tau$$

$$\dim \rho = [ML^{-3}]$$

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

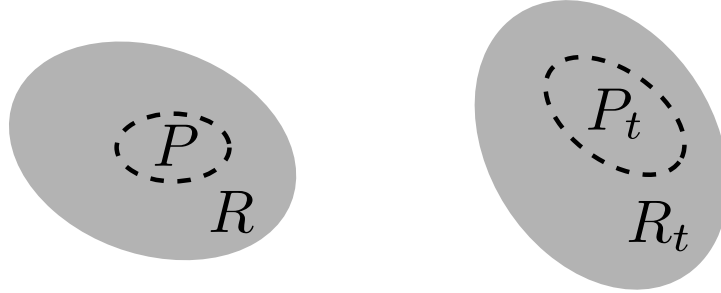


Figure 7.2: Scheme of deformation of P

7.2 Postulate of mass conservation (Global Version)

$$\boxed{m(P_t) = m(P_{t_0}), t_0 \leq t \leq t_1, \forall \text{ compact } P \subset R} \quad (*)$$

i.e., $m(P_{t_0})$ is time independent.

Claim 7.2.1.

Define

$$\phi(\underline{x}, t) = \rho(\hat{y}(\underline{x}, t), t) J(\underline{x}, t) \quad \forall (\underline{x}, t) \in R \times \tau \quad (a)$$

Then from ()*

$$\phi(\underline{x}, t) = \phi(\underline{x}, t_0) \equiv {}^* \rho_0(\underline{x}) \quad \forall (\underline{x}, t) \in R \times \tau \quad (b)$$

Also

*no longer depend on time

7.2. POSTULATE OF MASS CONSERVATION (GLOBAL VERSION)

$$m(P_t) = \int_P \rho_0(\underline{x}) dV_{\underline{x}} \quad (c)$$

Finally

$$\rho(y_{\underline{z}}^{\dagger}, t) J(\underline{x}, t) = \rho_0(\underline{x}) \quad \forall (\underline{x}, t) \in R \times \tau \quad (d)$$

Proof.

$$\begin{aligned} (ii) \Rightarrow m(P_t) &= \int_P \rho(\hat{y}(\underline{x}, t), t) J(\underline{x}, t) dV_{\underline{x}} \\ &\stackrel{(a)}{=} \int_P \phi(\underline{x}, t) dV_{\underline{x}} \end{aligned} \quad (1)$$

$$\begin{aligned} (1) \& (*) \Rightarrow m(P_t) - m(P_{t_0}) &= \int_P \{ \phi(\underline{x}, t) - \phi(\underline{x}, t_0) \} dV_{\underline{x}} = 0 \\ \Rightarrow \phi(\underline{x}, t) &= \phi(\underline{x}, t_0) \equiv \rho_0(\underline{x}) \end{aligned}$$

From (ii)

$$m(P_t) = \int_{P_t} \rho(y, t) dV_y = \int_P \phi(\underline{x}, t) dV_{\underline{x}} = \int_P \rho_0(\underline{x}) dV_{\underline{x}}$$

$$\Rightarrow \boxed{\begin{array}{l} \rho(y, t) J(\underline{x}, t) = \rho_0(\underline{x}) \\ \text{or } \rho(\hat{y}(\underline{x}, t), t) J(\underline{x}, t) = \rho_0(\underline{x}) \end{array}} \quad (**)$$

${}^{\dagger}y = \hat{y}(\underline{x}, t)$

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

(**) ... Local version of conservation of mass

$\rho_0(\underline{x}) \cdots$ *nominal* or *referential* mass density

□

Remark.

- (1) incompressible body ($J(\underline{x}, t) = 1$)
 (can only sustain locally volume preserving deformation)

$$\Rightarrow \rho(\underline{y}, t) = \rho_0(\underline{x})$$

- (2) infinitesimal deformation $|\nabla \underline{u}| \ll 1$,
 $\underline{E} \sim \underline{\varepsilon} = \frac{1}{2} \left(\nabla \underline{u} + (\nabla \underline{u})^T \right)$

$$\Rightarrow J = \det(\underline{1} + \nabla \underline{u}) \approx 1 + \text{tr } \underline{\varepsilon} = 1 + \cancel{u_{k,k}}^{\text{neglected}}$$

$$\Rightarrow \rho(\underline{y}, t) = \rho_0(\underline{x})$$

7.3 Velocity and acceleration fields

The material (Lagrangian, referential) velocity and acceleration defined as

7.3. VELOCITY AND ACCELERATION FIELDS

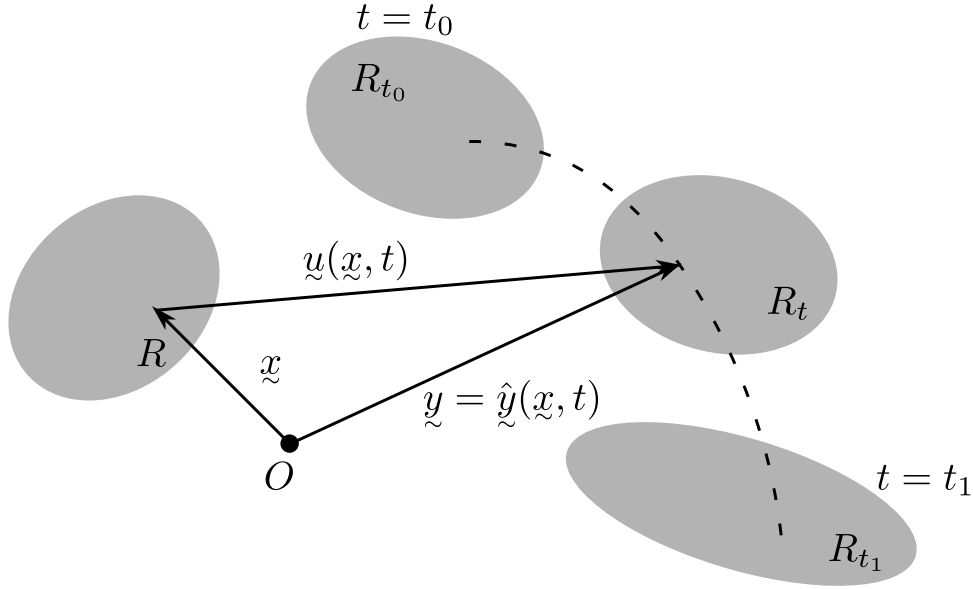


Figure 7.3: the motion of a body

$$\dot{\underline{y}}(\underline{x}, t) = \frac{\partial}{\partial t} \hat{\underline{y}}(\underline{x}, t) = \underline{v}(\underline{x}, t)$$

$$\ddot{\underline{y}}(\underline{x}, t) = \frac{\partial^2}{\partial t^2} \hat{\underline{y}}(\underline{x}, t) = \underline{a}(\underline{x}, t)$$

where $\underline{v}(\underline{x}, t) = \dot{\underline{y}}(\underline{x}, t) \cdots$ velocity at time t of a particle with reference position $\underline{x} \in R$.

Then the spatial (Eulerian) velocity or acceleration as

$$\left. \begin{aligned} \underline{q}(\underline{y}, t) = \bar{\underline{v}}(\underline{y}, t) &= \dot{\underline{y}}(\underline{x}, t) \Big|_{\underline{x} = \hat{\underline{x}}(\underline{y}, t)} \\ \bar{\underline{a}}(\underline{y}, t) &= \ddot{\underline{y}}(\underline{x}, t) \Big|_{\underline{x} = \hat{\underline{x}}(\underline{y}, t)} \end{aligned} \right\} \Rightarrow \begin{aligned} \bar{\underline{v}}(\hat{\underline{y}}(\underline{x}, t), t) &= \dot{\underline{y}}(\underline{x}, t) \\ \bar{\underline{a}}(\hat{\underline{y}}(\underline{x}, t), t) &= \ddot{\underline{y}}(\underline{x}, t) \end{aligned}$$

Caution

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

In general

$$\frac{\partial}{\partial t} \bar{\psi}(\underline{y}, t) \neq \bar{a}(\underline{y}, t)$$

However

$$\frac{\partial}{\partial t} \left\{ \bar{\psi}(\hat{\underline{y}}(\underline{x}, t), t) \right\} \Big|_{\underline{x}=\hat{\underline{x}}(\underline{y}, t)} = \bar{a}(\underline{y}, t)$$

This motivates the notion of material time derivative.

7.4 Material time derivative

Consider a motion $\hat{\underline{y}}: R \rightarrow R_t, t \in \tau$. Let $\psi(\underline{y}, t)$ be a spatial field and assume $\Psi(\underline{x}, t)$ is the corresponding referential (material) field, i.e. $\Psi(\underline{x}, t) = \psi(\hat{\underline{y}}(\underline{x}, t), t)$. Then the material time derivative of ψ is an other spatial field $\dot{\psi}(\underline{y}, t)$ such that

$$\begin{aligned} \dot{\psi}^\dagger(\underline{y}, t) &= \left[\frac{\partial}{\partial t} \Psi(\underline{x}, t) \right] \Big|_{\underline{x}=\hat{\underline{x}}(\underline{y}, t)} \\ &= \left[\frac{\partial}{\partial t} \psi(\hat{\underline{y}}(\underline{x}, t), t) \right] \Big|_{\underline{x}=\hat{\underline{x}}(\underline{y}, t)} \end{aligned}$$

Notation.

$$(1) \quad \frac{\partial}{\partial t} \psi(\underline{y}, t) \neq \dot{\psi}(\underline{y}, t)$$

[†]familiar notation $\frac{D\psi}{Dt}$

7.4. MATERIAL TIME DERIVATIVE

$$(2) \quad \dot{\Psi}(\underline{x}, t) = \frac{\partial}{\partial t} \Psi(\underline{x}, t)$$

Lemma 7.4.1.

Let $\phi(\underline{y}, t)$ and $\underline{g}(\underline{y}, t)$ be two spatial fields (scalar, vector).
Then

$$\frac{D\phi}{Dt} = \dot{\phi}(\underline{y}, t) = \frac{\partial}{\partial t} \phi(\underline{y}, t) + \nabla \phi(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t)$$

Similarly

$$\frac{D\underline{g}}{Dt} = \dot{\underline{g}}(\underline{y}, t) = \frac{\partial}{\partial t} \underline{g}(\underline{y}, t) + \nabla \underline{g}(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t)$$

Proof.

Let $\Phi(\underline{x}, t) = \phi(\hat{\underline{y}}(\underline{x}, t), t)$

Then

$$\begin{aligned} \frac{\partial}{\partial t} \Phi(\underline{x}, t) &= \left[\frac{\partial}{\partial t} \phi(\hat{\underline{y}}(\underline{x}, t), \tau) \right]_{\tau=t} + \left[\frac{\partial}{\partial t} \phi(\hat{\underline{y}}(\underline{x}, \tau), t) \right]_{\tau=t} \\ &= \left[\frac{\partial}{\partial t} \phi(\hat{\underline{y}}(\underline{x}, \tau), t) \right]_{\tau=t} + \nabla \phi(\underline{y}, t) \Big|_{\underline{y}=\hat{\underline{y}}(\underline{x}, t)} \cdot \frac{\partial \hat{\underline{y}}}{\partial t} \end{aligned}$$

$$\frac{\partial}{\partial t} \Phi(\underline{x}, t) = \frac{\partial}{\partial t} \phi(\underline{y}, t) + \nabla \phi(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t)$$

Let $\underline{x} = \hat{\underline{x}}(\underline{y}, t)$

$$\dot{\phi}(\underline{y}, t) = \frac{\partial}{\partial t} \phi(\underline{y}, t) + \nabla \phi(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t)$$

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

$$\begin{aligned}
 \frac{\partial}{\partial t} \left[g_i(\hat{y}(\underline{x}, t), t) \right] &= \frac{\partial g_i}{\partial y_j} \frac{\partial y_j(\underline{x}, t)}{\partial t} + \frac{\partial g_i}{\partial t} \\
 &= \left(\nabla_{\underline{g}} \right)_{ij} v_j + \frac{\partial g_i}{\partial t} \\
 \therefore \dot{\underline{g}}(\hat{y}(\underline{x}, t), t) &= \left(\nabla_{\underline{g}} \right) \bar{v} + \frac{\partial \underline{g}(\hat{y}(\underline{x}, t), t)}{\partial t}
 \end{aligned}$$

□

Lemma 7.4.2.

Let $\underline{\tau} \in C^1[t_0, t_1]$ be a function whose values are non-singular second order tensors. Then

$$\begin{aligned}
 \frac{d}{dt} \det \underline{\tau}(t) &= \det \underline{\tau}(t) \left\{ \dot{\underline{\tau}}(t) \cdot \underline{\tau}^{-T}(t) \right\} \\
 &= \det \underline{\tau}(t) \dot{\tau}_{ij} \tau_{ji}^{-1}(t) \quad (t_0 \leq t \leq t_1)
 \end{aligned} \tag{★}$$

Proof.

Recall

$$\det \underline{\tau}(t) = \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} \tau_{ip}(t) \tau_{jq}(t) \tau_{kr}(t)$$

$$\begin{aligned}
 \therefore \frac{d}{dt} \left\{ \det \underline{\tau}(t) \right\} &= \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} \left\{ \dot{\tau}_{ip}(t) \tau_{jq}(t) \tau_{kr}(t) \right. \\
 &\quad \left. + \tau_{ip}(t) \dot{\tau}_{jq}(t) \tau_{kr}(t) + \tau_{ip}(t) \tau_{jq}(t) \dot{\tau}_{kr}(t) \right\}
 \end{aligned} \tag{1}$$

Recall also that

7.4. MATERIAL TIME DERIVATIVE

$$\boxed{\left(\mathcal{T}^{-1}\right)_{ij} \equiv \tau_{ij}^{-1} = \frac{1}{2\det\mathcal{T}} \epsilon_{ipq} \epsilon_{jrs} \tau_{rp} \tau_{sq}} \quad (2)$$

From (1) and (2), we obtain

$$\begin{aligned} \frac{d}{dt} \left\{ \det \mathcal{T}(t) \right\} &= \frac{1}{6} \left\{ 2 \det \mathcal{T} \dot{\tau}_{ip} \tau_{pi}^{-1} + 2 \det \mathcal{T} \dot{\tau}_{jq} \tau_{qj}^{-1} \right. \\ &\quad \left. + 2 \det \mathcal{T} \dot{\tau}_{kr} \tau_{rk}^{-1} \right\} \end{aligned}$$

or

$$\frac{d}{dt} \left\{ \det \mathcal{T}(t) \right\} = \det \mathcal{T}(t) \dot{\tau}_{ij}(t) \tau_{ji}^{-1}(t)$$

□

Claim 7.4.1.

Let $\hat{y}(\bullet, t): R \rightarrow R_t$, one has the identity

$$\boxed{\dot{J}(\underline{x}, t) = J(\underline{x}, t) \nabla_{\underline{y}} \cdot \underline{q}(\underline{y}, t), \bar{\underline{v}} = \underline{q}(\underline{y}, t)} \quad (\S)$$

Proof.

Let $\mathcal{T}(t) = \nabla \hat{y}(\underline{x}, t)$, $\det \mathcal{T} = \det \nabla \hat{y}(\underline{x}, t) = J(\underline{x}, t)$

$$\frac{d}{dt} \left\{ \det \nabla \hat{y}(\underline{x}, t) \right\} = J(\underline{x}, t) \stackrel{(\star)}{\parallel} J(\underline{x}, t) \dot{\hat{y}}_{i,j}(\underline{x}, t) \hat{x}_{j,i}(\underline{y}, t)$$

recall $\nabla \hat{\underline{x}}(\underline{y}, t) = \left[\nabla \hat{y}(\underline{x}, t) \right]^{-1}$

$\parallel \frac{d}{dt}, \frac{\partial}{\partial x_i}$ interchangeable

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

$$\begin{aligned}
 \dot{J}(\underline{x}, t) &\stackrel{**}{=} J(\underline{x}, t) \frac{\partial}{\partial x_j} v_i(\underline{x}, t) \frac{\partial \hat{x}_j(\underline{y}, t)}{\partial y_i} \\
 &\stackrel{\dagger\dagger}{=} J(\underline{x}, t) \frac{\partial}{\partial y_i} \bar{v}_i(\underline{y}, t) \\
 &= J(\underline{x}, t) q_{i,i} \\
 &= J(\underline{x}, t) \nabla_{\underline{y}} \cdot \underline{q}(\underline{y}, t)
 \end{aligned}$$

□

Claim 7.4.2.

For any admissible motion mass balance (local) implies

$$\frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \rho(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t) + \rho(\underline{y}, t) \nabla \cdot \bar{\underline{v}}(\underline{y}, t) = 0$$

or

$$\frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \cdot [\rho(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t)] = 0$$

Proof.

Recall (**) and (§)

$$\rho(\underline{y}, t) J(\underline{x}, t) = \rho_0(\underline{x})$$

$$\dot{J}(\underline{x}, t) = J(\underline{x}, t) \nabla_{\underline{y}} \cdot \bar{\underline{v}}(\underline{y}, t)$$

Differentiate with respect to t ($\underline{x}$ fixed)

$\dagger\dagger \frac{\partial \hat{y}}{\partial t} = \underline{v}$ the Lagrangian velocity

$\dagger\dagger$ Chain rule and $\underline{q}(\underline{y}, t) = \bar{\underline{v}}(\underline{y}, t) = \underline{v}(\underline{x}(\underline{y}, t), t)$

7.5. FOR LOCALLY VOLUME PRESERVING (ISOCHORIC MOTION)

$$\begin{aligned}
&\rightarrow \left[\frac{\partial}{\partial t} \rho(\hat{y}(\underline{x}, t), t) \right] J(\underline{x}, t) + \rho(\hat{y}(\underline{x}, t), t) \frac{\partial}{\partial t} J(\underline{x}, t) = 0 \\
&\rightarrow \dot{\rho}(\underline{y}, t) J(\underline{x}, t) + \rho(\underline{y}, t) J(\underline{x}, t) \nabla \cdot \bar{\underline{v}}(\underline{y}, t) = 0 \\
&\rightarrow \dot{\rho}(\underline{y}, t) + \rho(\underline{y}, t) \nabla \cdot \bar{\underline{v}}(\underline{y}, t) = 0 \\
&\xrightarrow[7.4.1]{\text{Lemma}} \frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \rho(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t) + \rho(\underline{y}, t) \nabla \cdot \bar{\underline{v}}(\underline{y}, t) = 0 \\
&\rightarrow \frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \cdot [\rho(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t)] = 0
\end{aligned}$$

□

7.5 For locally volume preserving (isochoric motion)

$$J(\underline{x}, t) = 1$$

$$\begin{aligned}
&\Rightarrow \dot{J}(\underline{x}, t) = 0 \stackrel{(\S)}{=} J(\underline{x}, t) \nabla \cdot \bar{\underline{v}}(\underline{y}, t) = 0 \\
&\Rightarrow \nabla \cdot \bar{\underline{v}} = 0 \\
&\Rightarrow \dot{\rho}(\underline{y}, t) = 0
\end{aligned}$$

OR

$$\frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \rho(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t) = 0$$

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

7.6 Spatial velocity gradient tensor

Let $\bar{\underline{v}}(\underline{y}, t)$ be the spatial or Eulerian velocity field. Define a tensor field $\underline{L}(\bullet, t): R_t \in \mathcal{L}^{\sharp\sharp}$ by

$$\underline{L}(\underline{y}, t) = \nabla_{\underline{y}} \bar{\underline{v}}(\underline{y}, t) \quad \forall \underline{y} \in R_t \quad \forall t \in \tau$$

called the spatial velocity gradient tensor.

Recall

$$\begin{aligned} \underline{F}(\underline{x}, t) &= \nabla_{\underline{x}} \hat{\underline{y}}(\underline{x}, t) \\ \Rightarrow \dot{\underline{F}}(\underline{x}, t) &= \left[\nabla_{\underline{x}} \dot{\hat{\underline{y}}}(\underline{x}, t) \right] = \nabla_{\underline{x}} \dot{\hat{\underline{y}}}(\underline{x}, t) \end{aligned}$$

However recall

$$\begin{aligned} \dot{\hat{\underline{y}}}(\underline{x}, t) &= \bar{\underline{v}}(\hat{\underline{y}}(\underline{x}, t), t) \\ \Rightarrow \nabla_{\underline{x}} \dot{\hat{\underline{y}}}(\underline{x}, t) &= \nabla_{\underline{y}} \bar{\underline{v}}(\underline{y}, t) \Big|_{\underline{y}=\hat{\underline{y}}(\underline{x}, t)} \nabla_{\underline{x}} \hat{\underline{y}}(\underline{x}, t) \\ \therefore \boxed{\dot{\underline{F}}(\underline{x}, t) &= \underline{L}(\hat{\underline{y}}(\underline{x}, t), t) \underline{F}(\underline{x}, t)} \end{aligned}$$

Definition 7.6.1.

The stretching and spin tensor fields as

$\sharp\sharp\mathcal{L}$ is set of all 2^{nd} order tensor

7.6. SPATIAL VELOCITY GRADIENT TENSOR

$$\begin{aligned}\underline{\underline{D}}(\underline{y}, t) &= \text{sym } \underline{\underline{L}}(\underline{y}, t) = \frac{1}{2} \left[\nabla \bar{v}(\underline{y}, t) + \left(\nabla \bar{v}(\underline{y}, t) \right)^T \right] \\ \underline{\underline{W}}(\underline{y}, t) &= \text{skew } \underline{\underline{L}}(\underline{y}, t) = \frac{1}{2} \left[\nabla \bar{v}(\underline{y}, t) - \left(\nabla \bar{v}(\underline{y}, t) \right)^T \right]\end{aligned}$$

so that $\underline{\underline{L}} = \underline{\underline{W}} + \underline{\underline{D}}$, $\underline{\underline{D}} = \underline{\underline{D}}^T$, $\underline{\underline{W}} = -\underline{\underline{W}}^T$

Note.

Recall $\underline{\underline{F}} = \underline{\underline{Q}}\underline{\underline{U}} = \underline{\underline{V}}\underline{\underline{U}}$ ($\underline{\underline{F}}$, $\underline{\underline{Q}}$, $\underline{\underline{U}}$, and $\underline{\underline{V}}$ are all functions of time) and for infinitesimal deformation we have

$$\begin{aligned}\underline{\underline{L}} &\sim \dot{\underline{\underline{F}}} \\ \underline{\underline{D}} &\sim \dot{\underline{\underline{U}}} \\ \underline{\underline{W}} &\sim \dot{\underline{\underline{Q}}}\end{aligned}$$

Consider two particles P and Q with material position $\underline{x}$ and $\underline{x} + d\underline{x}$. At time t they are to be found at $\underline{y}(\underline{x}, t)$ and $\underline{y}(\underline{x} + d\underline{x}, t)$

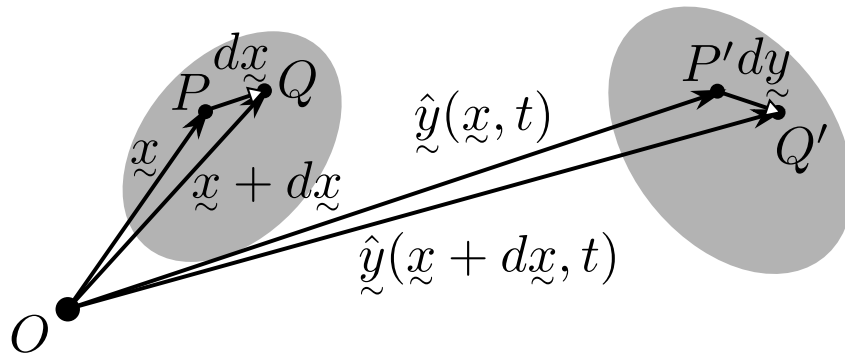


Figure 7.4: Scheme of deformation of P and Q

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

$$y(\underline{x} + d\underline{x}, t) = y(\underline{x}, t) + \nabla_{\underline{x}} y(\underline{x}, t) d\underline{x} + \text{higher order term (H.O.T.)}$$

$$\therefore dy = \underbrace{\nabla_{\underline{x}} y}_{F} d\underline{x} \text{ or } dy_i = \frac{\partial y_i}{\partial x_j} dx_j$$

Velocity of P

$$\underline{v} = \dot{\underline{y}}(\underline{x}, t)$$

Relative velocity of P' and Q'

$$\begin{aligned} d\underline{v} &= \left(\dot{dy} \right) = \left(\nabla_{\underline{x}} \dot{y} \right) d\underline{x} \\ &= \nabla_{\underline{x}} \underline{v}(\underline{x}, t) d\underline{x} \\ &= \frac{\partial v_i(\underline{x}, t)}{\partial x_j} dx_j \underline{e}_i \\ &= \frac{\partial \bar{v}_i(\hat{\underline{y}}(\underline{x}, t), t)}{\partial y_m} \underbrace{\frac{\partial y_m(\underline{x}, t)}{\partial x_j}}_{dy_m} dx_j \underline{e}_i \\ &= \underbrace{\left(\nabla_{\underline{y}} \bar{\underline{v}} \right)}_{\substack{\text{velocity} \\ \text{gradient tensor}}} d\underline{y} \end{aligned}$$

where relative velocity in terms of *current* relative position.

$$\Rightarrow dv_i = \frac{\partial v_i}{\partial x_j} dx_j = \frac{\partial \bar{v}_i}{\partial y_j} dy_j \quad (\blacktriangle)$$

7.7. PHYSICAL INTERPRETATION OF D_{ij}

The stretching (or rate of strain) and spin tensors are defined as

$$\begin{aligned} \underline{D} &\triangleq \text{sym } \nabla_{\underline{y}} \bar{v} = \frac{1}{2} \left[\nabla \bar{v} + (\nabla \bar{v})^T \right] \\ \underline{W} &\triangleq \text{skew } \nabla_{\underline{y}} \bar{v} = \frac{1}{2} \left[\nabla \bar{v} - (\nabla \bar{v})^T \right] \end{aligned}$$

Note. Recall that exists a vector ω such that

$$\omega_i = -\frac{1}{2} \epsilon_{ijk} W_{jk}$$

7.7 Physical interpretation of D_{ij}

Let the length of the line segment $P'Q'$ is ds , then

$$\begin{aligned} ds^2 &= dy_i dy_i = \frac{\partial y_i}{\partial x_j} \frac{\partial y_i}{\partial x_k} dx_j dx_k \\ \left(\dot{ds}^2 \right) &= \left(\frac{\partial \dot{y}_i}{\partial x_j} \frac{\partial y_i}{\partial x_k} + \frac{\partial y_i}{\partial x_j} \frac{\partial \dot{y}_i}{\partial x_k} \right) dx_j dx_k \\ &= \left(\frac{\partial v_i}{\partial x_j} \frac{\partial y_i}{\partial x_k} + \frac{\partial v_i}{\partial x_k} \frac{\partial y_i}{\partial x_j} \right) dx_j dx_k \\ &\stackrel{(\blacktriangle)}{=} \frac{\partial \bar{v}_i}{\partial y_j} dy_j dy_i + \frac{\partial \bar{v}_i}{\partial y_k} dy_k dy_i \\ &= 2 \frac{\partial \bar{v}_i}{\partial y_j} dy_i dy_j \\ &= 2 (D_{ij} + W_{ij}) dy_i dy_j = \S\S 2 D_{ij} dy_i dy_j \end{aligned}$$

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$$\therefore 2ds(\dot{ds}) = 2D_{ij}dy_i dy_j$$

or

$$\frac{1}{ds}(\dot{ds}) = D_{ij} \frac{dy_i}{ds} \frac{dy_j}{ds} \cdot \text{rate of stretching} \quad (*)$$

Let $\frac{dy}{ds} = \underline{e}_1$, then D_{11} represents the rate of strain of a fluid element parallel to $\underline{e}_1$ axis. Similar to D_{22} and D_{33} .

Note.

$$D_{11} + D_{22} + D_{33} = \text{rate of volume dilatation}$$

Again consider two segment PQ and PR , R is the particle with material position $\underline{x} + d\underline{x}'$

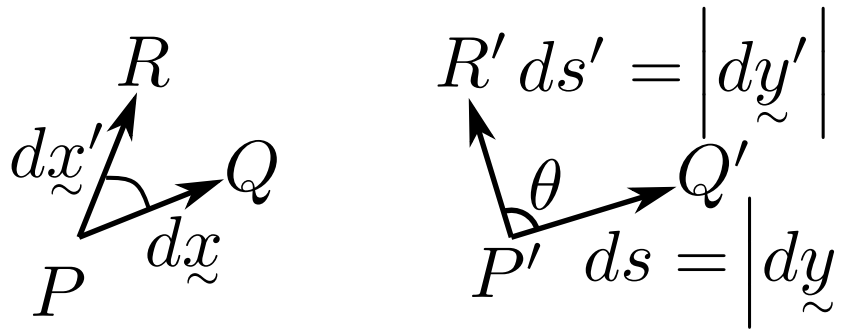


Figure 7.5: Scheme of change of two segment

$$ds ds' \cos \theta = dy_i dy'_i$$

$W_{ij} = -W_{ji}$

7.8. CONSERVATION OF MASS FOR A CONTROL VOLUME

Take material time derivative

$$\begin{aligned} (ds ds' \cos \theta)^\bullet &= dv_i dy'_i + dv'_i dy_i \\ &\stackrel{(\blacktriangle)}{=} \frac{\partial \bar{v}_i}{\partial y_j} dy_j dy'_i + \underbrace{\frac{\partial \bar{v}_i}{\partial y_j} dy'_j dy_i}_{i \leftrightarrow j} \end{aligned}$$

$$\begin{aligned} \Rightarrow \cos \theta \left[(\dot{ds}) ds' + ds (\dot{ds}') \right] - (\dot{\theta}) \sin \theta ds ds' &= \left(\frac{\partial \bar{v}_i}{\partial y_j} + \frac{\partial \bar{v}_j}{\partial y_i} \right) dy'_i dy_j \\ \Rightarrow \cos \theta \left[\frac{1}{ds} (\dot{ds}) + \frac{1}{ds'} (\dot{ds}') \right] - \dot{\theta} \sin \theta &= 2D_{ij} \frac{dy'_i}{ds'} \frac{dy_j}{ds} \end{aligned}$$

Let $\frac{dy'}{ds'} = \underline{e}_1$, $\frac{dy}{ds} = \underline{e}_2$
then $\theta = \frac{\pi}{2}$

$$\therefore -\dot{\theta} = 2D_{12}$$

D_{12} is one-half the rate of decrease of the angle between two segments originally parallel to $\underline{e}_1$ and $\underline{e}_2$ respectively.

7.8 Conservation of mass for a control volume

If D is a control volume independent of time, then

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$$\frac{d}{dt} \int_D \rho(\underline{y}, t) dV_{\underline{y}} + \int_{\partial D} \rho(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t) \cdot \underline{n}(\underline{y}, t) dA_{\underline{y}} = 0$$

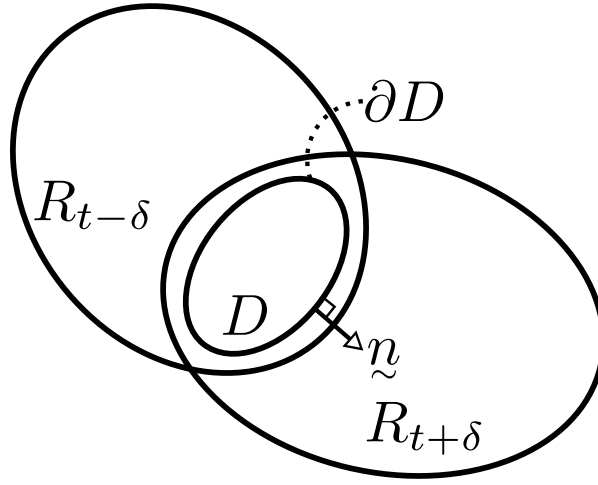


Figure 7.6: Scheme of control volume

Proof.

Recall D is time independent

$$\int_D \frac{\partial}{\partial t} \rho(\underline{y}, t) dV_{\underline{y}} = \frac{d}{dt} \int_D \rho(\underline{y}, t) dV_{\underline{y}}$$

also

$$\int_D \nabla \cdot [\rho \bar{\underline{v}}] dV_{\underline{y}} = \int_{\partial D} \rho \bar{\underline{v}} \cdot \underline{n} dA_{\underline{y}} \quad (\text{Divergence theorem})$$

so

$$\int_D \left\{ \frac{\partial}{\partial t} \rho(\underline{y}, t) + \nabla \cdot [\rho(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t)] \right\} dV_{\underline{y}} = 0$$

7.9. REYNOLDS TRANSPORT THEOREM (RTT)

□

7.9 Reynolds transport theorem (RTT)

Let $P \in R$, $P_t \in R_t$, $P_t = \hat{y}(P, t)$
and define

$$\phi(t) = \int_{P_t} \rho(\underline{y}, t) \psi(\underline{y}, t) dV_{\underline{y}}$$

Then

$$\frac{d}{dt} \phi(t) = \int_{P_t} \rho(\underline{y}, t) \dot{\psi}(\underline{y}, t) dV_{\underline{y}}$$

or

$$\frac{d\phi}{dt} = \int_{P_t} \rho \left[\frac{\partial \psi}{\partial t} + \nabla \psi \cdot \bar{\underline{v}} \right] dV_{\underline{y}}$$

Proof.

$$\begin{aligned} \phi(t) &= \int_P \rho(\hat{\underline{y}}(\underline{x}, t), t) \psi(\hat{\underline{y}}(\underline{x}, t), t) J(\underline{x}, t) dV_{\underline{x}} \\ &\stackrel{(**)}{=} \int_P \rho_0(\underline{x}) \psi(\hat{\underline{y}}(\underline{x}, t), t) dV_{\underline{x}} \end{aligned}$$

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$$\begin{aligned}\frac{d\phi}{dt} &= \int_P \rho_0(\mathfrak{x}) \frac{\partial}{\partial t} \psi(\hat{y}(\mathfrak{x}, t), t) dV_{\mathfrak{x}} \\ &= \int_P \left[\rho(\mathfrak{y}, t) \dot{\psi}(\mathfrak{y}, t) \right] \Big|_{\mathfrak{y}=\hat{y}(\mathfrak{x}, t)} J(\mathfrak{x}, t) dV_{\mathfrak{x}}\end{aligned}$$

$$\therefore \frac{d\phi}{dt} = \int_{P_t} \rho(\mathfrak{y}, t) \dot{\psi}(\mathfrak{y}, t) dV_{\mathfrak{y}}$$

□

If $\Psi(\mathfrak{x}, t) = \psi(\hat{y}(\mathfrak{x}, t), t)$ then

$$\frac{d\phi}{dt} = \int_P \rho_0(\mathfrak{x}) \frac{\partial}{\partial t} \Psi(\mathfrak{x}, t) dV_{\mathfrak{x}}$$

7.10 Body forces, contact forces, surface tractions

Let $\hat{y}(\bullet, t): R \rightarrow R_t$. Let $P \subset R$, $P_t = \hat{y}(P, t)$.

Admissible forces:

- (a) body forces (extrinsic, assigned), e.g. gravity, magnetic field.
- (b) contact forces. (contact of material in P_t with surrounding material)

7.10. BODY FORCES, CONTACT FORCES, SURFACE TRACTIONS



Figure 7.7: Scheme of deformation of P

7.10.1 Body force density

Assume that exists a vector valued function $\underline{b}(\bullet, t)$ defined on R_t called the body force per unit current volume such that

- (i) $\underline{b}(\bullet, t) \in C(R_t)$
- (ii) The resultant force and resultant moment about O due to the body forces acting on the material occupying R_t are given by

$$\int_{R_t} \underline{b}(\underline{y}, t) dV_{\underline{y}}, \int_{R_t} \underline{y} \wedge \underline{b}(\underline{y}, t) dV_{\underline{y}}$$

$$\dim \underline{b} = [FL^{-3}]$$

7.10.2 Contact forces, surface tractions

Assume there is a vector valued function $\underline{t}(\bullet, \bullet, t)$ defined on $R_t \times u$ ($t_0 \leq t \leq t_1$), $u \cdots$ set of all unit vector, called the surface force per unit current area or the actual surface traction or Cauchy traction with the properties:

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- (i) $\underline{t}(\bullet, \bullet, t) \in C(R_t \times u)$
- (ii) If S is a regular subsurface of P and if $S_t = \hat{y}(S, t) \quad \forall t \in \tau$, then the resultant force and resultant moment about O due to the contact forces acting over S_t on the material occupying P_t at time t are given by

$$\int_{S_t} \underline{t}(\underline{y}, \underline{n}(\underline{y}), t) dA_{\underline{y}} \cdots \text{resultant force}$$

$$\int_{S_t} \underline{y} \wedge \underline{t}(\underline{y}, \underline{n}(\underline{y}), t) dA_{\underline{y}} \cdots \text{resultant moment}$$

$$\dim \underline{t} = [FL^{-2}]$$

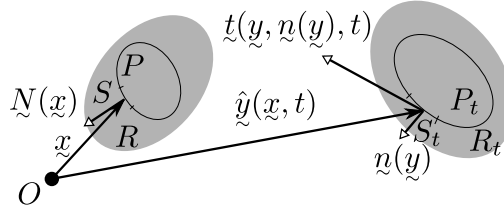


Figure 7.8: Scheme of contact forces, surface tractions

7.11 Conservation of linear momentum

$$\int_{\partial P_t} \underline{t}(\underline{y}, \underline{n}(\underline{y}), t) dA_{\underline{y}} + \int_{P_t} \underline{b}(\underline{y}, t) dV_{\underline{y}} = \frac{d}{dt} \int_{P_t} \rho(\underline{y}, t) \bar{\underline{v}}(\underline{y}, t) dV_{\underline{y}} \\ \stackrel{RTT}{=} \int_{P_t} \rho(\underline{y}, t) \bar{\underline{a}}(\underline{y}, t) dV_{\underline{y}}$$

7.12. CONSERVATION OF ANGULAR MOMENTUM

7.12 Conservation of angular momentum

$$\begin{aligned}
 & \int_{\partial P_t} \underline{y} \wedge \underline{t}(\underline{y}, \underline{n}(\underline{y}), t) dA_{\underline{y}} + \int_{P_t} \underline{y} \wedge \underline{b}(\underline{y}, t) dV_{\underline{y}} \\
 &= \frac{d}{dt} \int_{P_t} \rho(\underline{y}, t) \underline{y} \wedge \underline{\bar{v}}(\underline{y}, t) dV_{\underline{y}} \\
 &\stackrel{RTT}{=} \int_{P_t} \rho [\underline{y} \wedge \dot{\underline{\bar{v}}}] dV_{\underline{y}} \\
 &\stackrel{**}{=} \int_{P_t} \rho \underline{y} \wedge \underline{\bar{a}}(\underline{y}, t) dV_{\underline{y}}
 \end{aligned}$$

Example. For $\underline{b}(\underline{y}, t)$

Let the center of the earth be O , $O \notin R_t$, mass of earth M , then the gravitational body force density

$$\underline{b}(\underline{y}, t) = - \frac{\rho(\underline{y}, t) G M \underline{y}}{|\underline{y}|^3} \quad \forall \underline{y} \in R_t, t \in \tau$$

where $G > 0 \dots$ Universal constant of Gravitation.

7.13 Existence of stress

The following celebrated theorem by Cauchy introduces and establishes the notion of stress based on Cauchy's postulate that

$$\underline{\bar{y}} \wedge \underline{\bar{v}} = \underline{\bar{v}} \wedge \underline{\bar{v}} = \underline{0}$$

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tractions depend on the unit normal, and by using linear momentum balance.

Theorem 7.13.1. *Cauchy*

Let $\hat{y}(\bullet, t): R \rightarrow R_t$ be a motion and let $\underline{t}(\bullet, \bullet, t)$ and $\underline{b}(\bullet, t)$ be a traction and a body force density field such that linear momentum balance holds. Then there exists a second order tensor field $\underline{\tau}(\bullet, t): R_t \in \mathcal{L}^{***}$ for each $t \in [t_0, t_1]$ such that

$$\boxed{\underline{t}(\underline{n}, \underline{y}, t) = \underline{\tau}(\underline{y}, t)\underline{n}} \quad \forall \underline{n} \in u, \forall \underline{y} \in R_t, \forall t \in \tau$$

or

$$t_i^X(\underline{n}, \underline{y}, t) = \tau_{ij}^X(\underline{y}, t)n_j^X$$

Proof.

Fix $t \in [t_0, t_1]$

Assertion 1

Given any $\underline{y} \in R_t$ for $X = \{0; \underline{e}_1, \underline{e}_2, \underline{e}_3\} \in \mathcal{F}$ and unit vector $\underline{n}$ with $\underline{n} \cdot \underline{e}_i > 0$ then

$$\underline{t}(\underline{n}, \underline{y}) = - \sum_{i=1}^3 n_i \underline{t}(-\underline{e}_i, \underline{y}), \quad n_i = \underline{n} \cdot \underline{e}_i$$

Proof.

*** $\mathcal{L}$ is set of all 2^{nd} order tensor

7.13. EXISTENCE OF STRESS

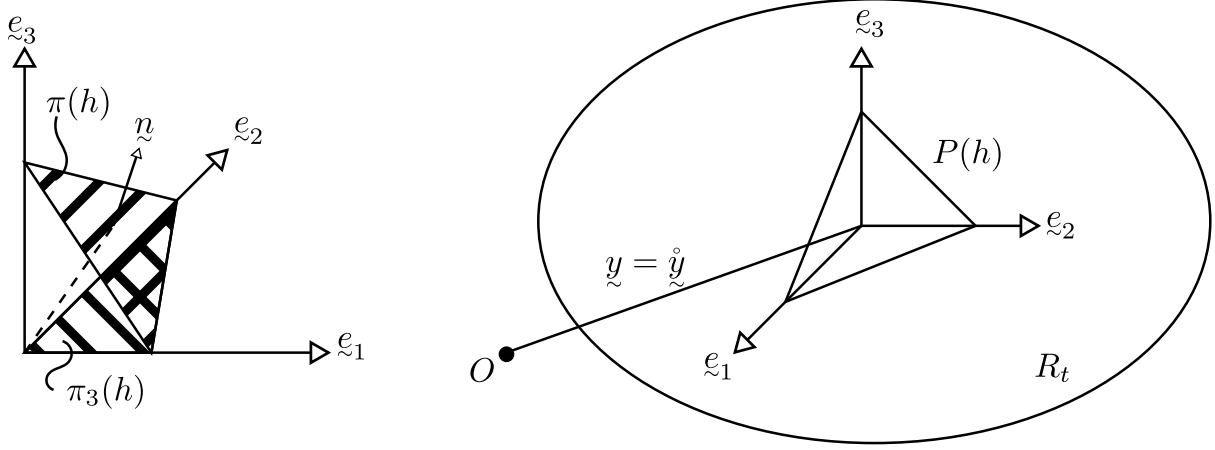


Figure 7.9: Scheme of tetrahedra

Fix $y = \dot{y}$ in R_t ($\dot{y}$ is arbitrary)

Let $h > 0$ and consider the h -parameter family of tetrahedra $P(h)$

$\pi_i(h)$ has $-e_i$ as normal

$\pi(h)$ has $\underline{n}$ as normal

Area $\pi(h) = A(h) = c_1 h^2$, where c_1 is a real number

Area $\pi_i(h) = A_i(h) = A(h)n_i$

Linear momentum for $P(h) \Rightarrow$

$$\int_{\partial P(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} = \int_{P(h)} [\rho \bar{a} - \underline{b}] dV_{\underline{y}} \quad , \text{ volume } P(h) = c_2 h^3 \quad (1)$$

the integral of R.H.S. of above is continuous on R_t so exists M such that $|\rho \bar{a} - \underline{b}| \leq M$ on $P(h) \quad \forall h \in [0, h_0]$

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$$\Rightarrow \left| \int_{P(h)} [\rho \bar{a} - b] dV_{\underline{y}} \right| \leq M \underbrace{c_2 h^3}_{\text{volume}}$$

from (1) $\Rightarrow$

$$\left| \frac{1}{A(h)} \int_{\partial P(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} \right| \leq \frac{M c_2 h^3}{A(h)} = \frac{M c_2 h^3}{c_1 h^2} \rightarrow 0 \text{ as } h \rightarrow 0$$

$$\Rightarrow \frac{1}{A(h)} \int_{\partial P(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} \rightarrow 0 \text{ as } h \rightarrow 0 \quad (2)$$

or

$$\begin{aligned} \frac{1}{A(h)} \int_{\partial P(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} &= \frac{1}{A(h)} \int_{\pi(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} \\ &\quad + \frac{1}{A(h)} \sum_{i=1}^3 \int_{\pi_i(h)} \underline{t}(-\underline{e}_i, \underline{y}) dA_{\underline{y}} \end{aligned}$$

also since traction field is continuous we have that

$$\left| \underline{t}(\underline{n}, \underline{y}) - \underline{t}(\underline{n}, \dot{\underline{y}}) \right| \leq m(h) \rightarrow 0 \text{ as } h \rightarrow 0$$

thus

$$\frac{1}{A(h)} \left| \int_{\pi(h)} \left[\underline{t}(\underline{n}, \underline{y}) - \underline{t}(\underline{n}, \dot{\underline{y}}) \right] dA \right| \leq \frac{m(h) A(h)}{A(h)} = m(h) \rightarrow 0$$

7.13. EXISTENCE OF STRESS

as $h \rightarrow 0$

hence

$$\begin{aligned} \frac{1}{A(h)} \int_{\pi(h)} \underline{t}(\underline{n}, \underline{y}) dA_{\underline{y}} &\rightarrow \frac{1}{A(h)} \underline{t}(\underline{n}, \dot{\underline{y}}) \underbrace{\int_{\pi(h)} dA_{\underline{y}}}_{=A(h)} \text{ as } h \rightarrow 0 \\ &\rightarrow \underline{t}(\underline{n}, \dot{\underline{y}}) \text{ as } h \rightarrow 0 \end{aligned}$$

Similarly

$$\frac{A_i(h)}{A(h)} \frac{1}{A_i(h)} \int_{\pi_i(h)} \underline{t}(-\underline{e}_i, \underline{y}) dA_{\underline{y}} \rightarrow \frac{A_i(h)}{A(h)} \underline{t}(-\underline{e}_i, \dot{\underline{y}})$$

where $A_i(h) = A(h)n_i$

$$(2) \Rightarrow \underline{t}(\underline{n}, \dot{\underline{y}}) + \sum_{i=1}^3 n_i \underline{t}(-\underline{e}_i, \dot{\underline{y}}) = 0 \text{ for arbitrary } \dot{\underline{y}}$$

$$\therefore \underline{t}(\underline{n}, \underline{y}) = - \sum_{i=1}^3 \underline{t}(-\underline{e}_i, \underline{y}) n_i \quad \forall \underline{y} \in R_t \quad (3)$$

□

Let $\underline{n} \rightarrow \underline{e}_1, \underline{e}_2$, and $\underline{e}_3$ respectively

$$\left. \begin{aligned} \underline{t}(\underline{e}_1, \underline{y}) &= -\underline{t}(-\underline{e}_1, \underline{y}) \\ \underline{t}(\underline{e}_2, \underline{y}) &= -\underline{t}(-\underline{e}_2, \underline{y}) \\ \underline{t}(\underline{e}_3, \underline{y}) &= -\underline{t}(-\underline{e}_3, \underline{y}) \end{aligned} \right\} \underline{t}(\underline{n}, \underline{y}) = -\underline{t}(-\underline{n}, \underline{y})$$

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... Newton's law of action and reaction

So (3) becomes

$$\underline{t}(\underline{n}, \underline{y}) = \sum_{i=1}^3 n_i \underline{t}(\underline{e}_i, \underline{y}) \quad \forall \underline{y} \in R_t \quad \forall \underline{n} \in u \quad (4)$$

If we repeat the above proof in other quadrants, i.e. n_i may not be positive, the result still holds. Thus one can remove the restriction $n_i > 0$

Define the following 9 real numbers as follows

$$\tau_{ij}^X = \underline{t}(\underline{e}_j, \underline{y}) \cdot \underline{e}_i \quad (5)$$

This is not consistent with the convention we use in solid mechanics $\tau_{ij}^X = \underline{t}(\underline{e}_i, \underline{y}) \cdot \underline{e}_j$

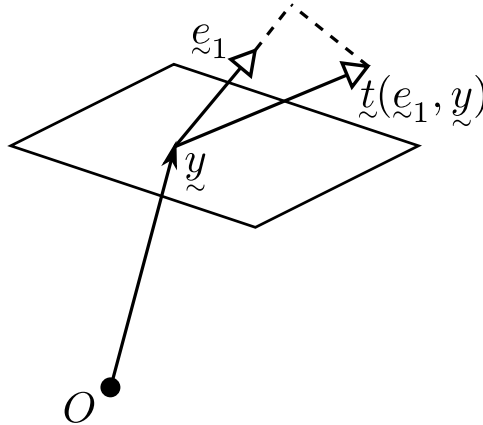


Figure 7.10: Scheme of orientation of stress

e.g.

$$\tau_{11}^X = \underline{t}(\underline{e}_1, \underline{y}) \cdot \underline{e}_1$$

7.13. EXISTENCE OF STRESS

$$\tau_{12}^X = \underline{t}(\underline{e}_2, \underline{y}) \cdot \underline{e}_1$$

dot multiply (4) by $\underline{e}_i$ and use definition (5)

$$t_i^X(\underline{n}, \underline{y}) = \underline{t}^X(\underline{n}, \underline{y}) \cdot \underline{e}_i = \sum_{j=1}^3 \underbrace{\underline{t}^X(\underline{e}_j, \underline{y}) \cdot \underline{e}_i}_{\tau_{ij}^X} n_j^X$$

$$\Rightarrow t_i^X(\underline{n}, \underline{y}) = \tau_{ij}^X n_j^X$$

or

$$t_i^X(\underline{n}, \underline{y}) = \tau_{ij}^X(\underline{y}) n_j^X \quad \forall X \in \mathcal{F}, \forall \underline{y} \in R_t, \forall \underline{n} \in u$$

by quotient rule τ_{ij}^X represent component in X of a tensor τ

$$\Rightarrow \underline{t}(\underline{n}, \underline{y}) = \tau(\underline{y})\underline{n}$$

$\tau(\underline{y}) \cdots$ Cauchy stress tensor field

□

Note.

In general, we release time so that

$$\underline{t}(\underline{n}^{\dagger\dagger\dagger}, \underline{y}, t) = \tau(\underline{y}, t)\underline{n}$$

$^{\dagger\dagger\dagger}\underline{n}(\underline{y}, t)$

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

7.14 Local version of balance laws

Theorem 7.14.1.

The traction field $\underline{t}$ and body force density $\underline{b}$ associated with the motion $\hat{y}(\bullet, t): R \rightarrow R_t$ satisfy linear and angular momentum balance if

- (i) That exists $\underline{\tau}(\bullet, t) \in C^1(R_t)$ such that
$$\underline{t}(\underline{n}, \underline{y}, t) = \underline{\tau}(\underline{y}, t)\underline{n}$$
- (ii) $\nabla \cdot \underline{\tau}(\underline{y}, t) + \underline{b}(\underline{y}, t) = \rho(\underline{y}, t)\bar{\underline{a}}(\underline{y}, t) \quad \forall \underline{y} \in R_t, \forall t \in \tau$
- (iii) $\underline{\tau}(\underline{y}, t) = \underline{\tau}^T(\underline{y}, t)$

Proof.

$$\int_{\partial P_t} \underline{t} dA_{\underline{y}} = \int_{\partial P_t} \underline{\tau} \underline{n} dA_{\underline{y}} = \int_{P_t} [\rho \bar{\underline{a}} - \underline{b}] dV_{\underline{y}}$$

by divergence theorem

$$\Rightarrow \int_{P_t} [\nabla \cdot \underline{\tau} + \underline{b} - \rho \bar{\underline{a}}] dV_{\underline{y}} = 0 \quad \forall P_t \subset R_t$$

by continuity of integration and localization argument

$$\Rightarrow \nabla \cdot \underline{\tau}(\underline{y}, t) + \underline{b}(\underline{y}, t) = \rho(\underline{y}, t)\bar{\underline{a}}(\underline{y}, t) \quad \forall \underline{y} \in R_t \quad (*)$$

or

$$\tau_{ij,j}(\underline{y}, t) + b_i(\underline{y}, t) = \rho(\underline{y}, t)\bar{a}_i(\underline{y}, t)$$

7.14. LOCAL VERSION OF BALANCE LAWS

Also from conservation of angular momentum and linear momentum

$$\int_{\partial P_t} \underline{y} \wedge \underline{\tau} \underline{n} dA_{\underline{y}} + \int_{P_t} \underline{y} \wedge [\underline{b} - \rho \underline{\bar{a}}] dV_{\underline{y}} = \underline{0}$$

Let $B_{il} = \epsilon_{ijk} y_j \tau_{kl}$,^{†††} $\underline{B} = B_{ij} \underline{e}_i \otimes \underline{e}_j$

$$\Rightarrow \int_{\partial P_t} \underline{B} \underline{n} dA_{\underline{y}} + \int_{P_t} \underline{y} \wedge [\underline{b} - \rho \underline{\bar{a}}] dV_{\underline{y}} = \underline{0}$$

by divergence theorem

$$\Rightarrow \int_{P_t} \left\{ \nabla \cdot \underline{B} + \underline{y} \wedge [\underline{b} - \rho \underline{\bar{a}}] \right\} dV_{\underline{y}} = \underline{0}$$

localization argument

$$\Rightarrow \nabla \cdot \underline{B} + \underline{y} \wedge [\underline{b} - \rho \underline{\bar{a}}] = \underline{0}$$

but

$$[\nabla \cdot \underline{B}]_i = \frac{\partial B_{il}}{\partial y_l} = \epsilon_{ijk} \frac{\partial \tau_{kl}}{\partial y_l} y_j + \epsilon_{ijk} \tau_{kl} \delta_{jl}$$

$$\Rightarrow \nabla \cdot \underline{B} = \underline{y} \wedge (\nabla \cdot \underline{\tau}) + \epsilon_{ijk} \tau_{kj} \underline{e}_i$$

$$\therefore \epsilon_{ijk} \tau_{kj} \underline{e}_i + \underline{y} \wedge [\cancel{\nabla \cdot \underline{\tau}} + \underline{b} - \rho \underline{\bar{a}}] \xrightarrow{0 \text{ from } (*)} = \underline{0}$$

^{†††}Define a vector $\underline{g} = \underline{\tau} \underline{n}$ or $g_k = \tau_{kl} n_l$, then $\underline{y} \wedge \underline{\tau} \underline{n} = \underline{y} \wedge \underline{g}$ or $\epsilon_{ijk} y_j \tau_{kl} n_l = \epsilon_{ijk} y_j g_k$.
If we define $\underline{B} = \underline{y} \wedge \underline{\tau}$ or $B_{il} = \epsilon_{ijk} y_j \tau_{kl}$ such that $\underline{y} \wedge \underline{\tau} \underline{n} = \underline{B} \underline{n}$ or $\epsilon_{ijk} y_j \tau_{kl} n_l = B_{il} n_l$

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$$\begin{aligned} \Rightarrow \epsilon_{ijk} \tau_{kj} \ell_i &= 0 \text{ or } \epsilon_{ijk} \tau_{kj} = 0 \\ \Rightarrow \underline{\tau} &= \underline{\tau}^T \cdots \text{symmetric tensor} \end{aligned}$$

□

However, if we analyze ferrofluid, $\underline{\tau} \neq \underline{\tau}^T$ because of the effect of *body moment density* $\underline{m}$. The balance of angular momentum would be rewritten as

$$\int_{P_t} \underline{m} dV_y + \int_{\partial P_t} \underline{y} \wedge \underline{\tau} \underline{n} dA_y + \int_{P_t} \underline{y} \wedge [\underline{b} - \rho \underline{\bar{a}}] dV_y = 0$$

Remark.

- (1) equation of motion (local version of conservation of linear momentum)

$$\nabla \cdot \underline{\tau} + \underline{b} = \rho \underbrace{\left[\frac{\partial \bar{\underline{v}}}{\partial t} + (\nabla \bar{\underline{v}}) \bar{\underline{v}} \right]}_{=\bar{\underline{a}}}$$

- (2) For *rest motions* where $\underline{y} = \hat{\underline{y}}(\underline{x})$ (independent of t) then $\bar{\underline{v}} = \bar{\underline{a}} = 0$.

$$\nabla \cdot \underline{\tau} + \underline{b} = 0 \text{ or } \tau_{ij,j} + b_i = 0 \cdots \text{equilibrium equation}$$

7.14. LOCAL VERSION OF BALANCE LAWS

(3) Since $\underline{\tau} = \underline{\tau}^T$ (symmetric)

$$\underline{\tau} = \sum_{i=1}^3 \tau_i \underline{e}_i \otimes \underline{e}_i$$

where

$\tau_i \cdots$ principal values of $\underline{\tau}$ or principal stresses

$\underline{e}_i \cdots$ principal (spatial) axes or stresses

(4) useful relation:

recall the formula $(\underline{u} \otimes \underline{v}) \underline{w} = \underline{u} (\underline{v} \cdot \underline{w})$

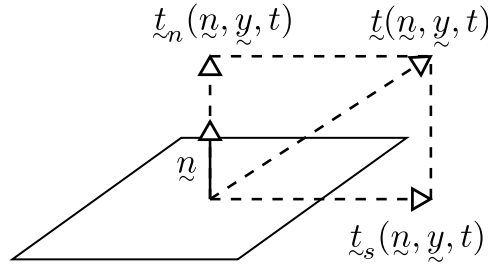


Figure 7.11: Scheme of orientation of traction

$$\begin{cases} \underline{t}_s = \underline{t}(\underline{n}) - (\underline{t}(\underline{n}) \cdot \underline{n}) \underline{n} \\ \underline{t}_n = (\underline{t}(\underline{n}) \cdot \underline{n}) \underline{n} \end{cases} \Rightarrow \begin{cases} \underline{t}_s = (\underline{1} - \underline{n} \otimes \underline{n}) \underline{t} \\ \underline{t}_n = (\underline{n} \otimes \underline{n}) \underline{t} = (\underline{n} \cdot \underline{\tau} \underline{n}) \underline{n} \end{cases}$$

Theorem 7.14.2. *Power theorem*

Let $\hat{y}(\bullet, t): R \rightarrow R_t$ be a motion with spatial velocity $\bar{v}$ and $\underline{D} = \text{sym } \nabla \bar{v}$. Then

CHAPTER 7. KINETICS OF CONTINUOUS MEDIA

$$\int_{\partial P_t} \underline{t}(\underline{n}) \cdot \bar{\underline{v}} dA_y + \int_{P_t} \underline{b} \cdot \bar{\underline{v}} dV_y = \int_{P_t} \underline{\tau} \cdot \underline{D} dV_y + \frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |\bar{\underline{v}}|^2 dV_y$$

Proof.

$$\underline{t}(\underline{n}) \cdot \bar{\underline{v}} = (\underline{\tau} \underline{n}) \cdot \bar{\underline{v}} = \underline{n} \cdot \underline{\tau}^T \bar{\underline{v}} = (\underline{\tau} \bar{\underline{v}}) \cdot \underline{n} \quad (1)$$

L.H.S. of identity to be proven.

$$r(t) \stackrel{(1)}{\underset{\text{divergence theorem}}{=}} \int_{P_t} \left\{ \nabla \cdot (\underline{\tau} \bar{\underline{v}}) + \underline{b} \cdot \bar{\underline{v}} \right\} dV_y$$

$$\nabla \cdot (\underline{\tau} \bar{\underline{v}}) = (\tau_{ij} \bar{v}_j)_{,i} = \tau_{ij,i} \bar{v}_j + \tau_{ij} \bar{v}_{j,i} = \tau_{ji,i} \bar{v}_j + \tau_{ij} \bar{v}_{j,i}$$

$$\begin{aligned} \nabla \cdot (\underline{\tau} \bar{\underline{v}}) &= (\nabla \cdot \underline{\tau}) \cdot \bar{\underline{v}} + \underline{\tau} \cdot \nabla \bar{\underline{v}} \\ &= (\nabla \cdot \underline{\tau}) \cdot \bar{\underline{v}} + \underline{\tau} \cdot [\underline{D} + \underline{W}] \\ &= [\nabla \cdot \underline{\tau}] \cdot \bar{\underline{v}} + \underline{\tau} \cdot \underline{D} \end{aligned}$$

where

$$\underline{D} = \text{sym } \nabla \bar{\underline{v}}$$

$$\underline{W} = \text{skew } \nabla \bar{\underline{v}}$$

$$\begin{aligned} \Rightarrow r(t) &= \int_{P_t} \left\{ \underbrace{[\nabla \cdot \underline{\tau} + \underline{b}]}_{=\rho \bar{\underline{q}}} \cdot \bar{\underline{v}} + \underline{\tau} \cdot \underline{D} \right\} dV_y \\ &= \int_{P_t} \{ \rho \bar{\underline{q}} \cdot \bar{\underline{v}} + \underline{\tau} \cdot \underline{D} \} dV_y \end{aligned}$$

7.14. LOCAL VERSION OF BALANCE LAWS

recall the formula $\frac{1}{2} \frac{D}{Dt} (\bar{\mathbf{v}} \cdot \bar{\mathbf{v}}) = \frac{1}{2} \frac{D}{Dt} |\bar{\mathbf{v}}|^2$

$$\bar{\mathbf{a}} \cdot \bar{\mathbf{v}} = \frac{1}{2} (\bar{\mathbf{v}} \cdot \dot{\bar{\mathbf{v}}}) = \left\{ \frac{1}{2} |\bar{\mathbf{v}}|^2 \right\}^\bullet$$

$$\begin{aligned} r(t) &= \int_{P_t} \boldsymbol{\tau} \cdot \underline{D} dV_{\underline{y}} + \int_{P_t} \rho \left\{ \frac{1}{2} |\bar{\mathbf{v}}|^2 \right\}^\bullet dV_{\underline{y}} \\ &\stackrel{RTT}{=} \int_{P_t} \boldsymbol{\tau} \cdot \underline{D} dV_{\underline{y}} + \frac{d}{dt} \int_{P_t} \frac{1}{2} \rho |\bar{\mathbf{v}}|^2 dV_{\underline{y}} \end{aligned}$$

□

Remark.

$$\begin{aligned} &\int_{\partial P_t} \underline{t} \cdot \bar{\mathbf{v}} dA_{\underline{y}} \cdots \text{power (rate of work) done by traction on } \partial P_t \\ &\int_{P_t} \underline{b} \cdot \bar{\mathbf{v}} dV_{\underline{y}} \cdots \text{power done by body force} \\ &\int_{P_t} \frac{1}{2} \rho |\bar{\mathbf{v}}|^2 dV_{\underline{y}} \cdots \text{kinetic energy} \\ &\int_{P_t} \boldsymbol{\tau} \cdot \underline{D} dV_{\underline{y}} \cdots \text{stress power} \end{aligned}$$

Chapter 8

Thermodynamics of Bodies

Up to this point: *purely mechanical* primitive notions.

Introduce new concepts:

1. Internal energy (specific internal energy)
2. Heating (heat flux and heat supply)

Those will be restricted together with other constituents of the early discussion of motions (dynamic processes) to obey the

- (A) Axiom of energy conservation (the first law of thermodynamics)
further introduce notions of

3. Entropy (or Calory) $\cdots$ specific entropy

8.1. ENERGETIC AXIOMS

4. Absolute temperature

and all of the above will obey a second postulate called the

- (B) Entropy inequality (Clausius-Duhem or Truesdell-Toupin) which is a particular version of the second law of thermodynamics.

8.1 Energetic Axioms

Let $\underline{y} = \hat{\underline{y}}(\underline{x}, t)$ be and admissible motions.

Principle 1.

There exists a scalar field $\varepsilon(\bullet, t): R_t \rightarrow R \quad \forall t \in \tau$ with $\varepsilon \in C^1(R_t \times \tau)$ called the specific internal energy (energy per unit mass) such that

$$\int_{P_t} \varepsilon(\underline{y}, t) \rho(\underline{y}, t) dV_{\underline{y}} = E(P_t)$$

$E(P_t) \cdots$ total internal energy of $P_t \subset R_t$

$\dim \varepsilon = \text{Energy/mass}$

Principle 2.

There exists for each $\underline{n} \in u$ a scalar field $\hbar(\underline{n}, \bullet, t) \in C^1(R_t)$ called *the heat flux* such that

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$$\begin{aligned} \int_{\partial P_t} \hbar(\underline{n}(\underline{y}, t), \underline{y}, t) dA_{\underline{y}} &= \text{total contact heating} \\ &\equiv Q^c \text{ (due to heat conduction)} \end{aligned}$$

$$\begin{aligned} \dim \hbar &= \frac{\text{Energy}}{(\text{unit area})(\text{unit time})} \\ \dim Q^c &= \frac{\text{Energy}}{\text{unit time}} \end{aligned}$$

Principle 3.

There exists a scalar field $r(\bullet, t) \in C(R_t)$ called the specific heat supply (per unit mass) such that

$$\begin{aligned} \int_{P_t} r(\underline{y}, t) \rho(\underline{y}, t) dV_{\underline{y}} &= \text{Body heating} \\ &\equiv Q^r \text{ (due to radiation)} \end{aligned}$$

Total heating of P_t

$$\begin{aligned} Q(P_t) &= Q^c + Q^r = \int_{\partial P_t} \hbar(\underline{n}, \underline{y}, t) dA_{\underline{y}} \\ &\quad + \int_{P_t} r(\underline{y}, t) \rho(\underline{y}, t) dV_{\underline{y}} \end{aligned}$$

For a motion recall also that

$$K(P_t) = \int_{P_t} \frac{1}{2} \rho(\underline{y}, t) \left| \bar{\underline{v}}(\underline{y}, t) \right|^2 dV_{\underline{y}}$$

While the mechanical power $P(P_t)$

8.1. ENERGETIC AXIOMS

$$P(P_t) = \int_{\partial P_t} \underline{t}(\underline{n}(\underline{y}, t), \underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t) dA_{\underline{y}} \\ + \int_{P_t} \underline{b}(\underline{y}, t) \cdot \bar{\underline{v}}(\underline{y}, t) dV_{\underline{y}}$$

Axiom 8.1.1. *Conservation (Balance) of Energy*

$\forall P_t \in R_t$

$$\frac{d}{dt} [E(P_t) + K(P_t)] = P(P_t) + Q(P_t)$$

Namely:

The rate of change of internal + kinetic energy of each subbody equals the mechanical power + the heating.

$$\Rightarrow \frac{d}{dt} \int_{P_t} \rho \left[\varepsilon + \frac{1}{2} \bar{\underline{v}} \cdot \bar{\underline{v}} \right] dV \\ = \int_{P_t} [\underline{b} \cdot \bar{\underline{v}} + r\rho] dV + \int_{\partial P_t} [\underline{h}(\underline{n}) + \underline{t}(\underline{n}) \cdot \bar{\underline{v}}] dA \quad (\textcircled{\text{S}})$$

Theorem 8.1.1. *Local Energy Balance, Heat flux vector*

Let $\hat{\underline{y}}: R \rightarrow R_t$ with mass density $\rho(\bullet, t)$ and spatial velocity $\bar{\underline{v}}(\bullet, t)$ on R_t . Then the specific internal energy $\varepsilon(\bullet, t)$, the heat flux $\underline{h}(\underline{n}, \bullet, t)$ and heat supply $r(\bullet, t)$ on R_t abiding by Principle 1, Principle 2, and Principle 3, are such that $(\textcircled{\text{S}})$ holds if and only if

(i) *There exists a vector field*

CHAPTER 8. THERMODYNAMICS OF BODIES

$$q(\bullet, t): R_t \rightarrow E_3, q(\bullet, t) \in C^1(R_t)$$

such that

$$h(\underline{n}, \underline{y}, t) = -q(\underline{y}, t) \cdot \underline{n} \quad \forall \underline{y} \in R_t \quad \forall \underline{n} \in u$$

(ii)

$$\begin{aligned} \tau(\underline{y}, t) \cdot \underline{D}(\underline{y}, t) - \nabla \cdot q(\underline{y}, t) + \rho(\underline{y}, t)r(\underline{y}, t) \\ = \rho(\underline{y}, t)\dot{\varepsilon}(\underline{y}, t) \quad \forall (\underline{y}, t) \in R_t \times \tau \end{aligned}$$

Proof.

First recall the power identity:

$$\frac{d}{dt}K(P_t) = P(P_t) - \int_{P_t} \tau \cdot \underline{D} dV \quad (1)$$

Also

$$\frac{d}{dt}E(P_t) = \frac{d}{dt} \int_{P_t} \rho \varepsilon dV \stackrel{RTT}{=} \int_{P_t} \rho(\underline{y}, t) \dot{\varepsilon}(\underline{y}, t) dV \quad (2)$$

where $\dot{\varepsilon} = \frac{\partial \varepsilon}{\partial t} + (\nabla \varepsilon) \cdot \bar{\underline{v}}$

Substitute (1) and (2) in (S)

$$\int_{P_t} \rho \dot{\varepsilon} dV = \int_{\partial P_t} h(\underline{n}) dA + \int_{P_t} [\rho r + \tau \cdot \underline{D}] dV \quad (3)$$

Repeat proof used in obtaining $\underline{t} = \tau \underline{n}$, i.e. choose P_t to be a tetrahedron $P_t(h)$, with height h and let $h \rightarrow 0$.

8.1. ENERGETIC AXIOMS

$$\Rightarrow \hbar(\underline{n}, \underline{y}, t) = -\underline{q}(\underline{y}, t) \cdot \underline{n} \quad (4)$$

$$(3) \& (4) \Rightarrow \int_{P_t} \rho \dot{\varepsilon} dV = - \int_{\partial P_t} \underline{q} \cdot \underline{n} dA + \int_{P_t} [\rho r + \underline{\tau} \cdot \underline{D}] dV$$

$$\xRightarrow[\text{theorem}]{\text{divergence}} \int_{P_t} [\rho \dot{\varepsilon} + \nabla \cdot \underline{q} - \rho r - \underline{\tau} \cdot \underline{D}] dV = 0$$

Smoothness of ρ , ε , $\hbar$ (or $\underline{q}$), r , $\dots$ and localization argument gives

$$\underline{\tau} \cdot \underline{D} - \nabla \cdot \underline{q} + \rho r = \rho \dot{\varepsilon}$$

□

Consequences of the

First law

$$\int_{P_t} \rho \dot{\varepsilon} dV = \int_{\partial P_t} \underbrace{\hbar(\underline{n})}_{\text{heat flux}} dA + \int_{P_t} \left[\underbrace{\rho r}_{\text{specific heat supply}} + \underline{\tau} \cdot \underline{D} \right] dV$$

$$\Rightarrow \hbar(\underline{n}, \underline{y}, t) = - \underbrace{\underline{q}(\underline{y}, t)}_{\text{heat flux vector}} \cdot \underline{n}$$

also

$$\boxed{\underline{\tau} \cdot \underline{D} - \nabla \cdot \underline{q} + \rho r = \rho \dot{\varepsilon}} \cdots \text{local version of 1}^{st} \text{ law} \quad (*)$$

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8.2 Motivation and comparison with classical thermodynamics

A thermodynamics consider processes which are homogeneous (independent of position $\underline{y}$).

Internal energy	:	$\varepsilon(t)$
Pressure	:	$p(t)$
Volume	:	$v(t)$
Heating	:	$\tilde{q}(t)$

(no distinction between conduction and radiation)

The first law (compare to (*))

$$\begin{aligned}\dot{\varepsilon} &= p\dot{v} + \tilde{q} \\ \dot{\varepsilon} \cdots \rho \dot{\varepsilon}(\underline{y}, t) \\ p\dot{v} \cdots \underline{\tau}(\underline{y}, t) \cdot \underline{D}(\underline{y}, t) \\ \tilde{q} \cdots - \nabla \cdot \underline{q}(\underline{y}, t) + \rho(\underline{y}, t)r(\underline{y}, t)\end{aligned}$$

For the second law one introduces an entropy $\eta(t)$ and an absolute temperature $\theta(t) > 0$. The second law assumes an upper bound for the quantity $\tilde{q}(t)/\theta(t)$, i.e.

$$\dot{\eta}(t) \geq \frac{\tilde{q}(t)}{\theta(t)}$$

(Entropy rate exceeds or equals heating over temperature)

First and second law

$$\Rightarrow \theta \dot{\eta} - (\dot{\varepsilon} - p\dot{v}) \geq 0$$

8.3. AXIOMS OF ENTROPY AND TEMPERATURE

(Entropy rate $\times$ temperature $\geq$ internal energy rate – mechanical power ^{*})

8.3 Axioms of entropy and temperature

Principle 4.

Given a motion $\hat{y}(\bullet, t): R \rightarrow R_t$, $t \in [t_0, t_1] = \tau$, define a scalar field $\eta(\bullet, t): R_t \rightarrow \mathbb{R}$ called the specific entropy $\eta \in C^1(R_t)$ such that

$$\int_{P_t} \rho(\underline{y}, t) \eta(\underline{y}, t) dV_{\underline{y}} = S(P_t) \cdots \text{total entropy, } P_t \subset R_t$$

A strictly positive scalar field $\theta(\bullet, t): R_t \rightarrow \mathbb{R}$ ($\theta(\underline{y}, t) > 0 \quad \forall \underline{y} \in R_t \quad \forall t \in \tau$), $\theta \in C^1(R_t)$ called the absolute temperature field. The following axiom is commonly accepted as a statement of the second law in continuous mechanics.

$$\frac{dS(P_t)}{dt} \geq - \int_{\partial P_t} \frac{1}{\theta(\underline{y}, t)} \underline{q}(\underline{y}, t) \cdot \underline{n}(\underline{y}, t) dA_{\underline{y}} + \int_{P_t} \frac{\rho(\underline{y}, t) r(\underline{y}, t)}{\theta(\underline{y}, t)} dV_{\underline{y}}$$

where

$$S(P_t) = \int_{P_t} \rho(\underline{y}, t) \eta(\underline{y}, t) dV_{\underline{y}}$$

^{*}rate of energy *not* produced by mechanical work which is reversible process

CHAPTER 8. THERMODYNAMICS OF BODIES

Remark.

Call the entropy production rate of P_t

$$\Gamma(P_t) \equiv \frac{d}{dt} \int_{P_t} \rho \eta dV + \int_{\partial P_t} \frac{\underline{q} \cdot \underline{n}}{\theta} dA - \int_{P_t} \frac{\rho r}{\theta} dV$$

Thus second law (Clausius-Duhem)

$$\Leftrightarrow \Gamma(P_t) \geq 0 \quad P_t \subset R_t \quad t \in \tau$$

recall that $\frac{d}{dt} \int_{P_t} \rho \eta dV = \int_{P_t} \rho \dot{\eta} dV$ by RTT

where $\dot{\eta} = \frac{\partial \eta}{\partial t} + \nabla \eta \cdot \underline{\bar{v}}$

by using the divergence theorem and the above

$$\Rightarrow \rho \dot{\eta} + \nabla \cdot \left(\frac{\underline{q}}{\theta} \right) - \frac{\rho r}{\theta} \geq 0$$

$$\Rightarrow \boxed{\rho \dot{\eta} + \frac{1}{\theta} \nabla \cdot \underline{q} - \frac{1}{\theta^2} \underline{q} \cdot \nabla \theta - \frac{\rho r}{\theta} \geq 0} \quad (**)$$

recall (*) (local version of the first law, $\nabla \cdot \underline{q} = -\rho \dot{\varepsilon} + \underline{\tau} \cdot \underline{D} + \rho r$)

$$(*), (**) \Rightarrow \boxed{\rho \theta \dot{\eta} - \rho \dot{\varepsilon} + \underline{\tau} \cdot \underline{D} - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0}$$

... reduced dissipation inequality

Theorem 8.3.1. (*Dissipation Inequalities*)(Local)

The Clausius-Duhem inequality is equivalent to

(i) *local entropy production inequality*

8.3. AXIOMS OF ENTROPY AND TEMPERATURE

$$\rho\dot{\eta} + \nabla \cdot \left(\frac{1}{\theta} \underline{q} \right) - \frac{\rho r}{\theta} \geq 0 \quad (\text{i})$$

(ii) *reduced dissipation inequality (the first law + the second law)*

$$\rho\theta\dot{\eta} - (\rho\dot{\varepsilon} - \underline{\tau} \cdot \underline{D}) - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0 \quad (\text{ii})$$

Note. equivalent in classical thermodynamics

$$\theta\dot{\eta} - (\dot{\varepsilon} - p\dot{v}) \geq 0$$

(i.e., $\nabla \theta = 0$, homogeneous assumption)

Remark.

- The continuum and classical versions of dissipation inequalities are equivalent as expected.
- The term $\frac{1}{\theta} \underline{q} \cdot \nabla \theta$ is missed in classical thermodynamics because of the homogeneity assumption.

Thermodynamics deal with *equilibrium system* only.

Define the *internal dissipation* rate δ as

$$\delta = \theta\dot{\eta} - \left[\dot{\varepsilon} - \frac{1}{\rho} \underline{\tau} \cdot \underline{D} \right]$$

and the *conductive entropy production* density ξ as

CHAPTER 8. THERMODYNAMICS OF BODIES

$$\xi = -\frac{1}{\theta} \underline{q} \cdot \nabla \theta$$

Entropy production

$$\Gamma(P_t) = \int_{P_t} \frac{\rho(\underline{y}, t) \delta(\underline{y}, t) + \xi(\underline{y}, t)}{\theta(\underline{y}, t)} dV_{\underline{y}} \geq 0$$

is the sum of the internally dissipation and heat conduction pants. The inequalities $\delta \geq 0$ and $\xi \geq 0$ are known as the Clausius-Plank and the Fourier Inequality (in heat transfer). We do not view each of them as valid separately in a general process where both mechanical and conductive effects occurs. These inequalities are not implied by the second law.

Specific *Helmholtz free energy* $\psi(\bullet, t): R_t \rightarrow \mathbb{R}$
define

$$\psi(\underline{y}, t) = \varepsilon(\underline{y}, t) - \theta(\underline{y}, t) \eta(\underline{y}, t) \quad \forall \underline{y} \in R_t \quad \forall t \in \tau$$

Take Lagrangian derivative

$$\dot{\psi} = \dot{\varepsilon} - \dot{\theta} \eta - \theta \dot{\eta} \tag{\star}$$

(i), (ii), and $(\star) \Rightarrow$

$$-\rho \left(\dot{\psi} + \eta \dot{\theta} \right) + \underline{\tau} \cdot \underline{D} - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0$$

Remark.

8.3. AXIOMS OF ENTROPY AND TEMPERATURE

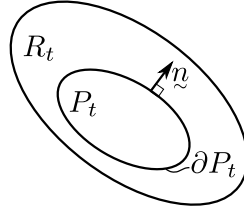


Figure 8.1: Scheme for P_t

- (1) We say that P_t is thermally isolated if $\underline{q} \cdot \underline{n} = 0$ on ∂P_t and $r = 0$ on P_t (closed) system.

$$\frac{d}{dt}S(P_t) \geq 0 \quad (\text{in classical thermodynamics, } \dot{\eta} \geq \frac{\tilde{q}}{\theta} = 0)$$

i.e., entropy cannot decrease under thermal isolation.

- (2) Special cases

$$\dot{\varepsilon} = \dot{\eta} = 0 \quad \text{or} \quad \dot{\psi} = \dot{\theta} = 0 \quad \rightarrow \text{steady state}$$

$$\Rightarrow \underline{\tau} \cdot \underline{D} - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0$$

case (i) Non-conductive material ($\underline{q} = \underline{0}$ or θ is uniform, $\nabla \theta = 0$)

$$\Rightarrow \underline{\tau} \cdot \underline{D} \geq 0 \quad (\text{non-negative stress-power density})$$

CHAPTER 8. THERMODYNAMICS OF BODIES

case (ii) For a rest motion $\underline{D} = \underline{0}$

$$\Rightarrow -\frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0 \Rightarrow \underline{q} \cdot \nabla \theta \leq 0 \dots \text{Fourier's inequality}$$

(3) If $\underline{q} = \underline{0}$ (adiabatic)

$$\begin{aligned} \delta &= \theta \dot{\eta} - \left(\dot{\varepsilon} - \frac{1}{\rho} \underline{\tau} \cdot \underline{D} \right) \geq 0 \\ &= - \left(\dot{\psi} + \eta \dot{\theta} \right) + \frac{1}{\rho} \underline{\tau} \cdot \underline{D} \geq 0 \end{aligned}$$

$$\text{If the process is } \textit{isentropic} \ (\dot{\eta} = 0) \quad \Rightarrow \quad \frac{1}{\rho} \underline{\tau} \cdot \underline{D} \geq \dot{\varepsilon}$$

$$\text{If the process is } \textit{isothermal} \ (\dot{\theta} = 0) \quad \Rightarrow \quad \frac{1}{\rho} \underline{\tau} \cdot \underline{D} \geq \dot{\psi}$$

8.4 Thermodynamic process

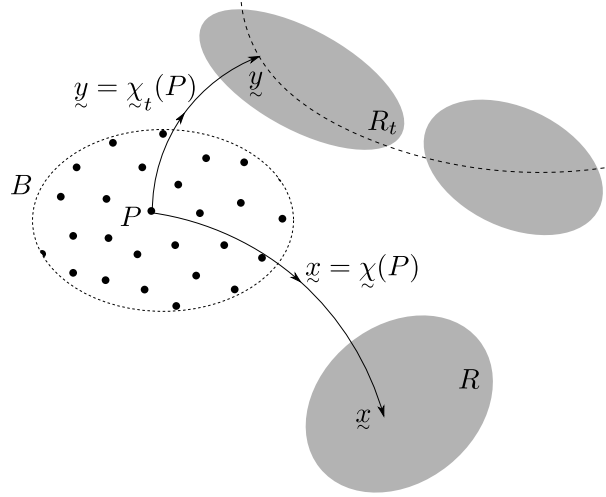


Figure 8.2: Scheme for χ_t

Definition 8.4.1.

Let $\chi_t: B \rightarrow E_3$, $t \in \tau(\chi_t \in \Omega(B))$ be a motion of body B with $R_t = \chi_t(B)$. The ordered array of 8 elements

$[\chi_t, \tau, \underline{b}, \varepsilon, \underline{q}, r, \eta, \theta]$ is a thermodynamic process undergone by B during τ if $[\chi_t, \tau, \underline{b}]$ is a dynamical process and $\varepsilon, \underline{q}, r, \eta, \theta$ satisfy balance of energy and the entropy inequality in which ρ is governed by mass balance.

8.5 Fundamental field equations for a thermodynamic process

Let $\bar{\mathbf{v}}(\bullet, t): R_t \rightarrow E_3$ is spatial (Eulerian) velocity field due to the motion and assume that the density $\rho \in C^1(\mathfrak{m})$,
 $\mathfrak{m} = \left\{ (\underline{y}, t) \mid \underline{y} \in R_t, t \in \tau \right\}$ then $\left[\chi_t, \tau, \underline{b}, \varepsilon, \underline{q}, r, \eta, \theta \right]$ is a thermodynamic process *if and only if*:

$$(0) \quad \dot{\rho} + \rho \nabla \cdot \bar{\mathbf{v}} = 0 \cdots (\text{mass balance})$$

$$(1) \quad \nabla \cdot \tau + \underline{b} = \rho \bar{\mathbf{a}} \cdots (\text{linear momentum balance})$$

$$(2) \quad \tau = \tau^T \cdots (\text{angular momentum balance})$$

$$(3) \quad \rho \dot{\varepsilon} = -\nabla \cdot \underline{q} + \tau \cdot \underline{D} + \rho r \cdots (\text{the first law: energy balance})$$

$$(4) \quad \rho \dot{\eta} + \nabla \cdot \left(\frac{1}{\theta} \underline{q} \right) - \frac{\rho r}{\theta} \geq 0 \cdots (\text{the second law: energy inequality})$$

in the above (4) is interchangeable with

$$(i) \quad \rho \theta \dot{\eta} - \rho \dot{\varepsilon} + \tau \cdot \underline{D} - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0$$

$$(ii) \quad -\rho \left(\dot{\psi} + \eta \dot{\theta} \right) + \tau \cdot \underline{D} - \frac{1}{\theta} \underline{q} \cdot \nabla \theta \geq 0$$

where $\bar{\mathbf{a}} = \dot{\bar{\mathbf{v}}}$, $\underline{D} = \text{sym } \nabla \bar{\mathbf{v}}$ on $\mathfrak{m}$

(0)-(4) are the fundamental field equations for thermodynamic process.

8.5. FUNDAMENTAL FIELD EQUATIONS FOR A THERMODYNAMIC PROCESS

Remark.

The above are valid for all bodies and admissible motions. Given a *motion* and a *temperature field*, that exists infinitely many choice of $\mathcal{T}$, ε , q , η , $\mathcal{b}$, and r such that (1)-(4) hold. Moreover for all different types of bodies subjected to the same motion and temperature fields, in general, will undergo different thermodynamic processes. Therefore, the constitutive laws are needed.

Note.

We assumed that ρ obeys mass balance.

Appendices

Appendix A

Some definitions concerning curves, surfaces, and regions

1. A curve $C: \vec{r} = \vec{r}(t)$, $t \in [a, b]$ is a *smooth curve* if $d\vec{r}(t)/dt$ exists, is continuous and is non-zero on $[a, b]$.
2. A curve C is *piecewise smooth* if it is continuous and consists of a finite number of smooth curves.
3. A curve C is *simple* if it is not self-intersecting.
4. A curve C is *regular* if it is simple and piecewise smooth.
5. A surface S is *smooth* if at each point of S there exists a unique unit normal vector which varies continuously along S . S is *piecewise smooth* if it consists of a finite number

of smooth pieces.

6. A smooth surface is *orientable* if it is two-side, i.e. it is possible on the entire surface to identify a positive normal direction in a unique and continuous way.
7. A surface is *regular* if it is orientable and piecewise smooth.
8. A set of points S is any collection of points in E_3 . The *complementary of S* is the set of all points in E_3 that are not in S .
9. S is *bounded* if all points of S lie within a sphere of finite radius. Otherwise, S is *unbounded*.
10. A δ neighborhood of a point P is the set of points lying inside a sphere of radius δ centered at P .
11. A point $P \in S$ is an *interior point* of S if there is a neighborhood of P lying entirely in S .
12. A set S is *open* if every point in S is an interior point.
13. A set S is *connected* if any two points in S can be connected by a finite-length path lying entirely in S .
14. A *domain* (or open region) is an open connected set.
15. A point P (not necessary in S) is a *limit point* (or *accumulation point*) if every neighborhood of P (no

APPENDIX A. SOME DEFINITIONS CONCERNING CURVES, SURFACES, AND REGIONS

matter how small it is) contains points of S distinct from P .

16. A set is *closed* if it contains all of its limit points.
17. A set is *compact* if it is both closed and bounded.
18. The *closure* of S , denoted by $\bar{S}$, is the union of S and all of its limit points.
19. A *boundary point* of S is a limit point which is not an interior point. The boundary of S is denoted by ∂S . The interior of S is denoted by $\overset{\circ}{S}$.
20. A region is a domain together with all, some, or none of its boundary points. In other words, a region may be neither open nor closed.
21. A region R is *simply-connected* if every closed curve *lying entirely in R* can be continuously deformed to a point without leaving R . Otherwise, it is *multiply-connected*.

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